

Uncertainty-aware Spatial Perception for Generalizable State Estimation

Yuheng Qiu

Committee Members:

- Sebastian Scherer, Carnegie Mellon University (Chair)
- Aaron Johnson, Carnegie Mellon University
- Wenshan Wang, Carnegie Mellon University
- Kostas Daniilidis, University of Pennsylvania

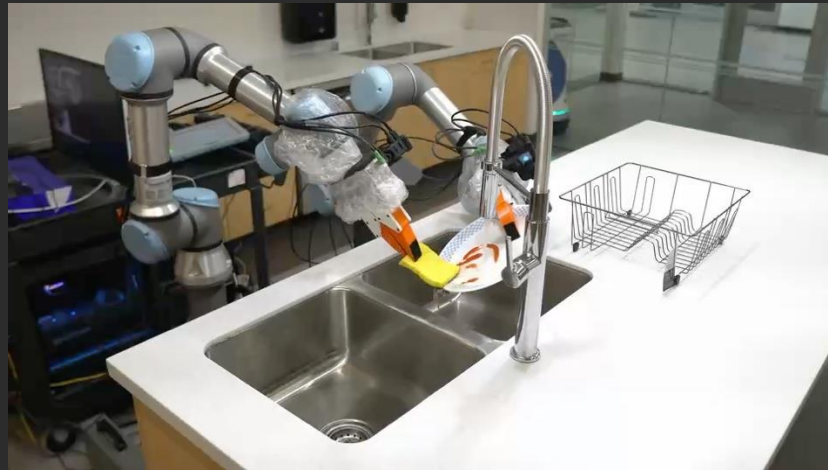
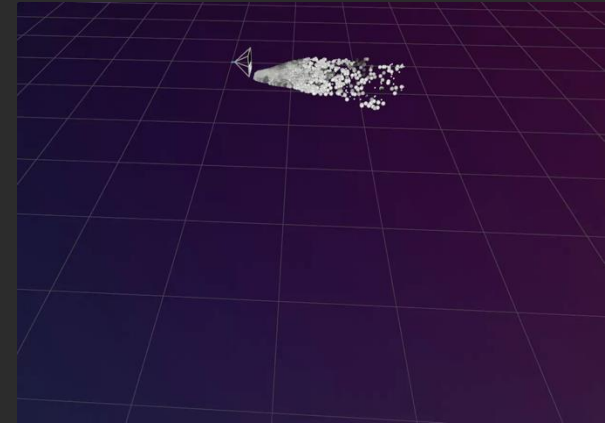
Spatial Perception for Robots



Drone Exploration and Navigation



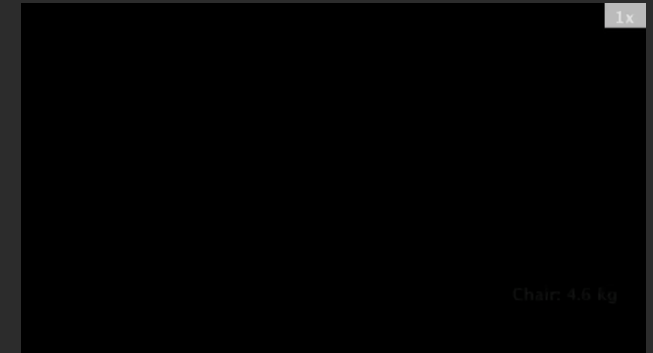
Space Robotics



Grasping and Acting



Autonomous Driving



Humanoid

Challenges in Uncertain World

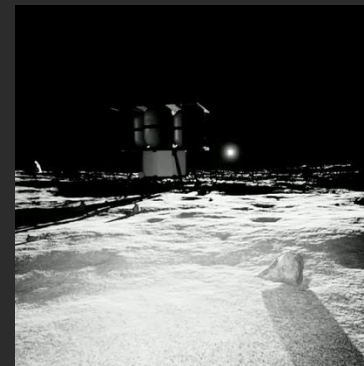
Highly Uncertain, Hard-to-model World



Dynamic Objects



Illumination Changes



Featureless



Rigorous Motion

Inconsistency between environments and payloads



Indoor office reflection

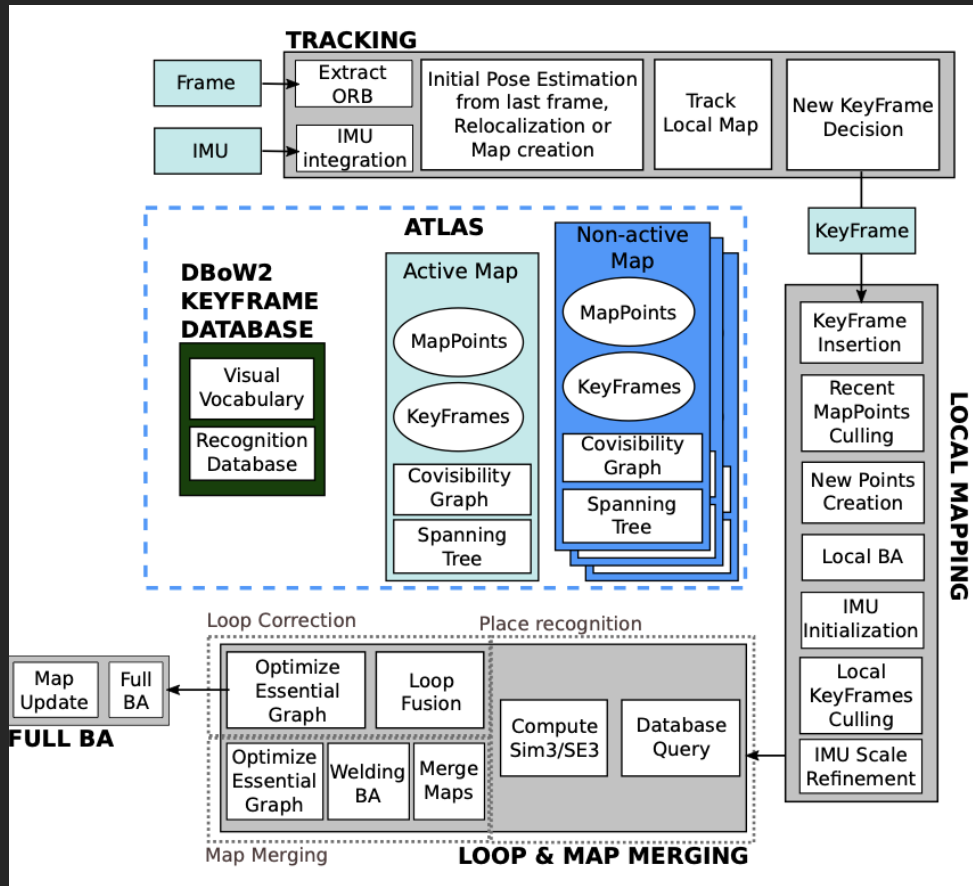


Occlusion



Thermal Payload

Background: Toward Generalizable SLAM



ORBSLAM3 (Campos, et al., 2020)

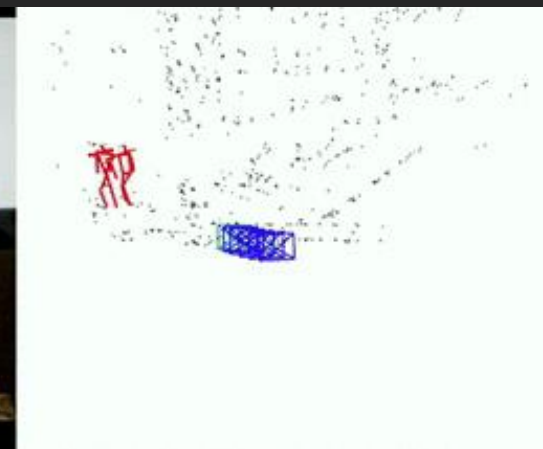
- Multi-stage Optimization
- RanSAC
- Filtering
- Loop Closure
- Multi-Sensor Fusion
- Semantic Information

AirDOS My Initial Attempt to Dynamic World

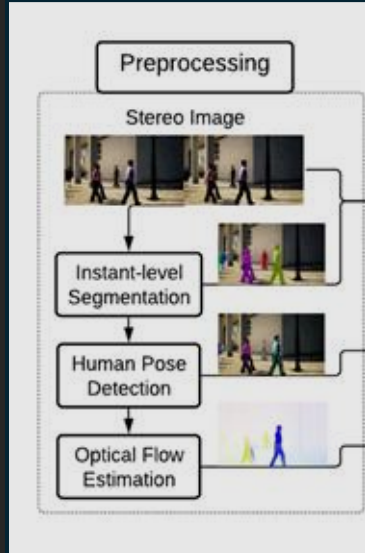
Motivation: To navigate through highly dynamic environments



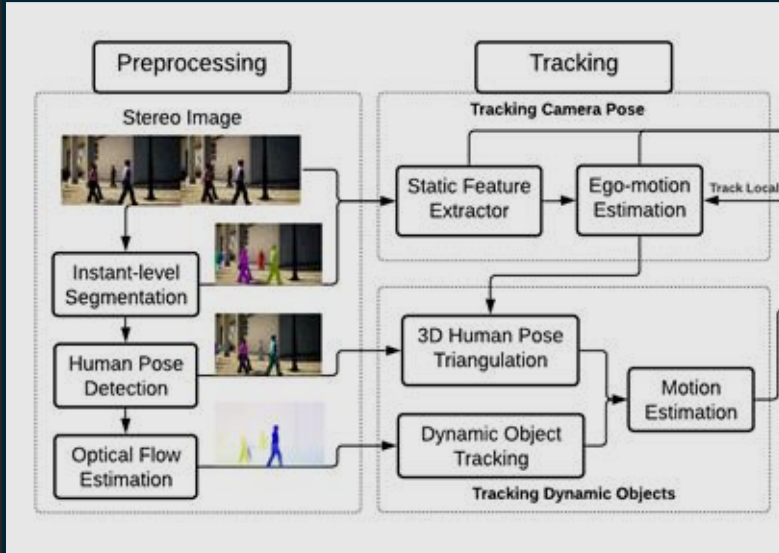
Modeling Dynamic Object as Articulated Object In Dynamic Environment



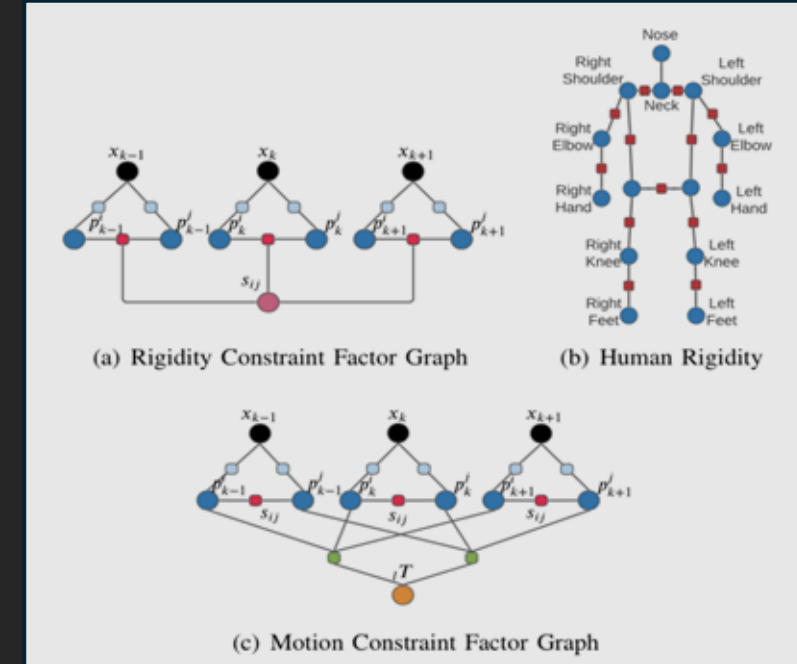
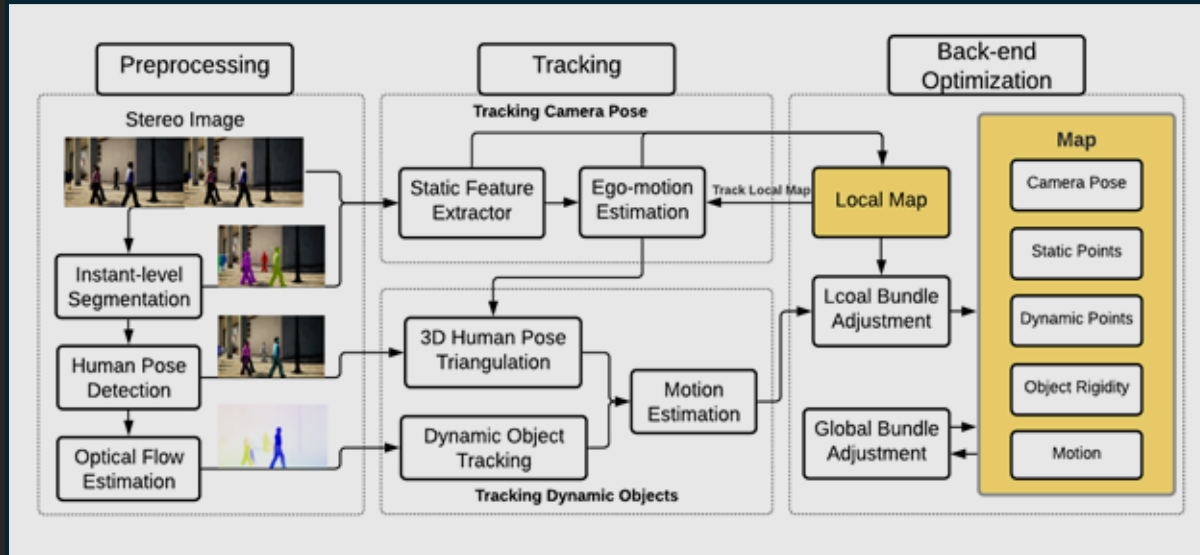
Bitter Lessons from AirDOS: Manual Tuning or Data Driven?



Bitter Lessons from AirDOS: Manual Tuning or Data Driven?



Bitter Lessons from AirDOS: Manual Tuning or Data Driven?



- We have to tune the uncertainty of different constraint to form a complicated back-end optimization.
- There is no guarantee that this system can generalize to any dynamic scenes without tuning.



Bitter Lessons from SLAM



Yuheng, is the odometry working?

Not yet.. I need to tune the hyper parameter.



Okay.. Can we deploy on drone next week?



No I need to collect more data for tuning..



One day Yuheng graduate....



Data-driven is the Key

- Consistent across Different Environment
- No need for Hyper Parameter Tuning
 - State Estimation that Generalizable



Uncertainty is all you need

- The covariance should be metrics-aware
- Guarantee Optimal Result
 - Multi-sensor Fusion

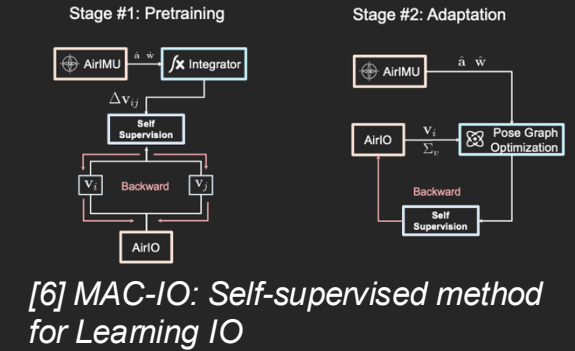
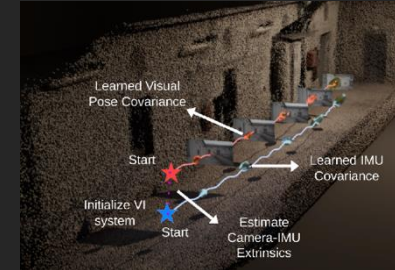
Outline for the Talks



[2] MACVO: Learning Stereo Visual Odometry



[5] MAC-I²: Metrics-aware covariance for IMU initialization and Calibration



Visual Perception

[Part. 1]



Learned Uncertainty



Visual-Inertial Fusion & Adaption

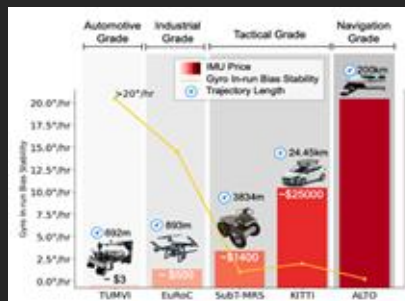
[Part. 3]



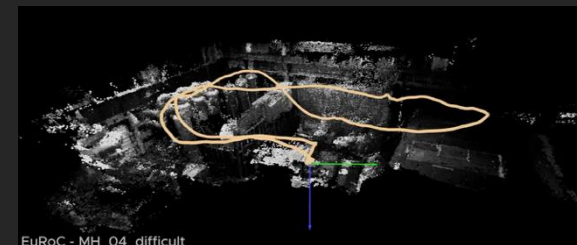
Inertial Perception

[Part. 2]

[3] AirIMU: Learning IMU sensor model

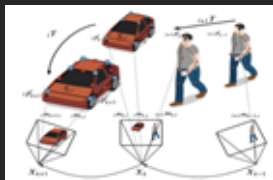


[4] AirIO: Learning IO with enhanced observability



[Part. I]: Visual Perception with learned metrics-aware covariance

[1] AirDOS: Dynamic Object SLAM



[6] TartanAir: Synthetics Datasets



motivate

train



[2] MACVO: Learning Stereo Visual Odometry

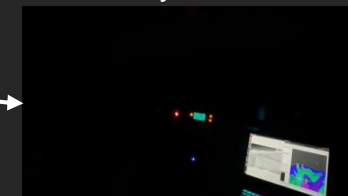
Accelerate

support

[3] QuantMACVO: Quantization for Efficient Learning VO



MAC-Thermal: Passive night-time Autonomy



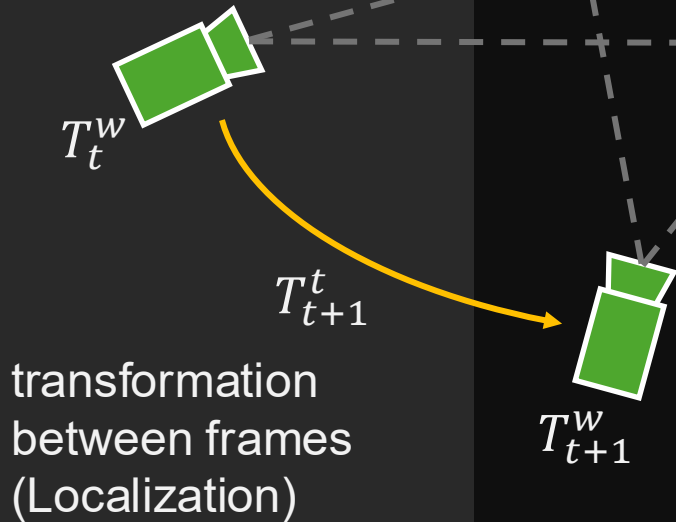
Problem

Simultaneously Optimize 3D Geometry and Camera Pose from Correspondence

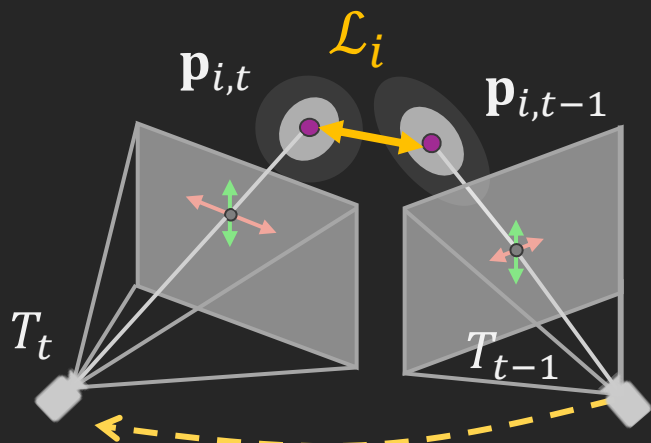
Landmark Feature
(Mapping)

Feature Correspondence is Perfect

Matched Features are Static

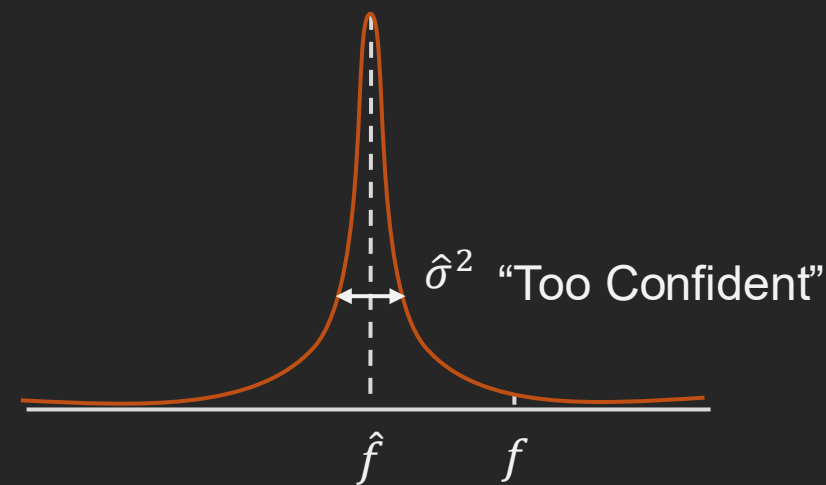
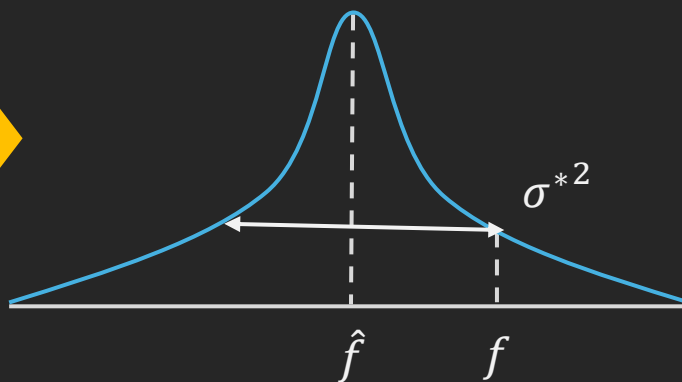


Metrics-aware Covariance

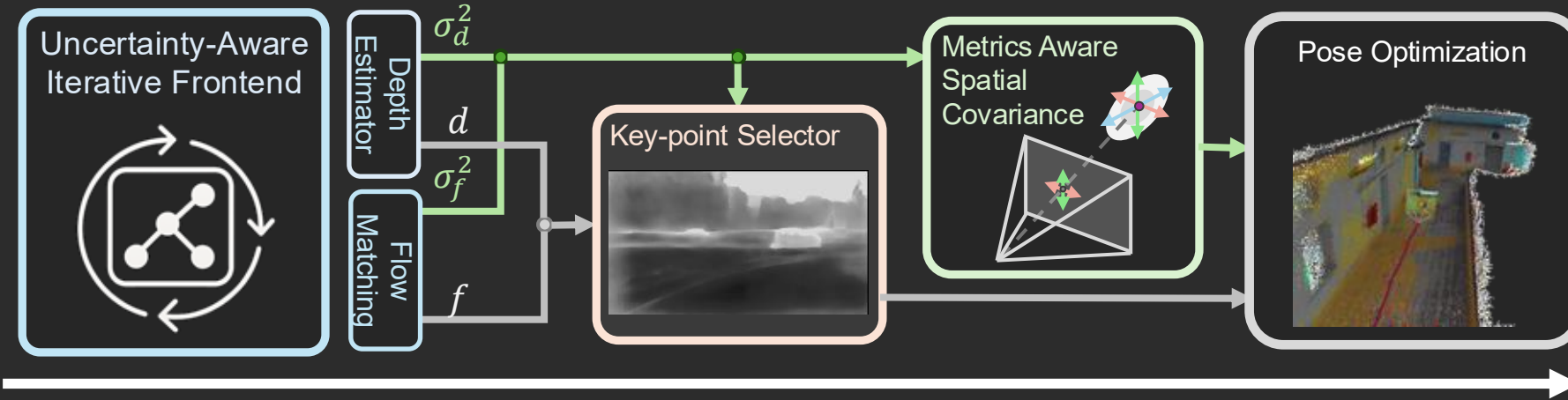


- Scale of the covariance should have physical meaning
- Assume Gaussian Distribution for the observation

“Too Uncertain”



MAC-VO: Metrics-Aware Covariance for Learning-based Stereo Visual Odometry



Front-end: Hard-to-model

Back-end: Deterministic



Modular System



Differentiable



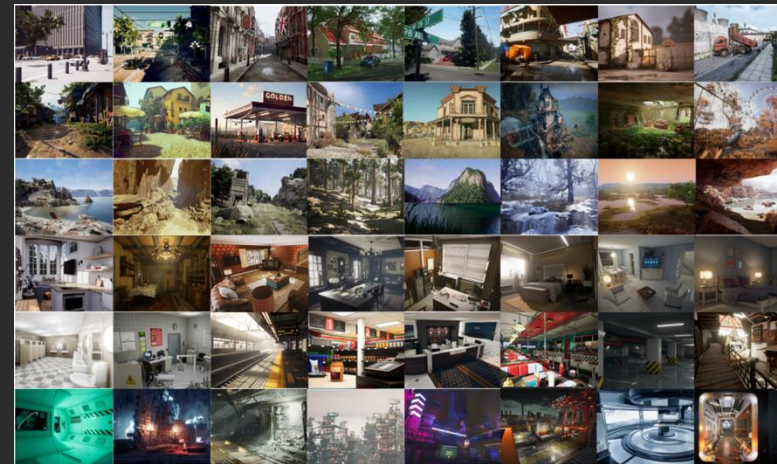
Efficient



Sim to Real

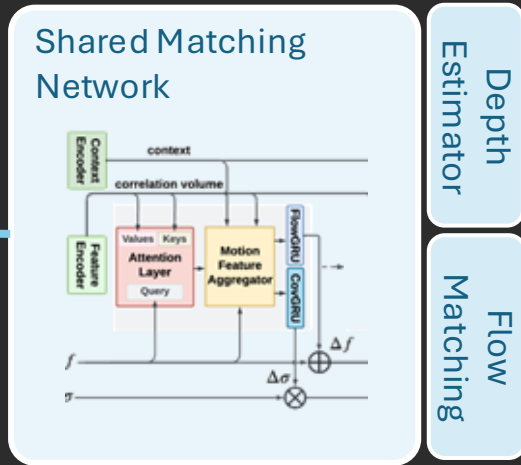
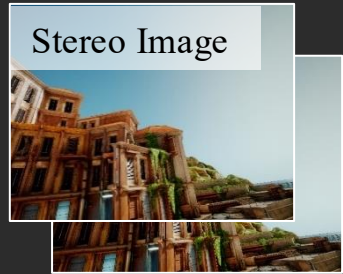


Scalable



Instead of tuning the feature matching model, we use large scale data synthetic data to tune the uncertainty head of matching front-end.

Uncertainty Training



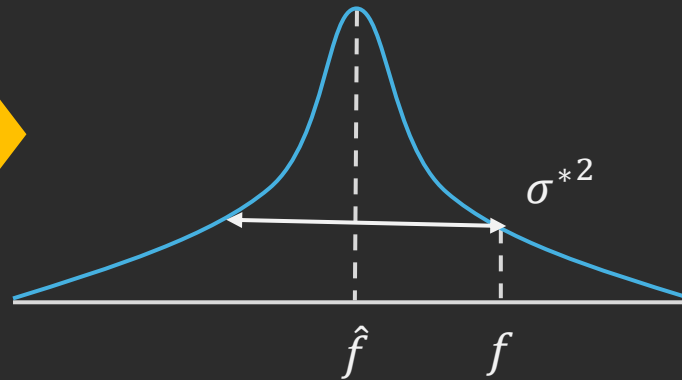
Loss Function: $\mathcal{L}_i = (f - \hat{f}_i)^T \widehat{\Sigma}_f^{-1} (f - \hat{f}_i) + \log(\det \widehat{\Sigma}_f)$

Motivation: Learn covariance that maximize the log-likelihood of ground truth flow f in $\mathcal{N}(\hat{f}, \widehat{\Sigma}_f)$

“Too Uncertain”

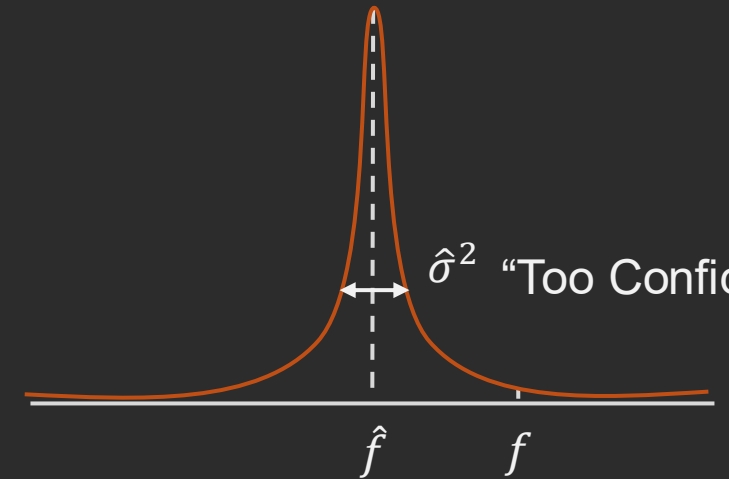


$$(f - \hat{f}_i)^T \widehat{\Sigma}_f^{-1} (f - \hat{f}_i) + \log(\det \widehat{\Sigma}_f)$$



Minimum Loss

$$\hat{\sigma}^2 = |f - \hat{f}|^2$$



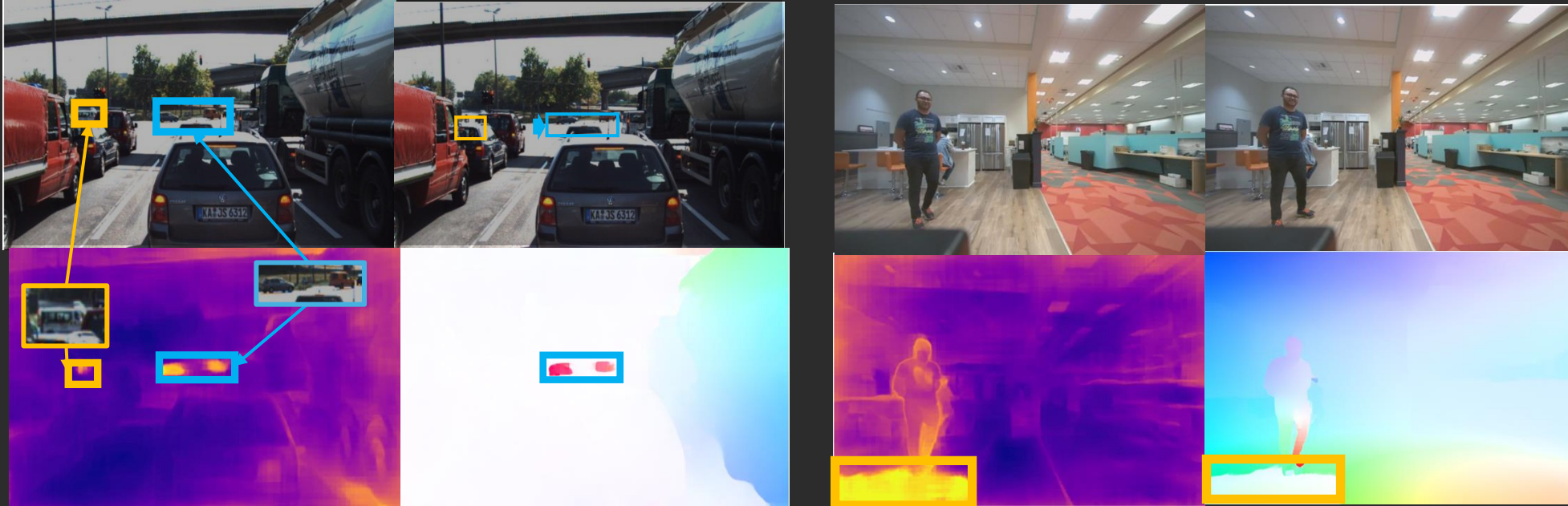
“Too Confident”

$$(f - \hat{f}_i)^T \widehat{\Sigma}_f^{-1} (f - \hat{f}_i) + \log(\det \widehat{\Sigma}_f)$$

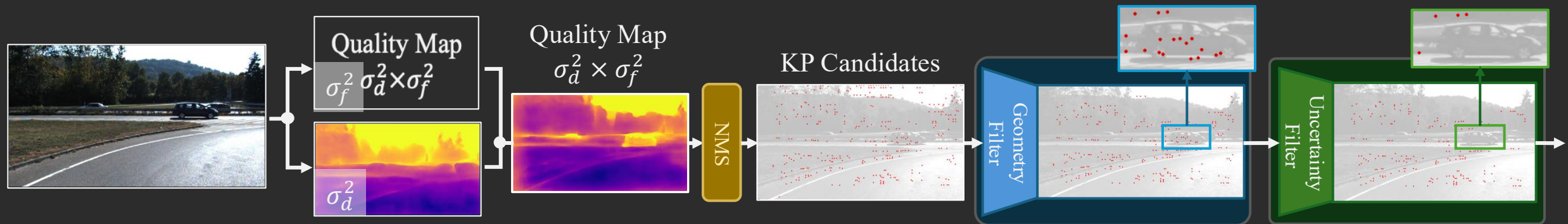
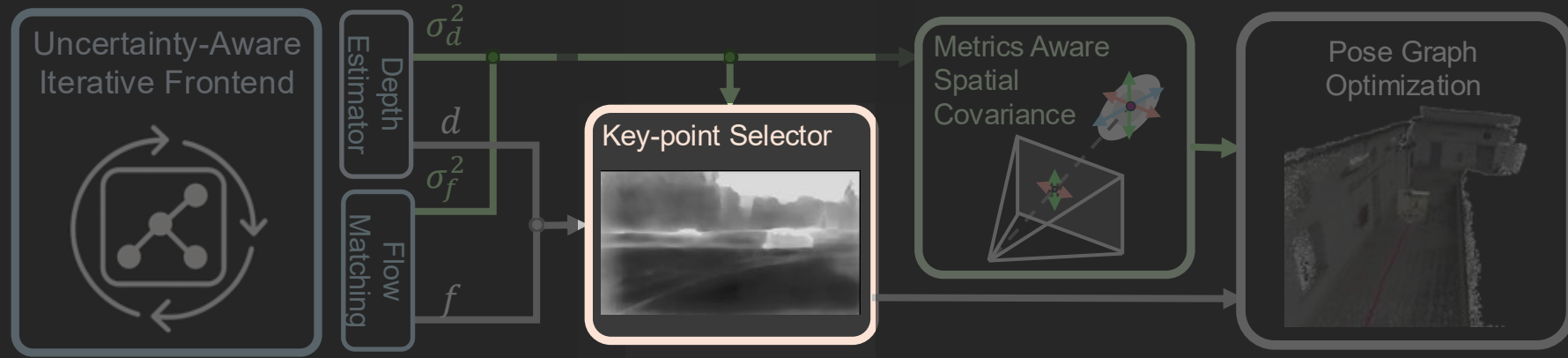
How does the Uncertainty Work?

Dynamic objects introduce higher flow with higher uncertainty

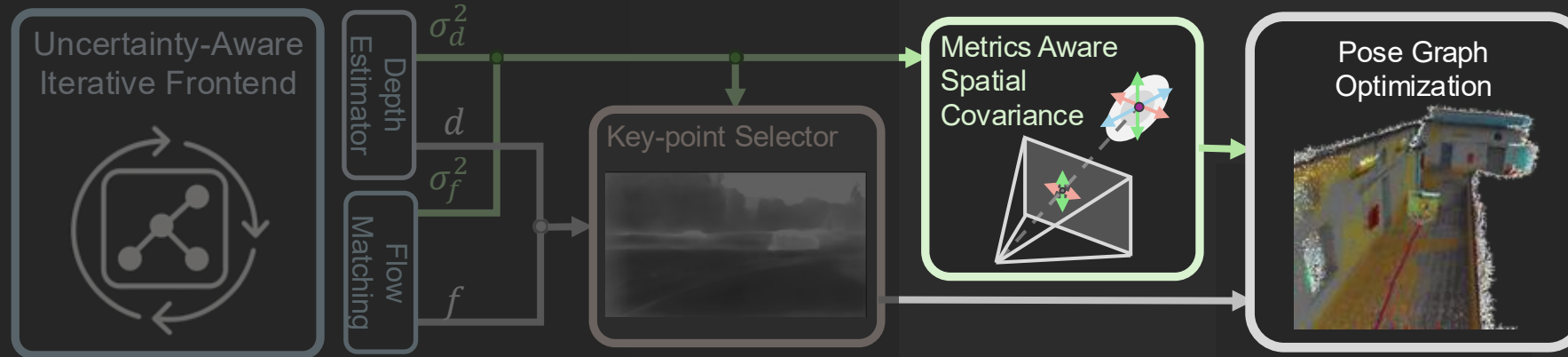
Uncertainty coming with the motion inconsistency



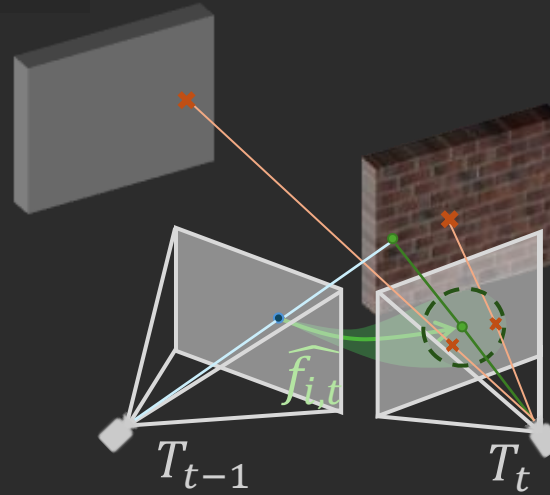
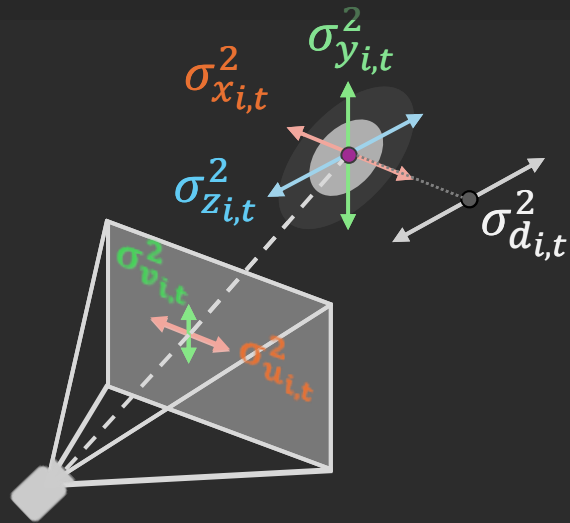
Uncertainty-Aware Keypoint Selector



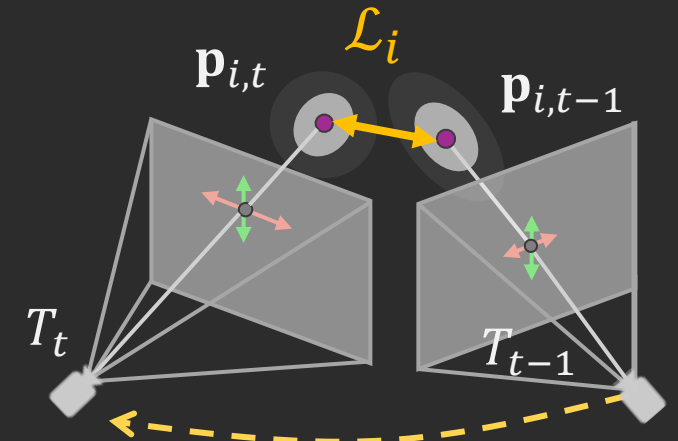
Metrics-Aware Covariance Model



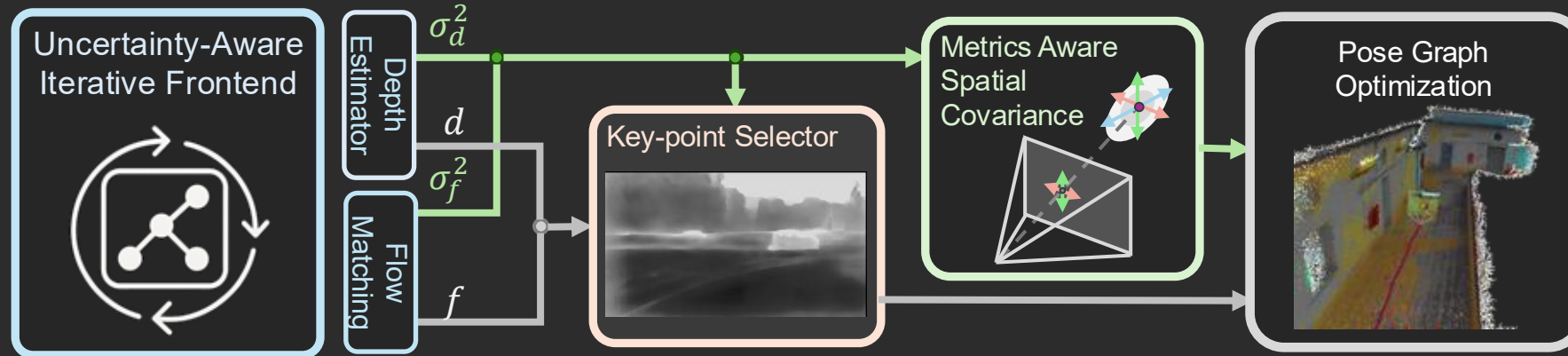
$$T^* = \operatorname{argmin}_{T_t} \sum_i |p_{i,t-1} - T_t p_{i,t}|_{\Sigma_i} = \operatorname{argmin}_{T_t} \sum_i \mathcal{L}_i$$



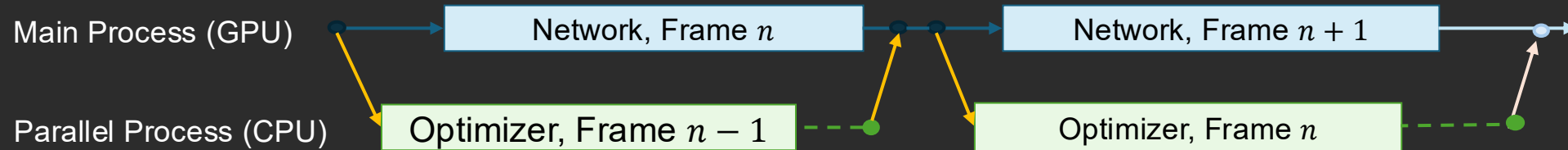
Matching Uncertainty Adjustment



System Design and Real-world Deployment



Running the network on the GPU in parallel with optimization on the CPU

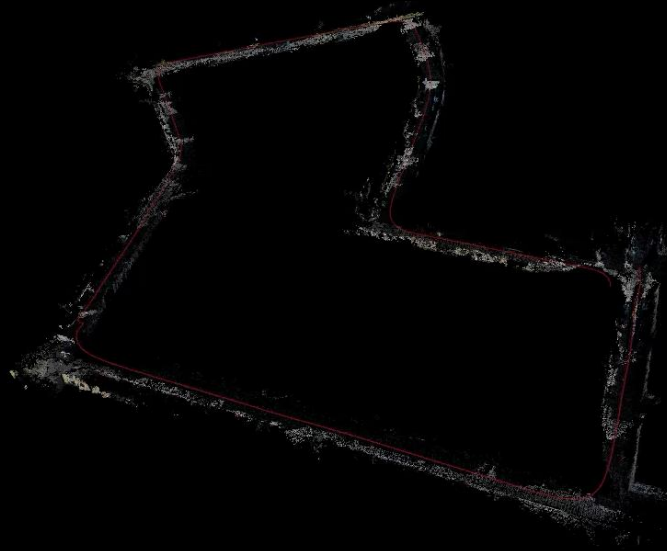


Generalization on Different Environments

EuRoC V102



KITTI Odometry 07



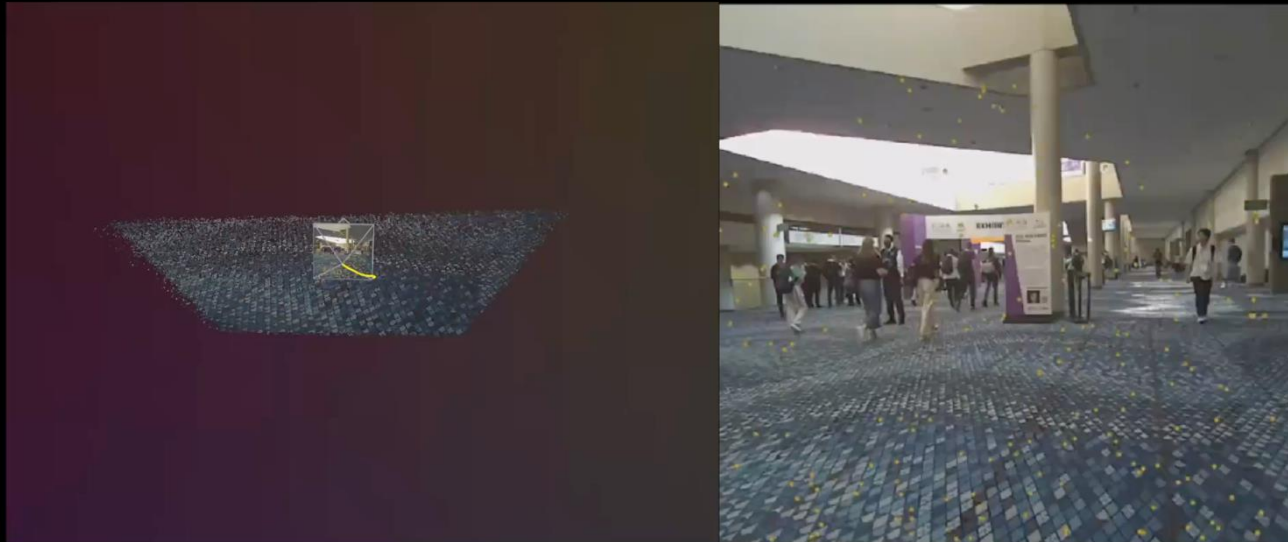
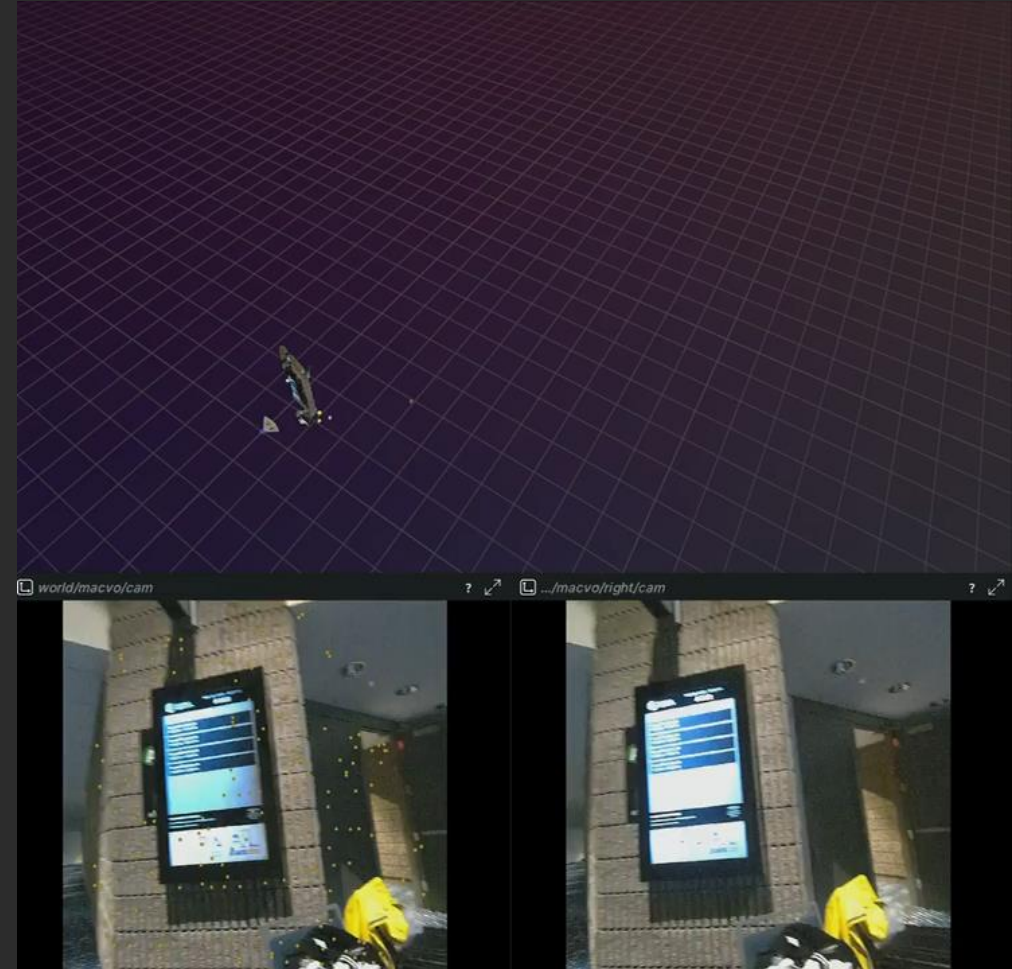
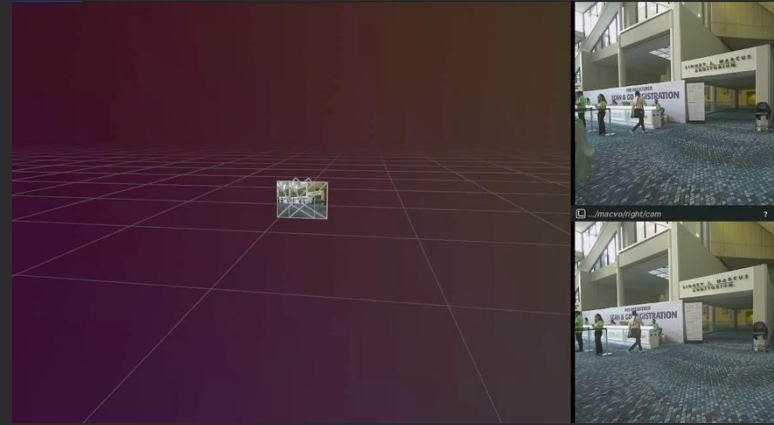
TartanAir
AbandonedFactory 001



MAC-VO generalize across diverse environments and payloads:

- Without model finetuning
- Without task-specific hyperparameter finetuning
- Without multi-frame optimization and loop closure

Generalize Anytime Anywhere during ICRA

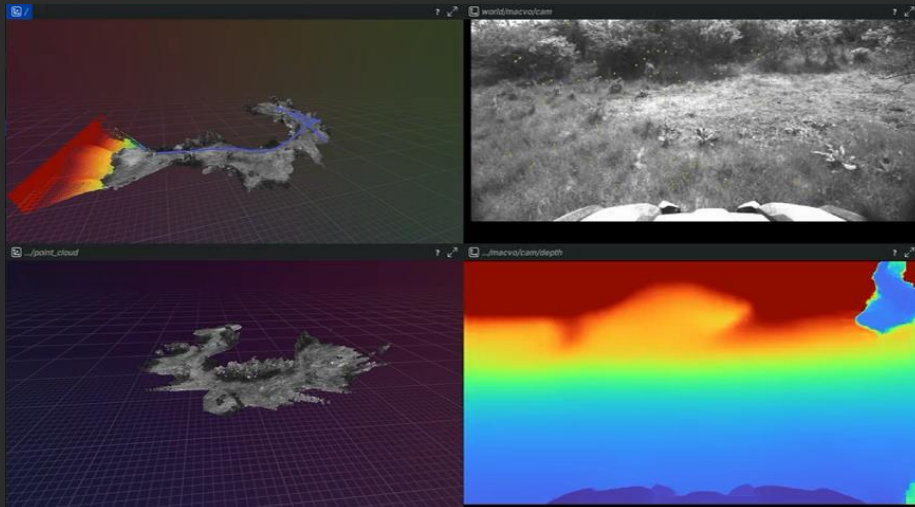


MAC-Thermal: Generalization on Different Payload

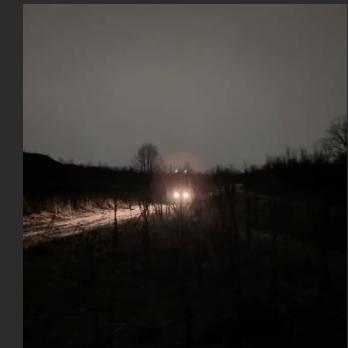
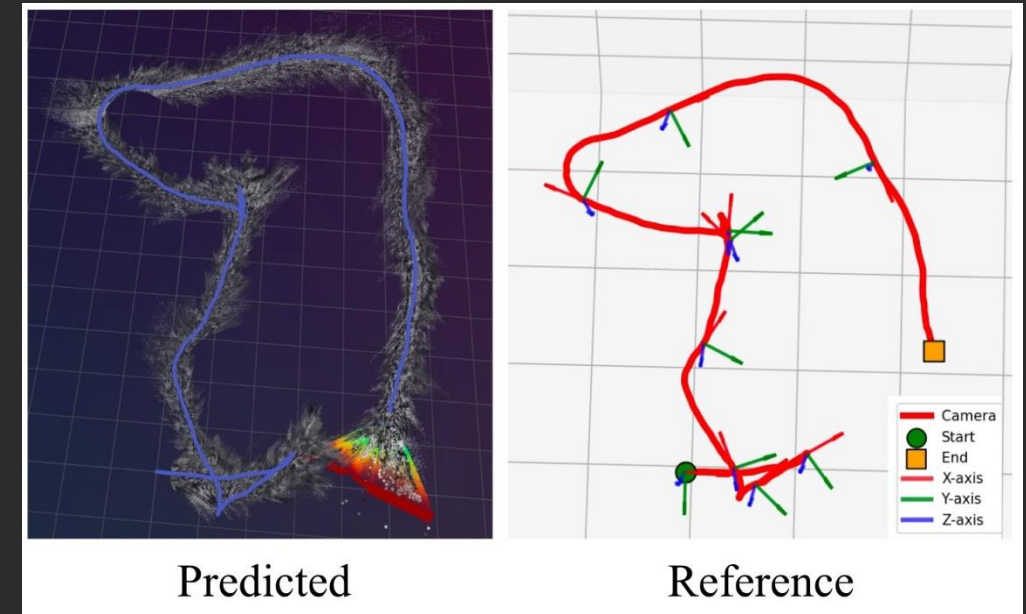
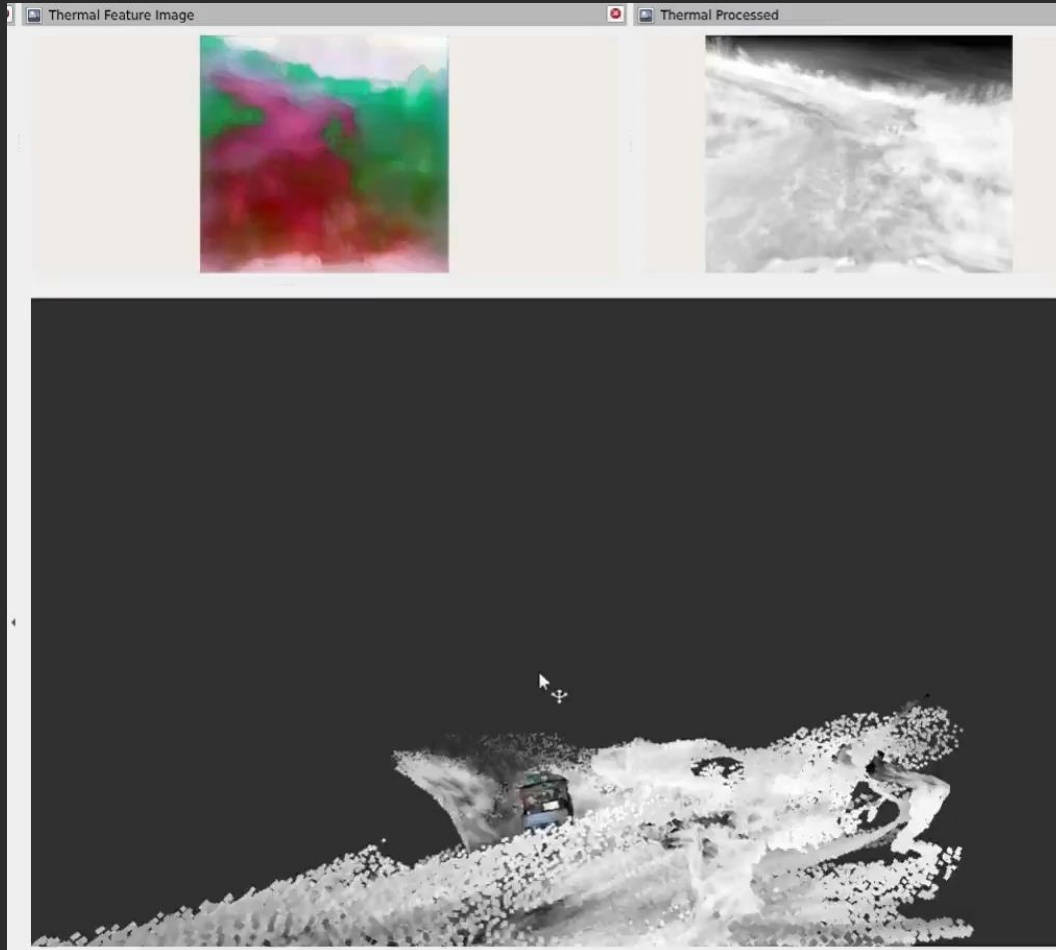


Thermal Camera

- MAC-VO zero-shot generate to thermal data
- Thermal Image introduce blurry
- **Uncertainty** is the key to select the feature correspondence



MAC-Thermal: Toward Fully Passive Night-time Off-Road Driving



Adapt the uncertainty for Mac-Thermal to support the passive night-time off-road driving

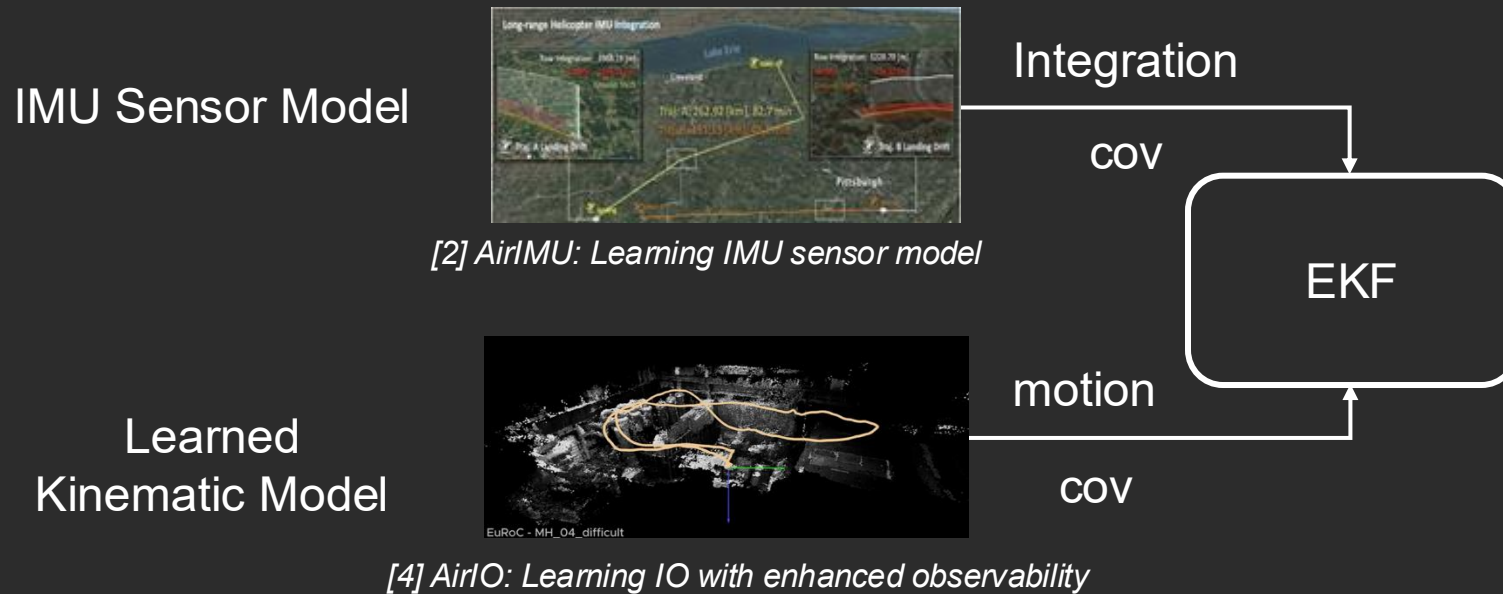
“Scaling law” in SLAM: Image Resolution

MAC-VO meets Jetson Thor



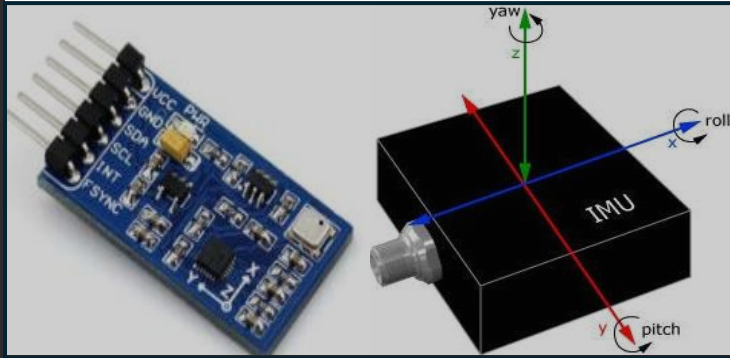
AIRLAB

[Part. II]: Inertial Perception with learned uncertainty



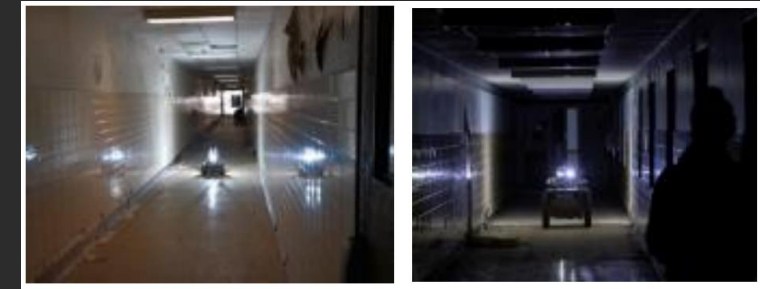
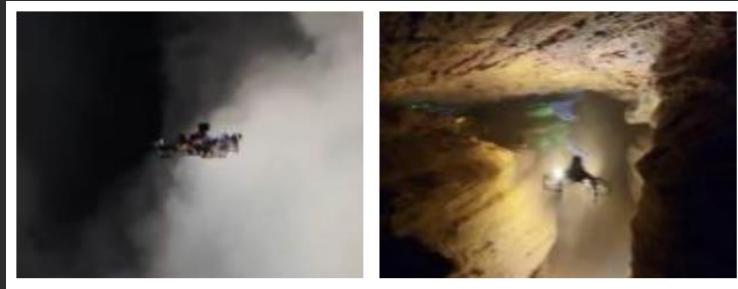
Background: The Advantage of IMU

Inertial Measurements Units (IMU)



- **Fundamental** Acceleration & Angular Velocity
- **Popular** Any smart device, cars, flights
- **Low-cost** Cheap IMU only cost \$2
- **Light-weight** Easier for UAV
- **Perceptive** Independent from External Environment

Camera may be failed, Lidar may be blocked, but IMU will not.



Problems: IMU Noise and Uncertainty

Discrete Form for IMU Pre-integration:

$$\Delta R_{ij} = \prod_{k=i}^{j-1} \text{Exp} \left({}^B \tilde{\omega}_k \Delta t \right)$$

$$\Delta v_{ij} = \sum_{k=i}^{j-1} \Delta R_{ik} {}^B \tilde{a}_k \Delta t$$

$$\Delta p_{ij} = \sum_{k=i}^{j-1} \left(\Delta v_{ik} \Delta t + \frac{1}{2} \Delta R_{ik} {}^B \tilde{a}_k \Delta t^2 \right)$$

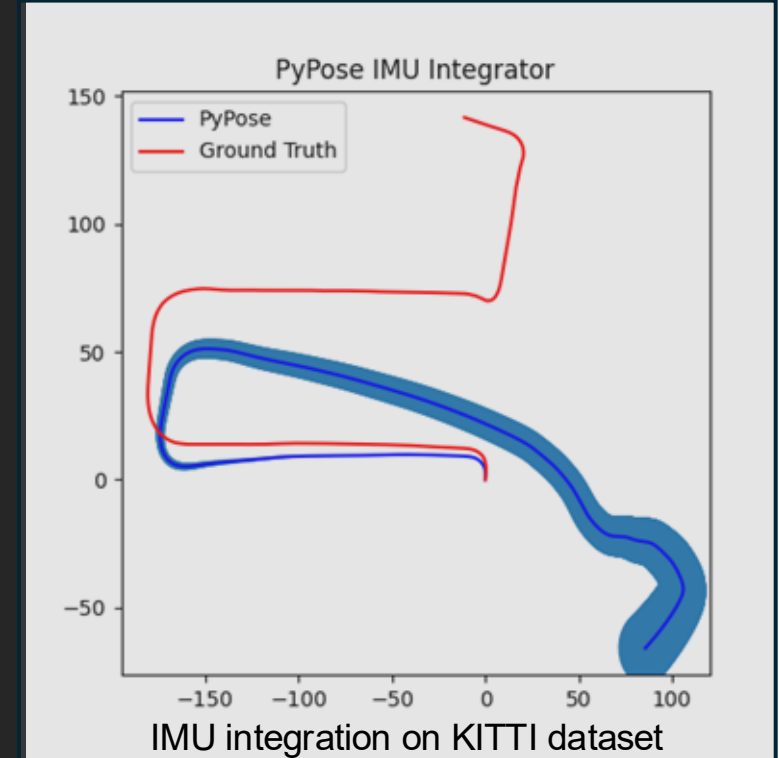
Problems: IMU Noise and Uncertainty

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Drift is unavoidable due to the integration and IMU noise.

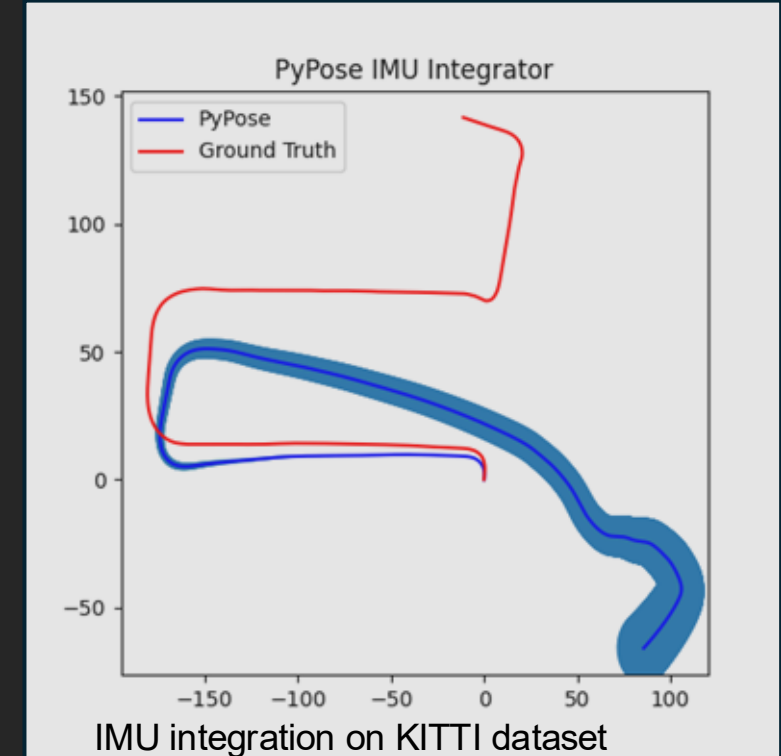
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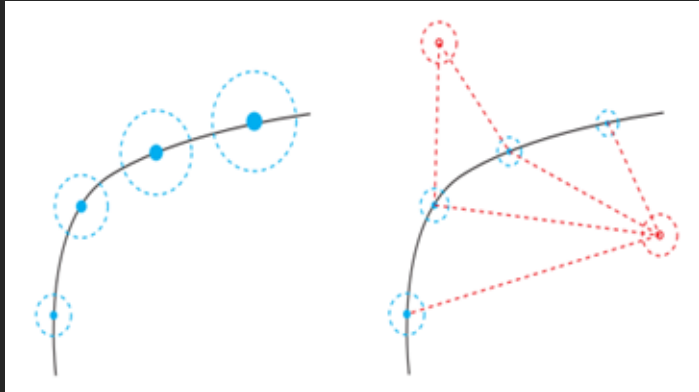
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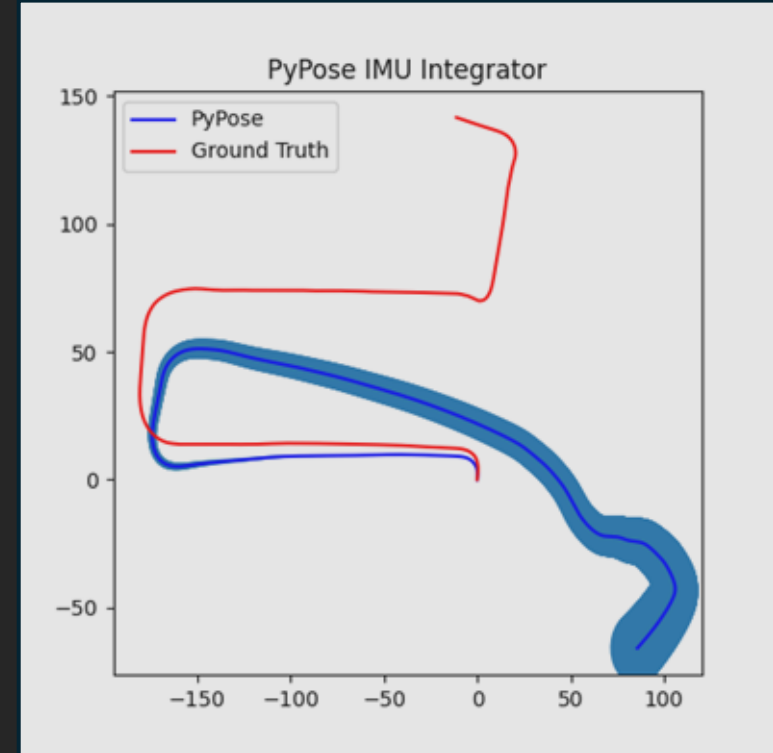


Drift is unavoidable due to the integration and IMU noise.

Problems: IMU Noise and Uncertainty

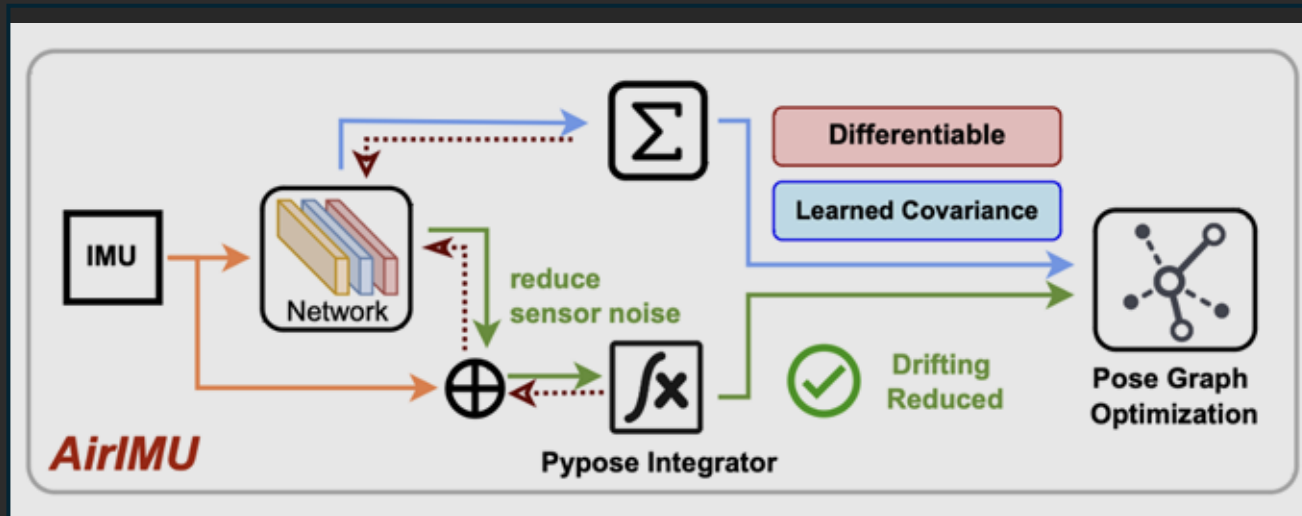


Uncertainty determines how long and how well you can trust the IMU



Drift is unavoidable due to the integration and IMU noise.

AirIMU: Model and Approach

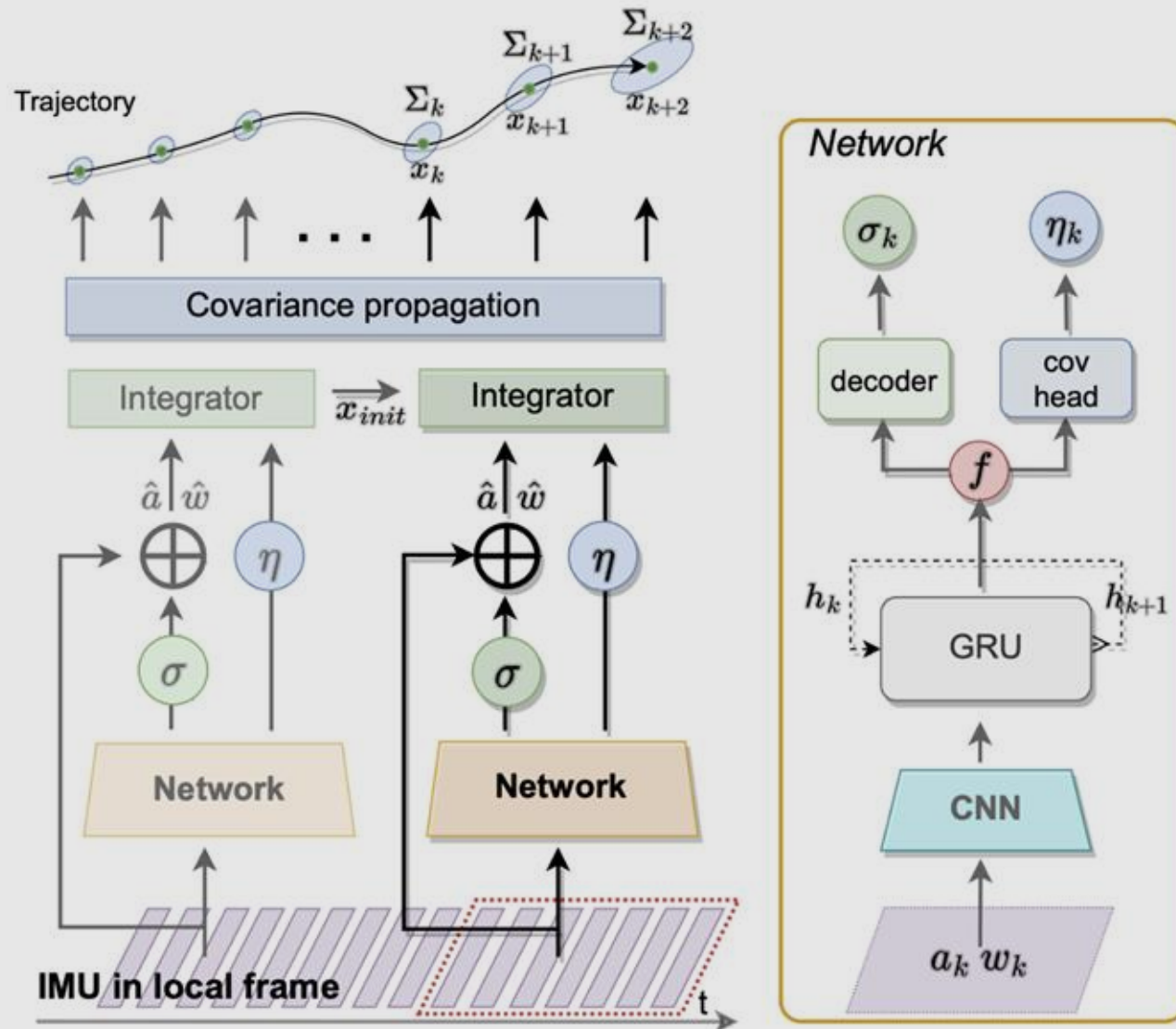


Key Idea:

Learning the IMU sensor noise and the uncertainty through Integration and covariance propagation



AirIMU: Model and Approach



1. We design a shared CNN-GRU encoder to encode raw IMU.
2. To supervise covariance model we build a differentiable covariance propagation method.

$$L = \underbrace{\frac{(y - f(x))^2}{2\Sigma(x)}}_{\text{Error Term}} + \underbrace{\frac{1}{2} \ln |\Sigma(x)|}_{\text{Covariance Term}} + \underbrace{\frac{1}{2} \ln |\Sigma(x)|}_{\text{Regularization Term}}$$

Datasets and Benchmarks

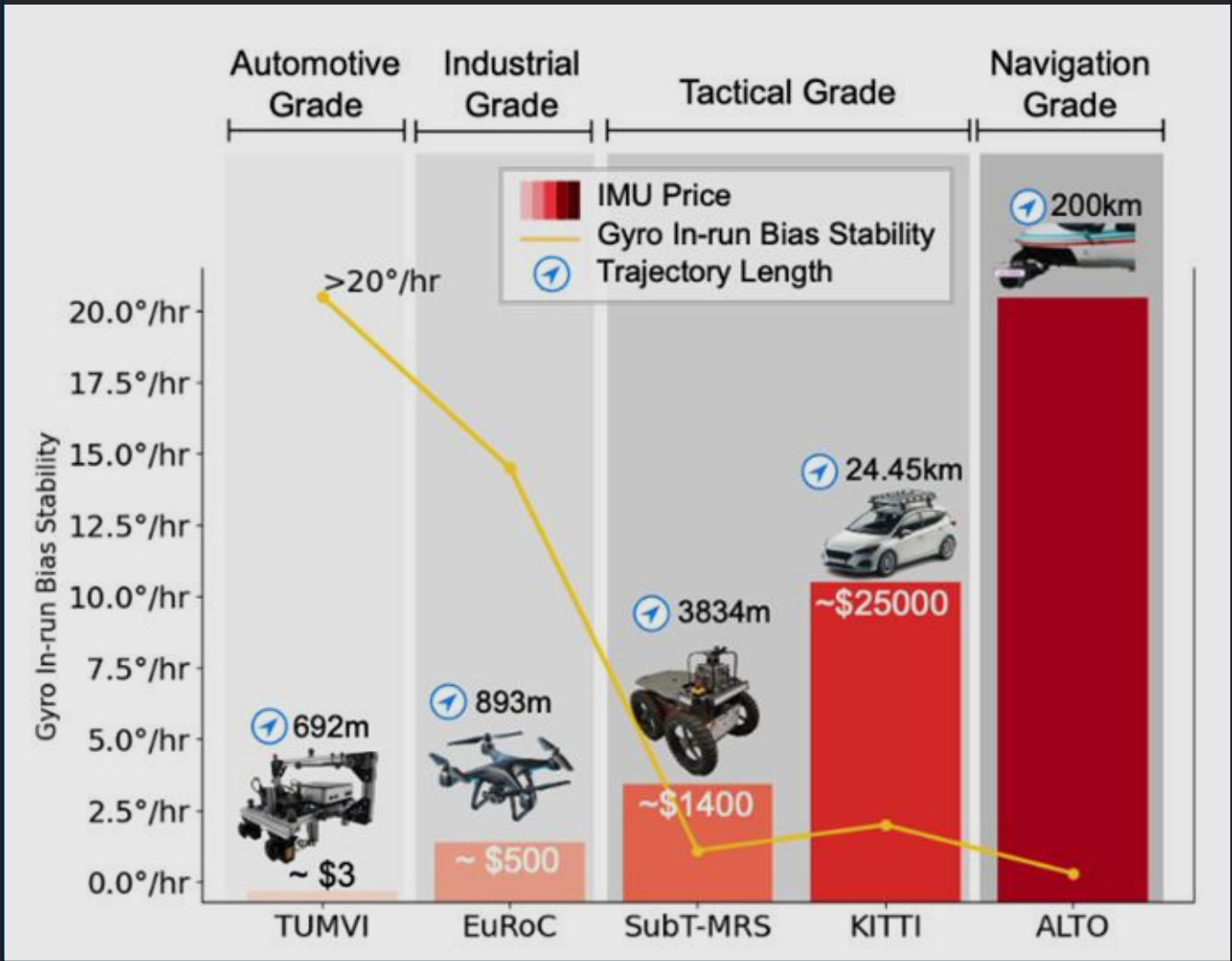


TABLE I: Datasets summary

Datasets	Duration	IMU	Modality
EuRoC [12]	22m29s	ADIS16448	Drone
TUM-VI [14]	13m31s	BMI160	Handheld
SubtMRS [15]	2h52m	Epson M-G365	Ground robot
KITTI [13]	43m44s	OXTS RT 3000	Vehicle
ALTO [29]	2h12m	NG LCI-1	Helicopter

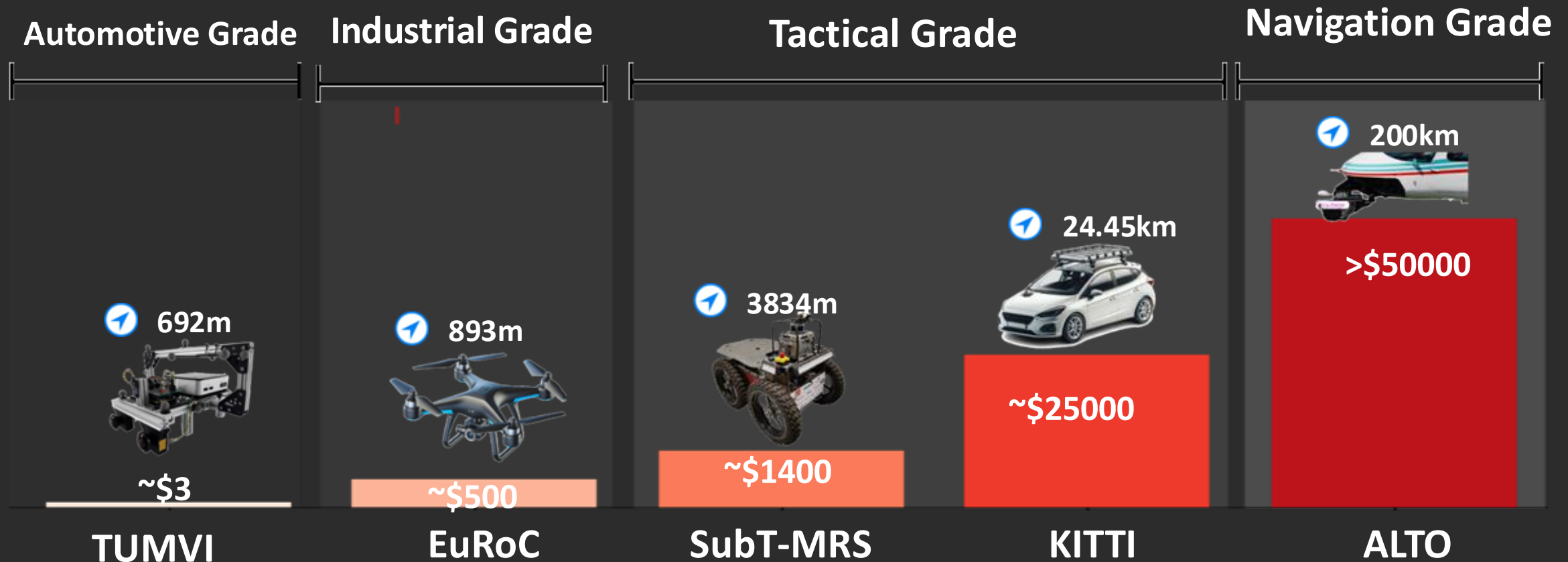
Integration Accuracy: TUMVI, EuRoC

Learning inertial odometry: KITTI

Ablation Study: Subt-MRS

GPS-denied Navigation: ALTO

Experiments



Improvement Compared to Baseline (Raw Data):

ROE(1s): 80.7%
RTE (1s): 77.2%

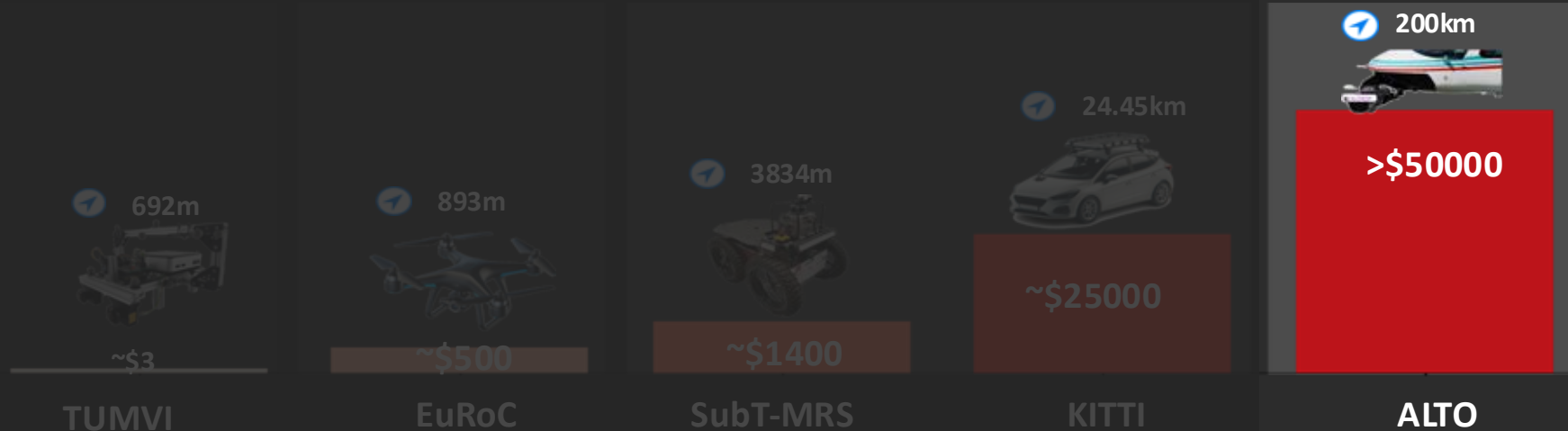
R-RMSE(1s): 97.1%
P-RMSE(1s): 73.8%

ROE(5s): 72.6%
RTE(5s): 42.1%

ATE: 14.7%

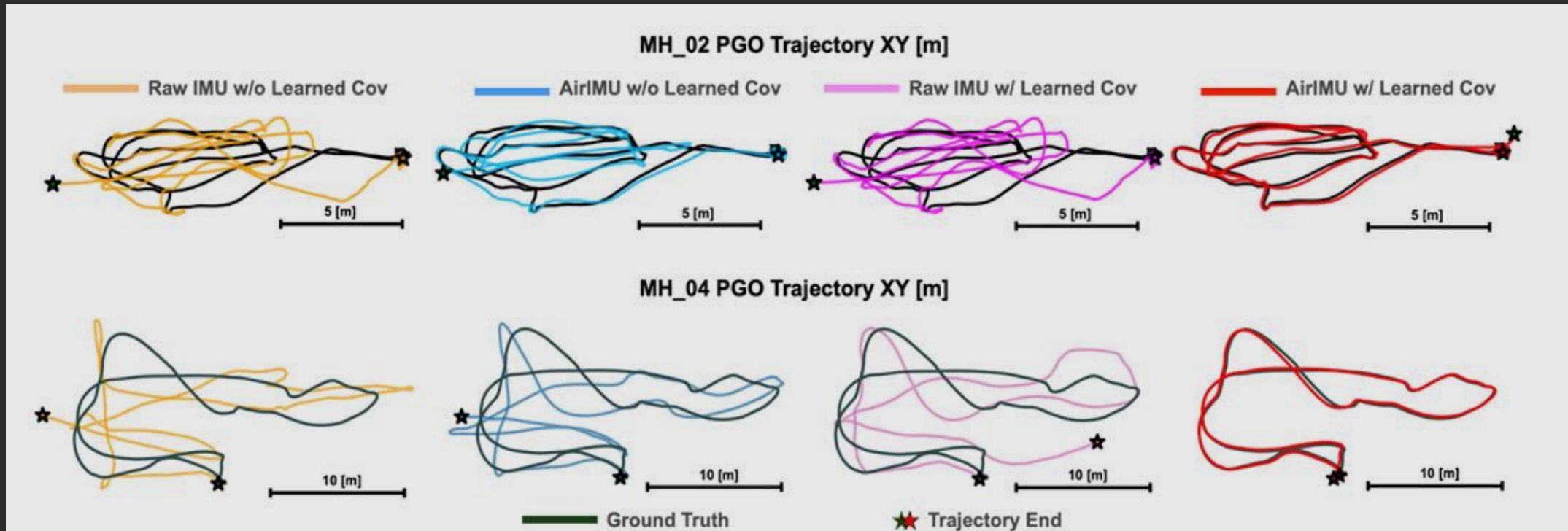
Accum Velo Err: 54.9%
Landing Drift: 73.2%

ALTO: Navigation-Grade IMU

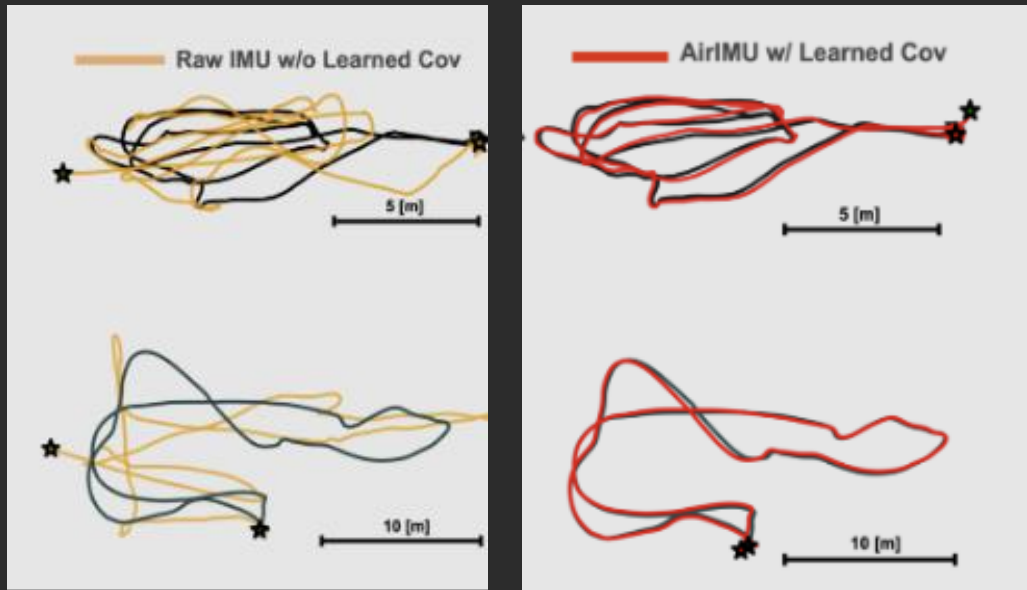


Simulated GPS on Pose Graph Optimization

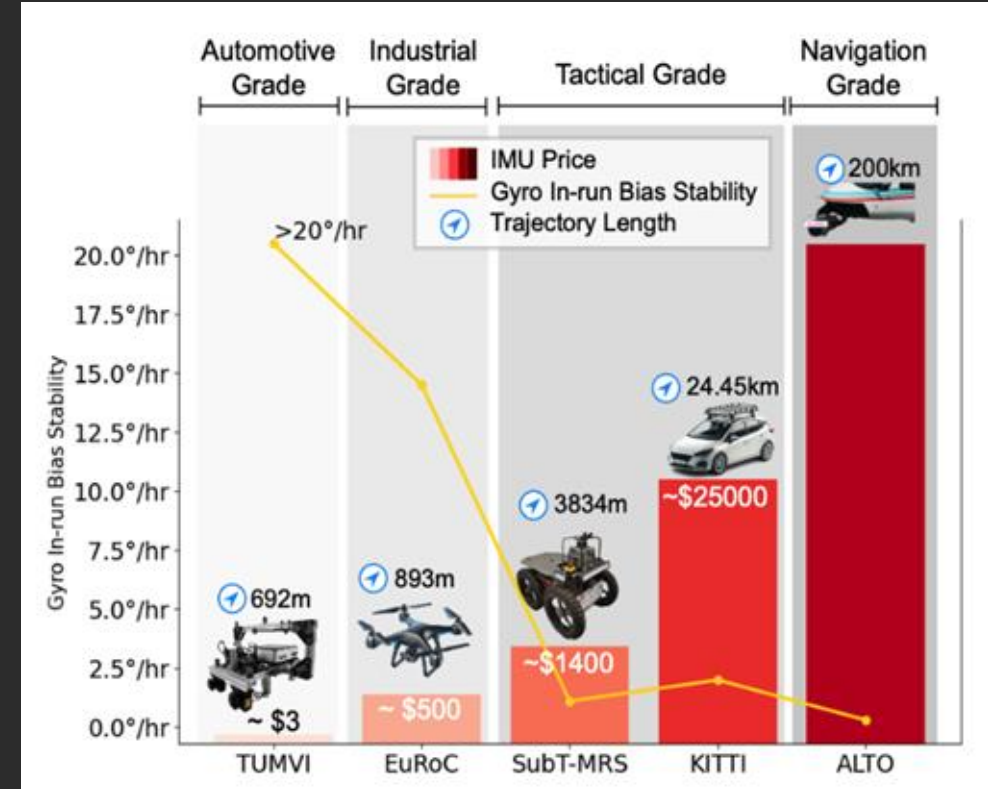
- The simulated IMU-GPS Graph optimization performed every 10s.



Summary



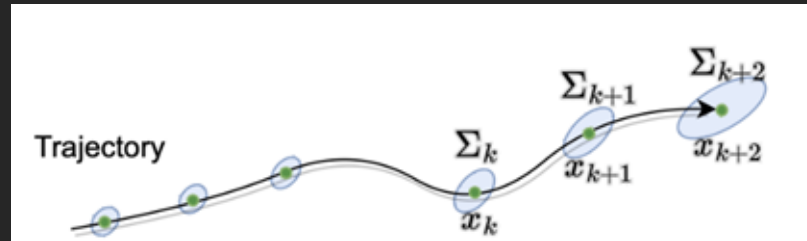
Better uncertainty empower the Pose Graph Optimization and sensor fusion



AirIMU can be deployed on all-spectrum IMU

Problem: Limitation of pure integration

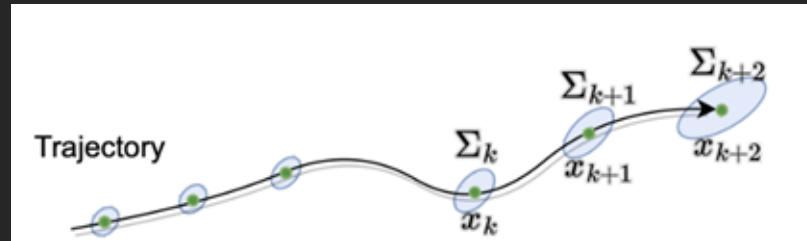
IMU integration will eventually drift



Background: Kinematics from Internal sensor

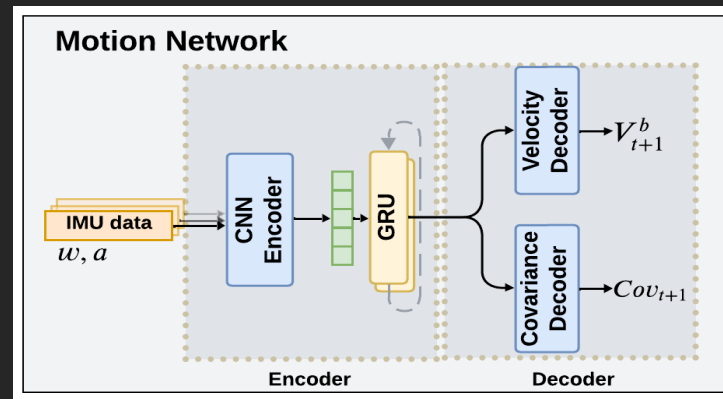


IMU integration will eventually drift



Inertial Information helps to determines the kinematics of the robots

Input
IMU data



Output
Motion
(velocity, position)

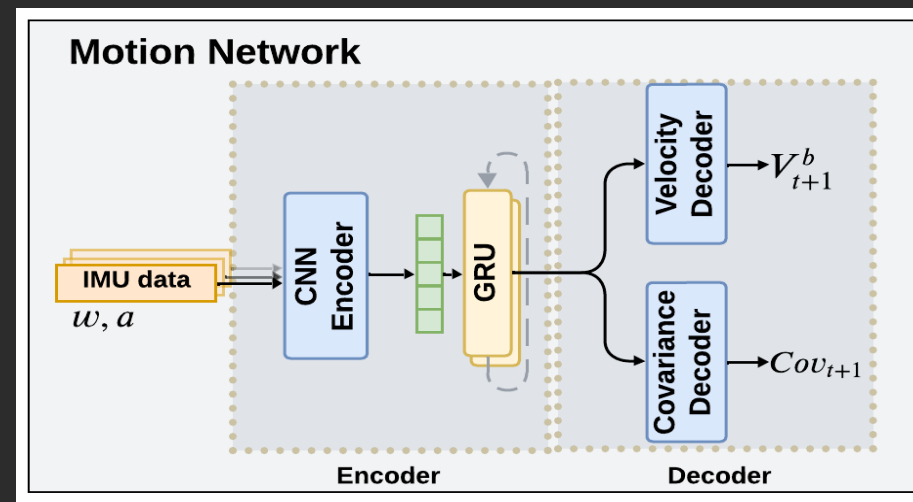


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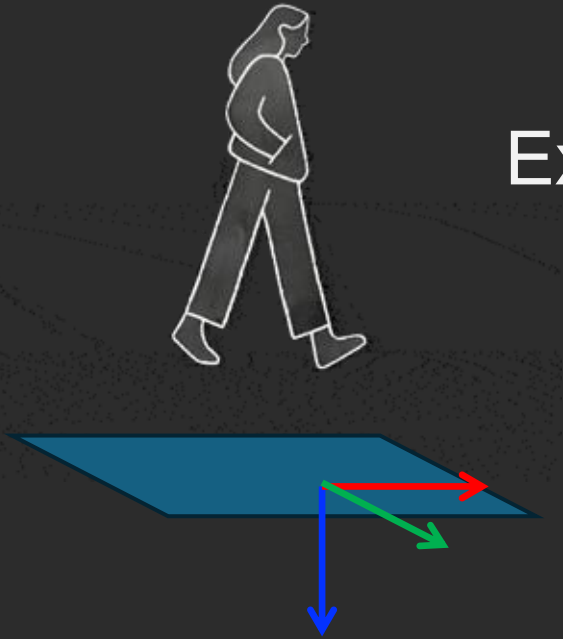
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Input
IMU data



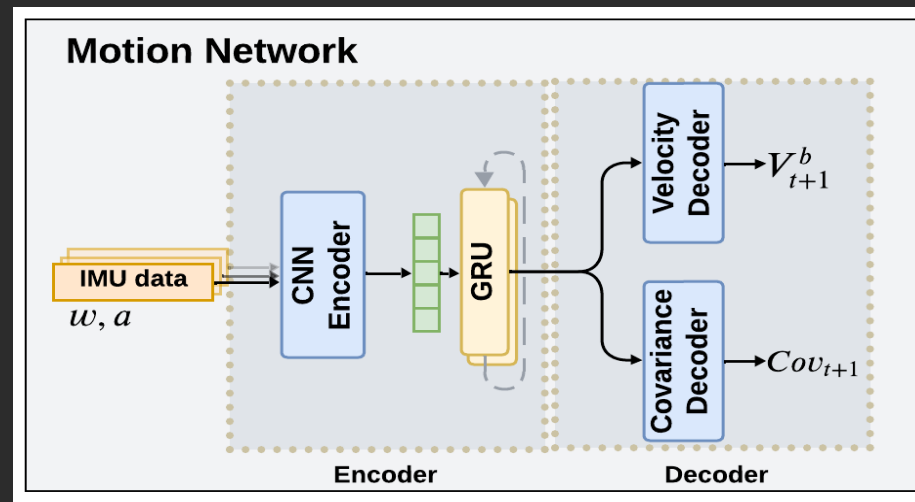
Output
Motion
(velocity, position)

Background: Challenging when Deploying on Drone



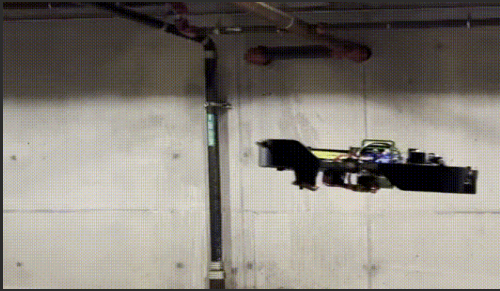
Existing IOs are designed for pedestrian

Input
IMU data



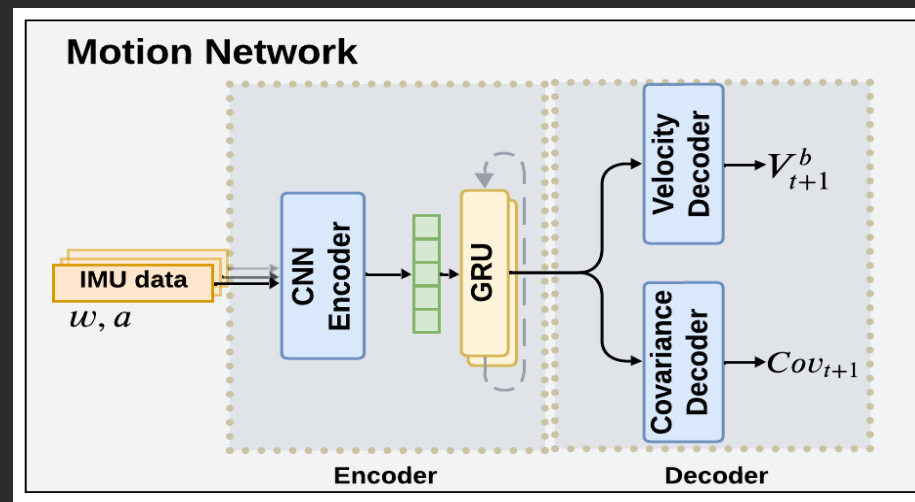
Output
Motion
(velocity, position)

Problems: Challenging when Deploying on Drone



Existing IOs are designed for pedestrian
UAVs' flight patterns are **highly dynamic and non-linear**

Input
IMU data



Output
Motion
(velocity, position)

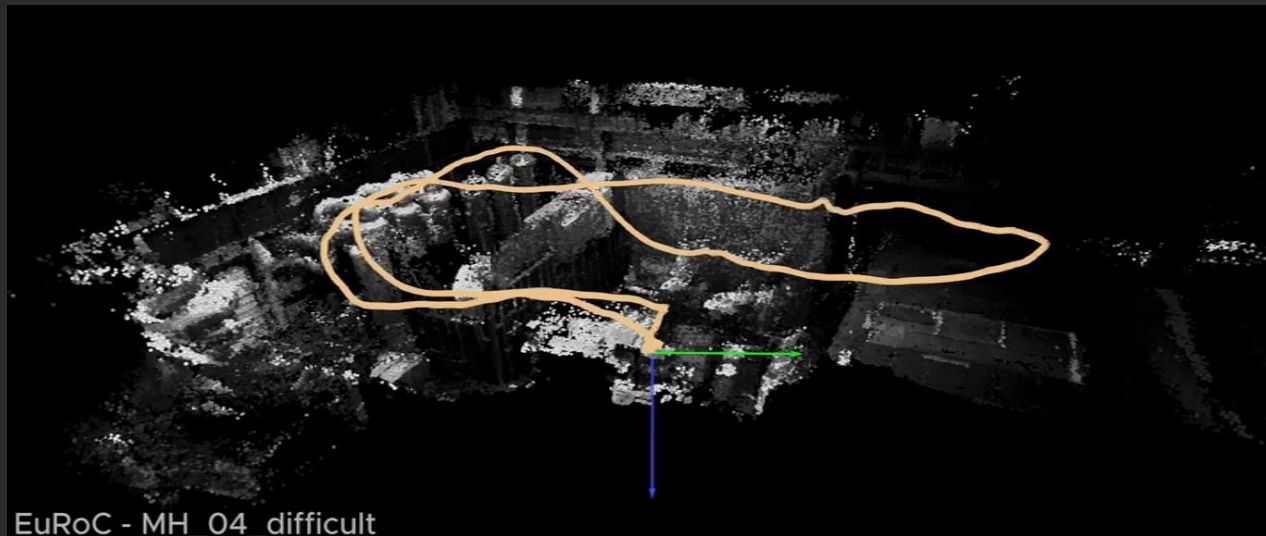
AirIO: Learning Inertial Odometry with Enhanced IMU Feature Observability



AirIO Outperforms SOTA methods e.g., IMO ( 86.6%)

Without Extra Sensor or Control Information

Generalizable to UNSEEN Trajectories



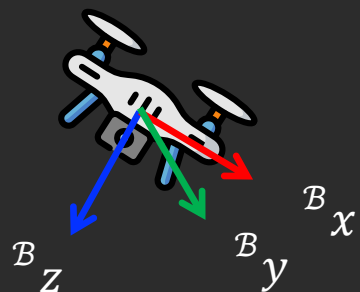
Global Coordinate in the Existing Methods is Suboptimal

Rotation Coupling

PCA: Explained Variance

t-SNE: Dimensionality Reduction

Attitude inherent in the IMU representation



Coupled with constant gravity force

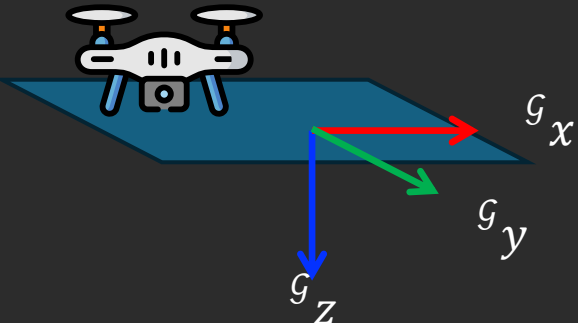
$${}^B a_i = \underbrace{\frac{{}^B F}{m}}_{\text{Linear Coupling}} + \underbrace{R_i^T \mathcal{G} g}_{\text{Nonlinear Coupling with motion force}}$$

Accelerometer Measurement in Canonical Body Coordinate Frame

Nonlinear Coupling with motion force

$$\mathcal{G} a_i = \hat{R}_i \frac{{}^B F}{m} + \hat{R}_i R_i^T \mathcal{G} g \approx \underbrace{\hat{R}_i \frac{{}^B F}{m}}_{\text{Nonlinear Coupling with motion force}} + \mathcal{G} g$$

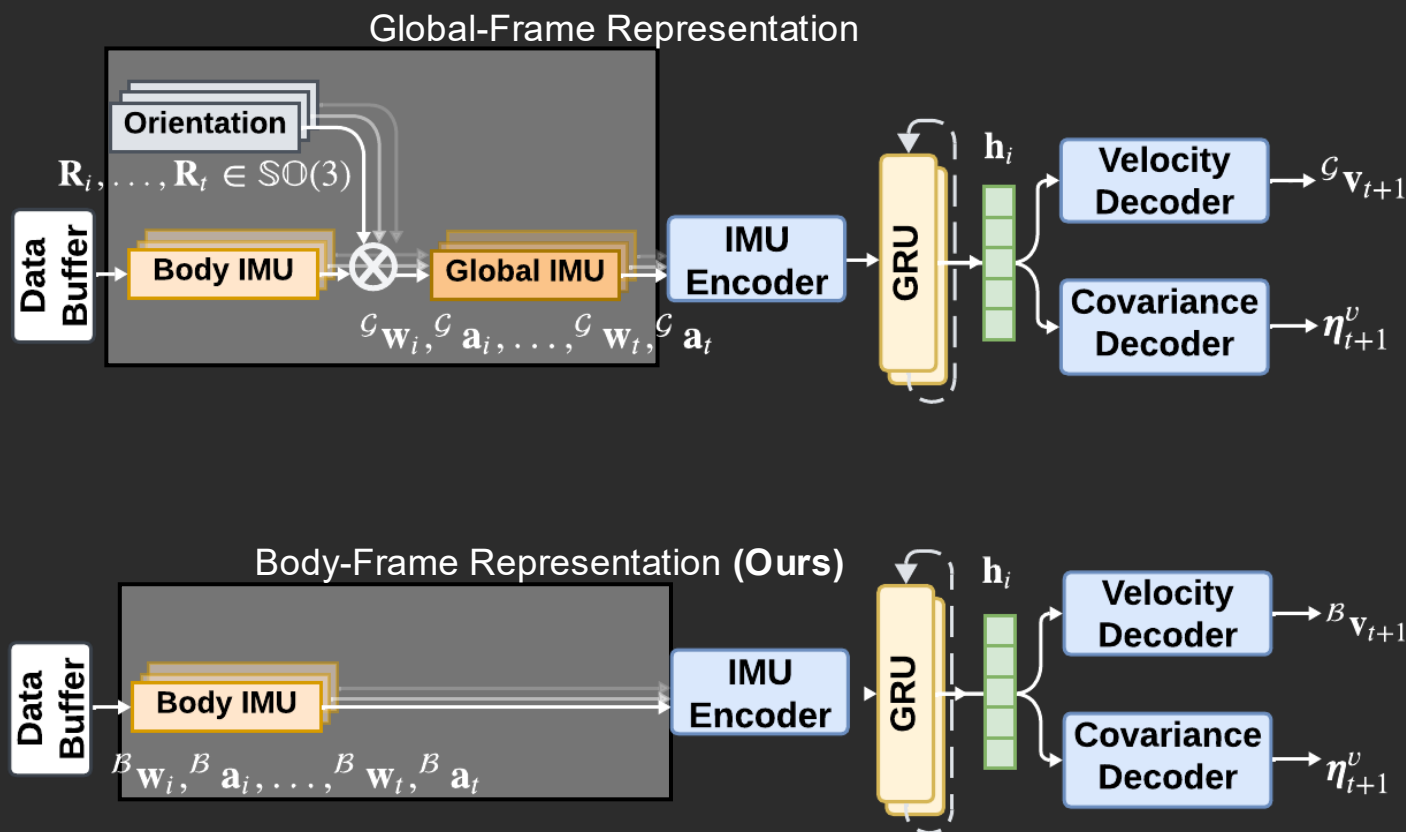
Accelerometer Measurement in Global Coordinate Frame



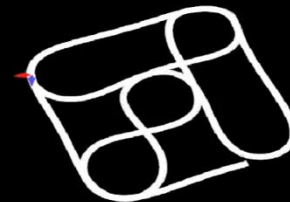
Methods: Body-frame IMU is more representative

Global coordinate frame is **suboptimal**

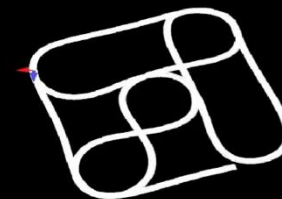
A single step achieves **66.7%** improvement (avg.)



Blackbird dataset
Unseen-bentDice

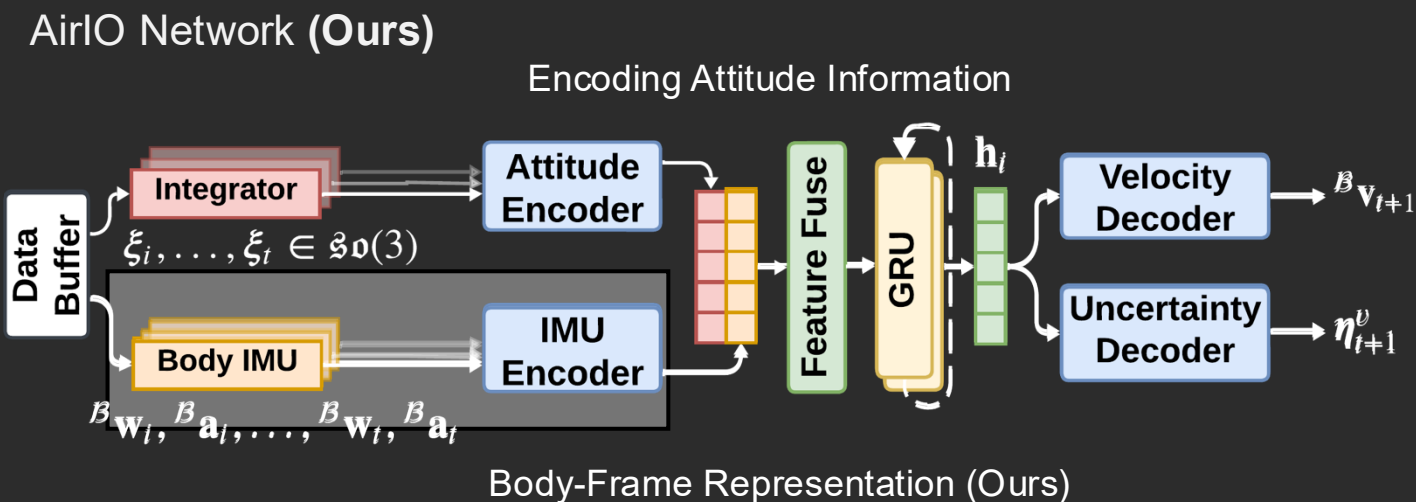


Global-Frame
ATE: 11.453

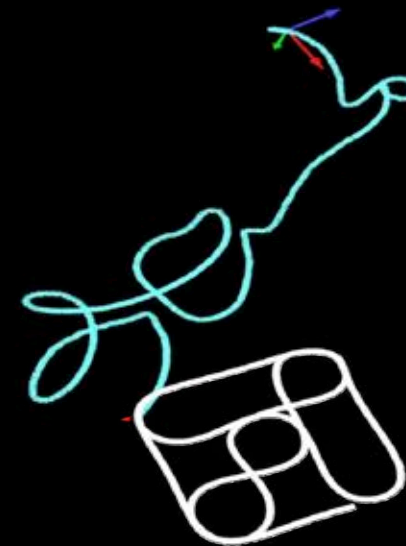


Body-Frame
ATE: 1.607

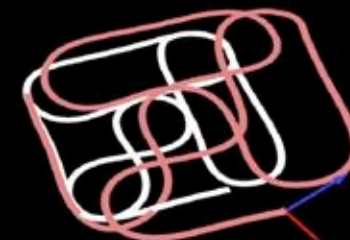
The key to learning-based IO of a quadrotor is **attitude information**
 Encoding Attitude yields an **additional 23.8%** improvement (avg.)



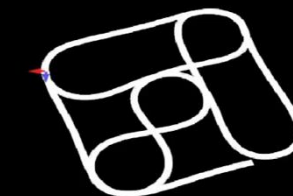
Blackbird dataset
 Unseen-bentDice



Global-Frame
 ATE: 11.453



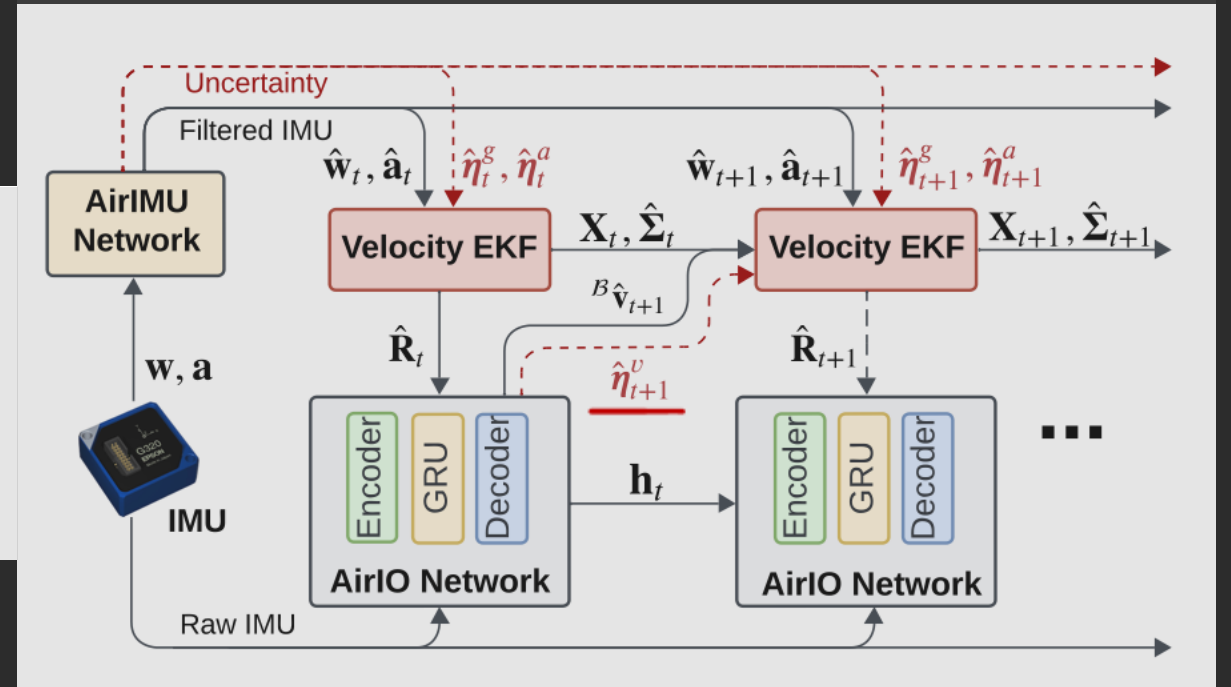
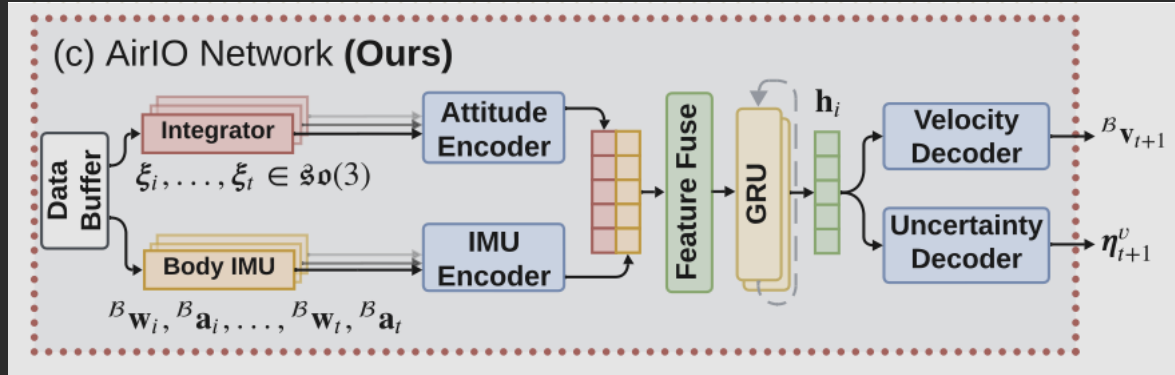
Body-Frame
 ATE: 1.607



Body-Frame
 +
 Attitude Encoding
 ATE: 1.156



AirIO System Pipeline



EKF fusion with uncertainty-aware IMU pre-integration model

AirIO outperforms SOTA methods

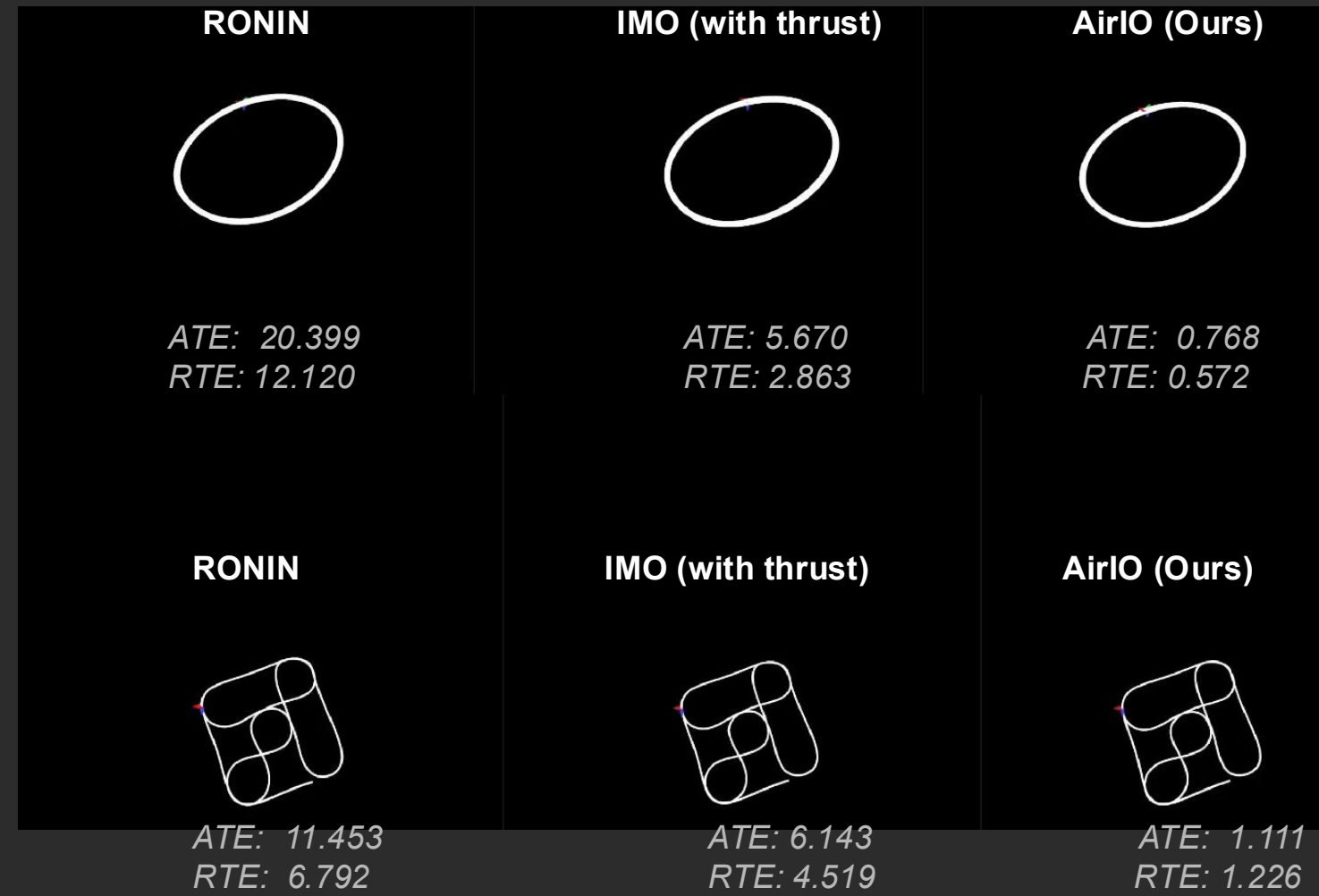
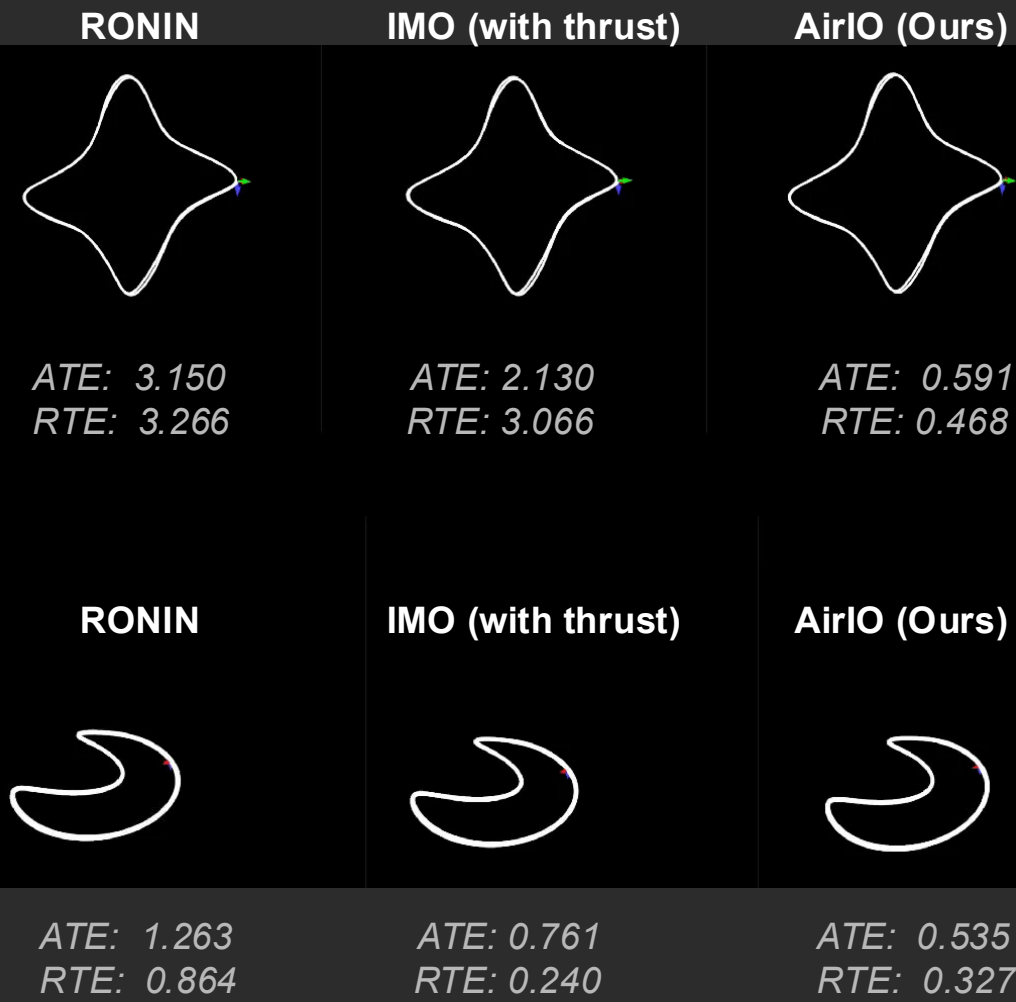


Without any extra information

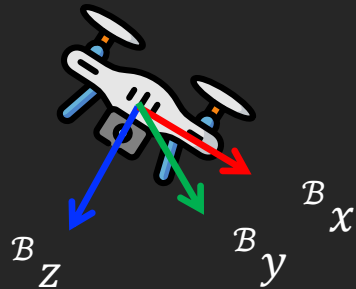
Generalizability for unseen trajectories

Blackbird dataset
SEEN Sequences

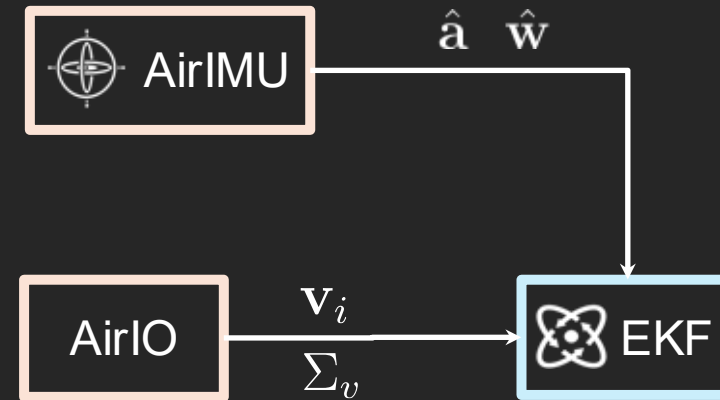
UNSEEN Sequences



Summary

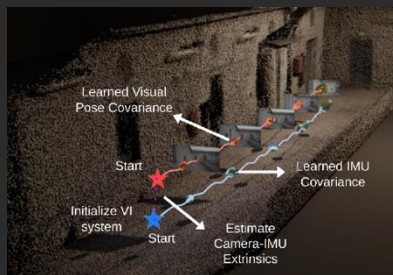


IMU feature in body-coordinate frame is most representative

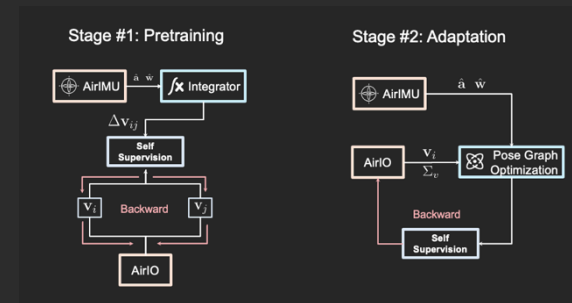


A Fusion framework with AirIMU and AirIO achieve optimal performance and robustness

[Part. III]: Sensor Fusion and Adaptation powered by learned uncertainty



Visual-Inertial Fusion & Adaption [Part. 3]



Background: Learning IO is Generic? Or Overfitting?

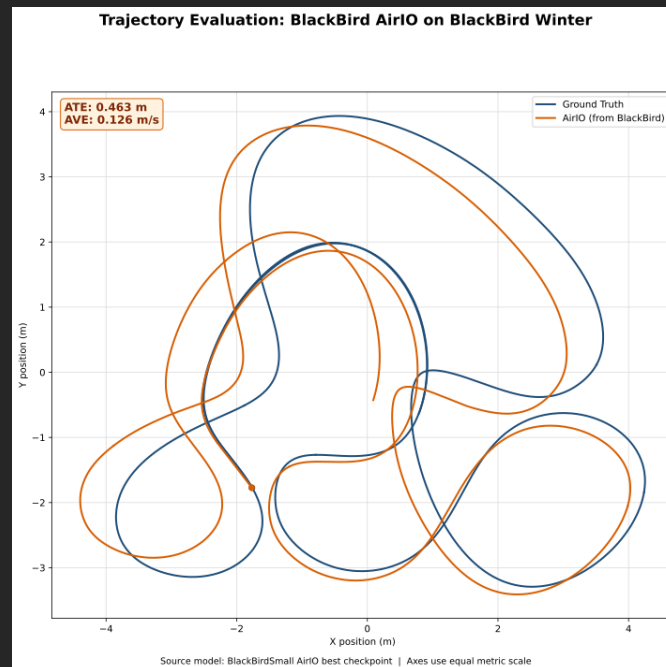


Training on Blackbird



Velocity Range:
0.178 ~ 7.708 m/s

Running on Blackbird



Learning IMU odometry will have
a strong performance in the in-
domain data.

Background: Generic? Or Overfitting?

Training on Blackbird

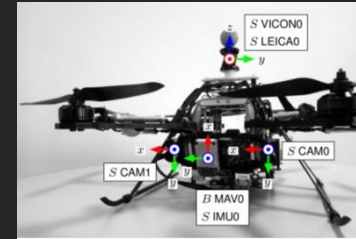
Average Velocity: 2.766 m/s



?

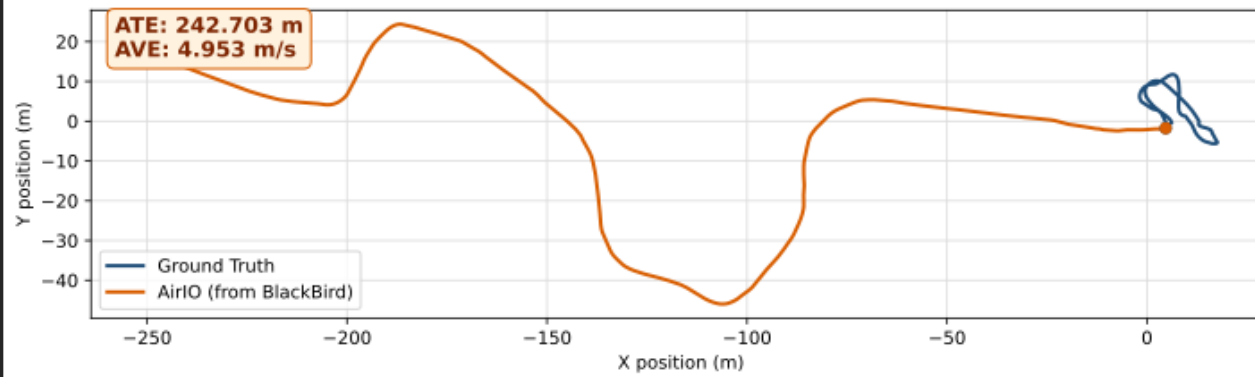
Running on EuRoC

Average Velocity: 0.926 m/s



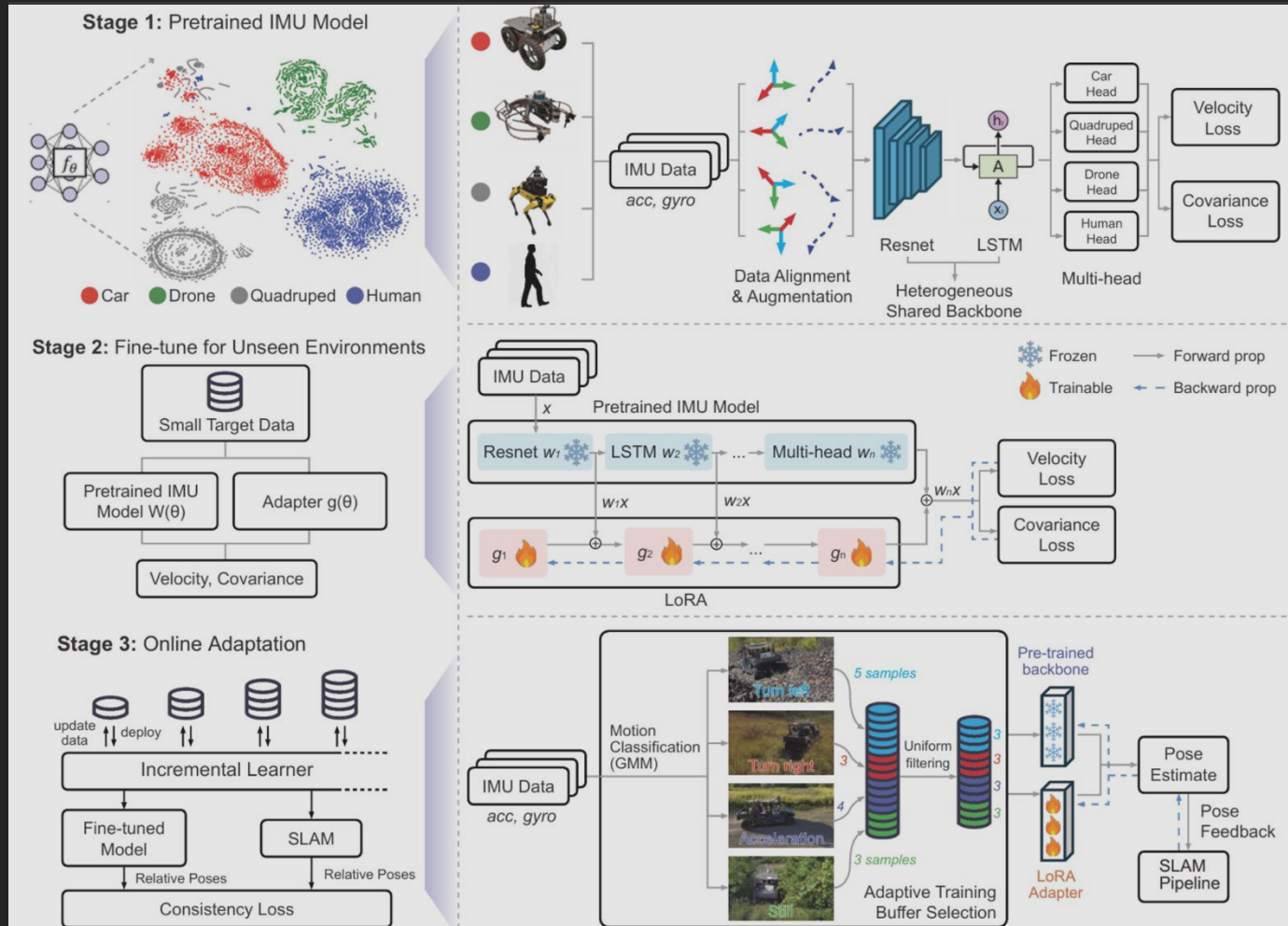
Trajectory Evaluation: AirIO (from BlackBird) on EuRoC MH04

It drift badly....



Motivation: Challenges in Generalization

TartanIMU



Challenge #1:

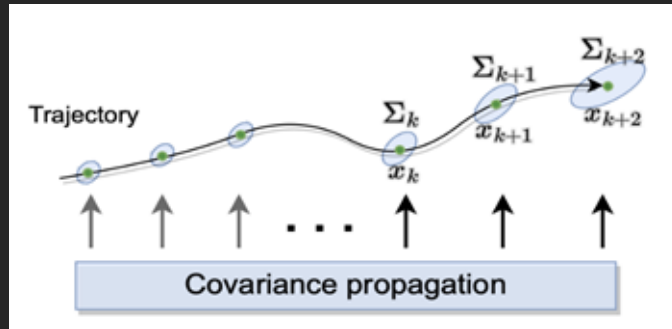
- How to get enough data with ground truth?

Challenge #2:

- We need ground-truth label for new datasets

Motivation: Learning IMU from IMU

What is General and Fundamental?



IMU integration follows generic physical principles.

Where is Accuracy Coming From?

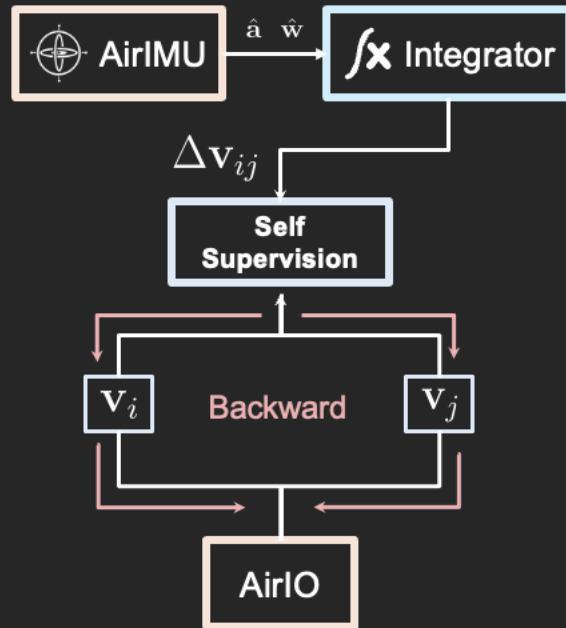


The Adaptation on the specific Robot or Platform

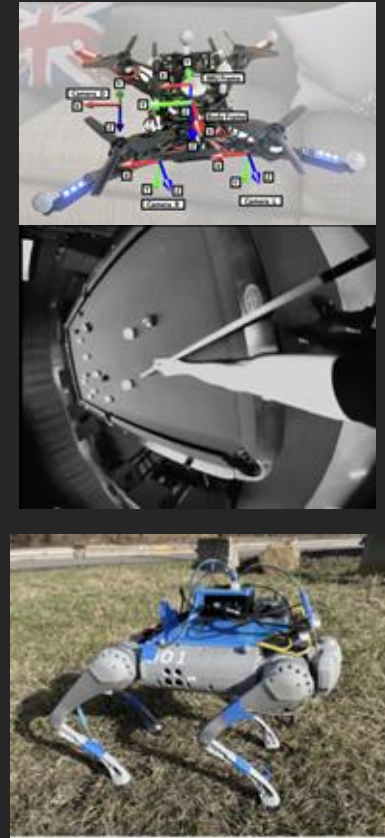
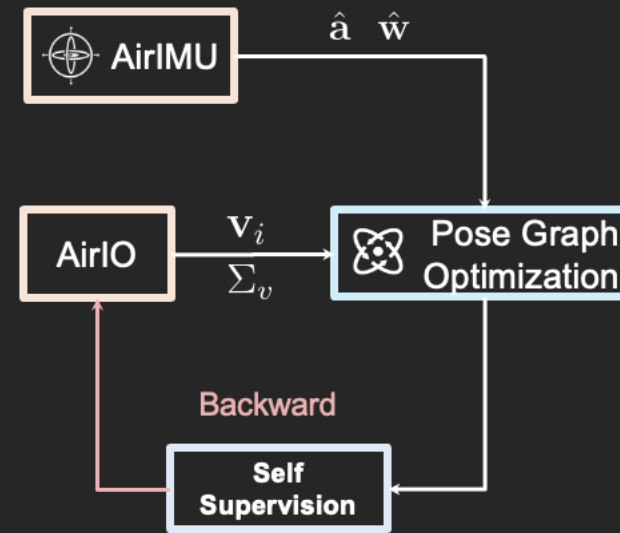
MAC-IO: Two Stage Self-supervision and adaptation for Multiple Modality



Stage #1: Pretraining

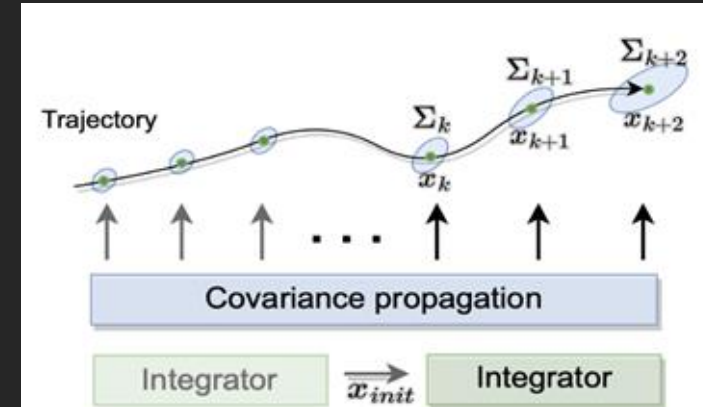
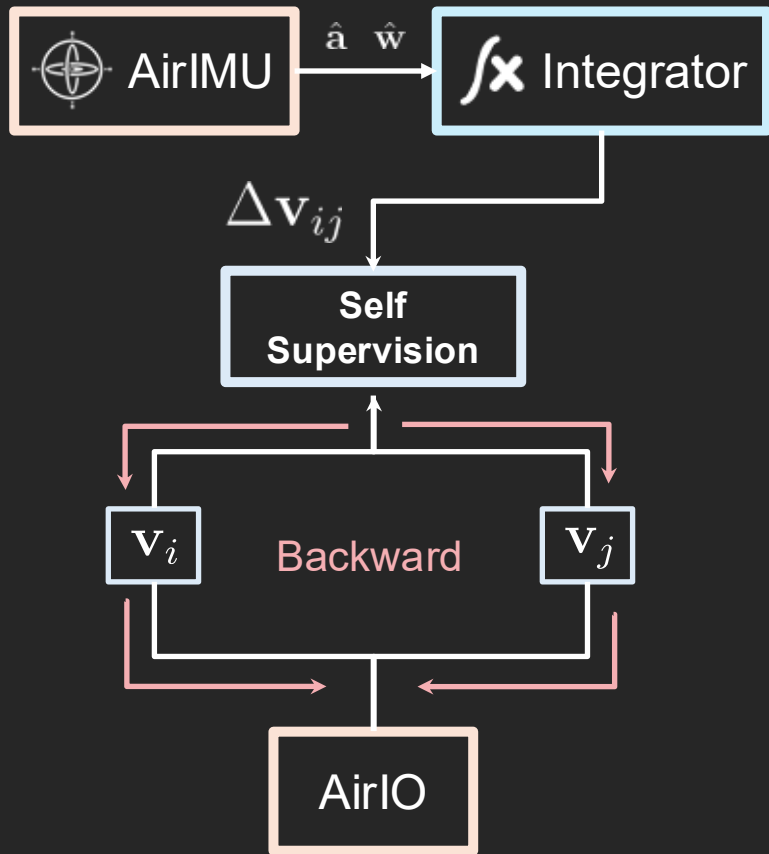


Stage #2: Adaptation



- Deployable for multiple modality and platform
- Don't need ground truth data

Method: Stage #1 – Pretraining



$$\Delta R_{ij} = \prod_{k=i}^{j-1} \text{Exp} \left({}^B \tilde{\omega}_k \Delta t \right)$$

$$\Delta v_{ij} = \sum_{k=i}^{j-1} \Delta R_{ik} {}^B \tilde{a}_k \Delta t$$

$$L_{\text{pretrain}} = \|v_j - v_i - \Delta v_{ij}\|^2$$

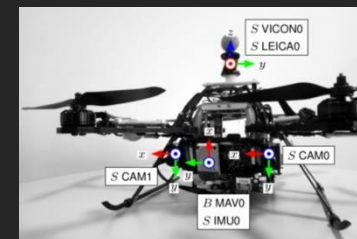
Results on Velocity estimation



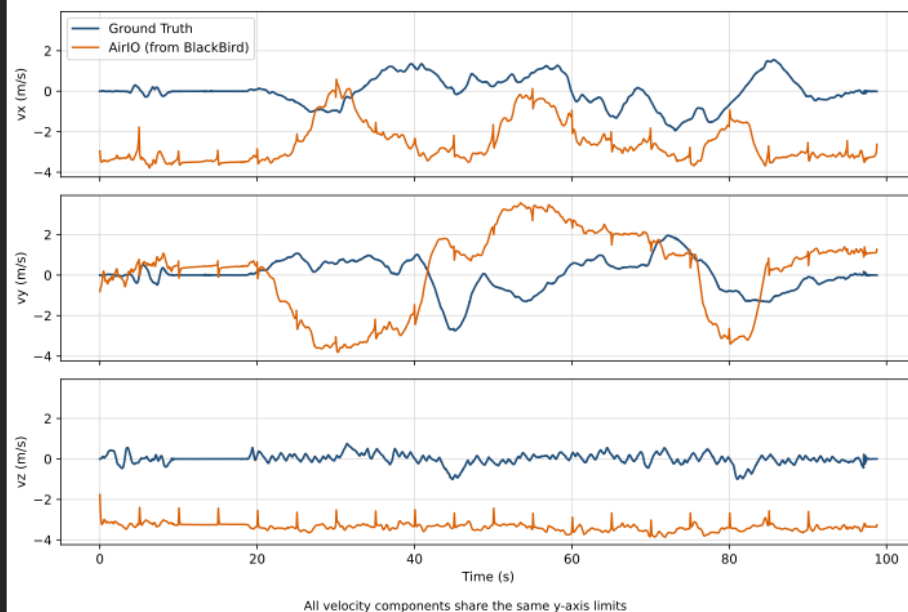
Model on BlackBird



Pretraining on EuRoC

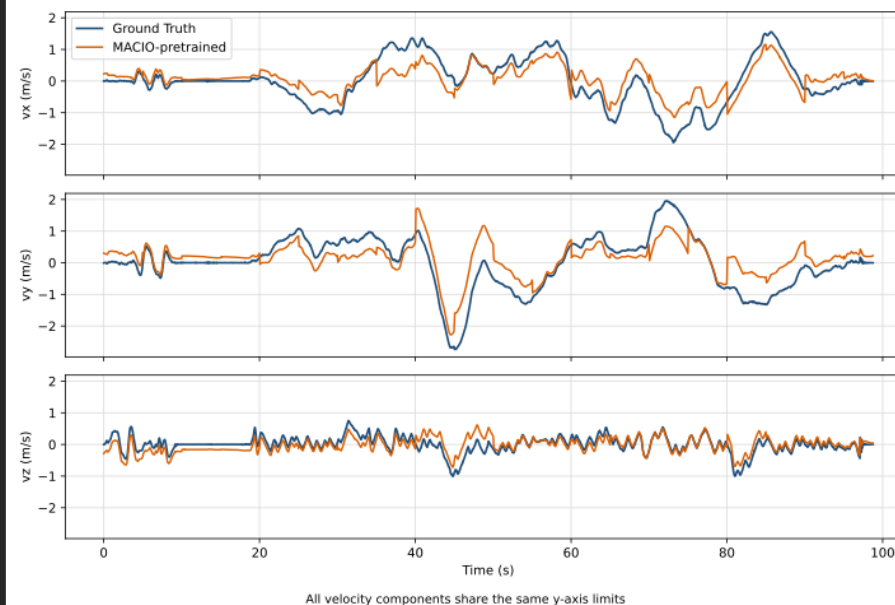


Trajectory Evaluation: BlackBird AirIO to EuRoC MH04: Velocity Comparison



➔
Pretraining
(w/o GT label)

Trajectory Evaluation: MACIO-pretrained to EuRoC MH04: Velocity Comparison

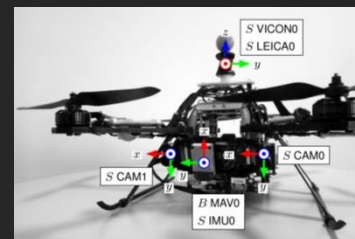


Results on Velocity estimation

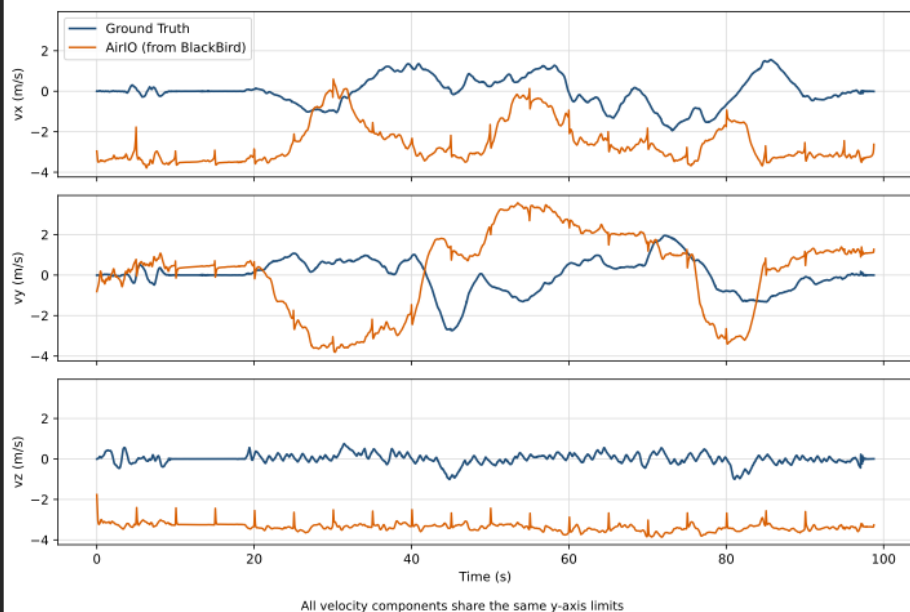
Model on BlackBird



Pretraining on EuRoC

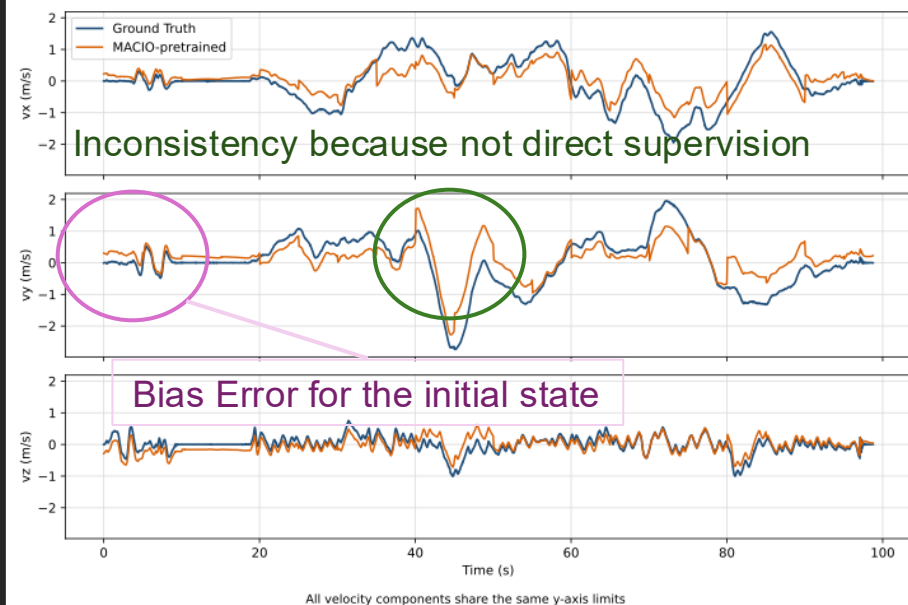


Trajectory Evaluation: BlackBird AirIO to EuRoC MH04: Velocity Comparison



Pretraining
(w/o GT label)

Trajectory Evaluation: MACIO-pretrained to EuRoC MH04: Velocity Comparison

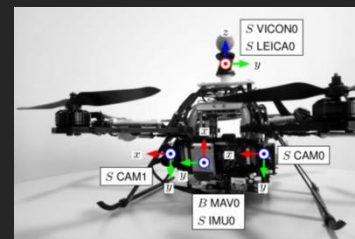


Results on Velocity estimation

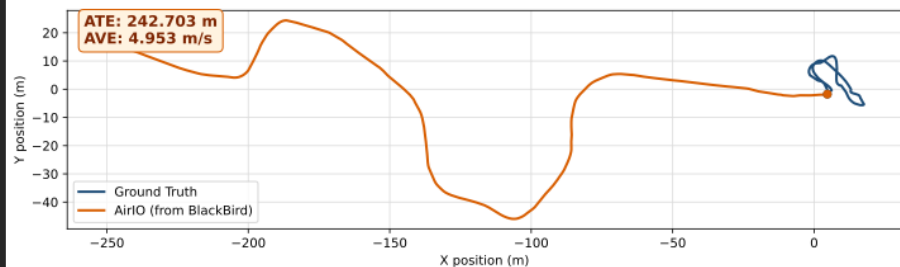
Model on BlackBird



Pretraining on EuRoC

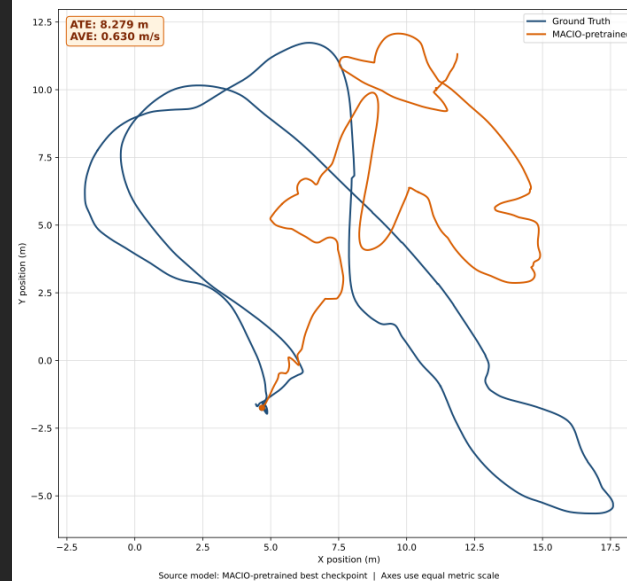


Trajectory Evaluation: AirIO (from BlackBird) on EuRoC MH04

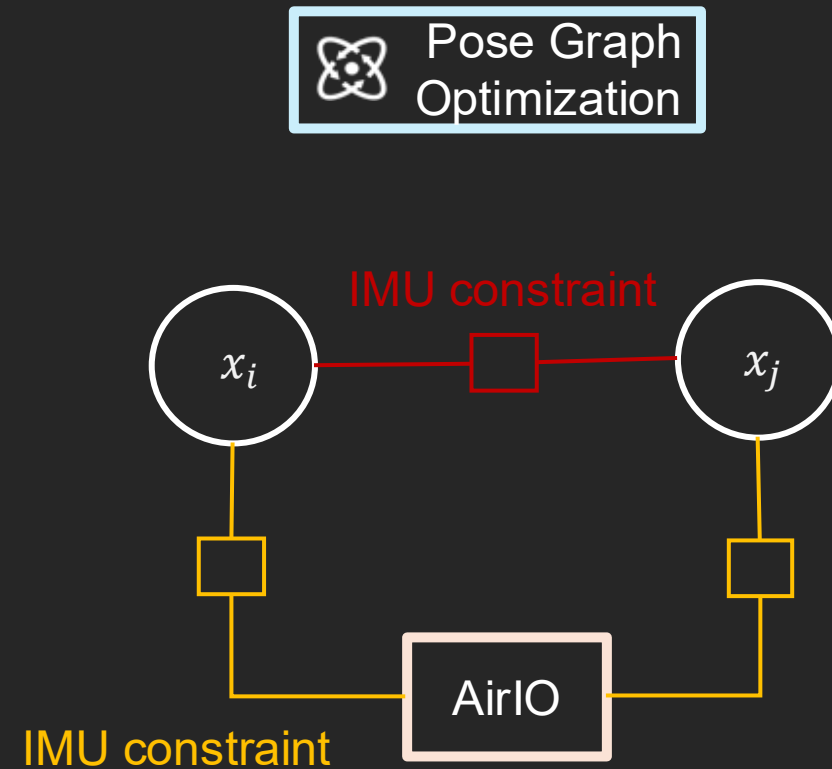
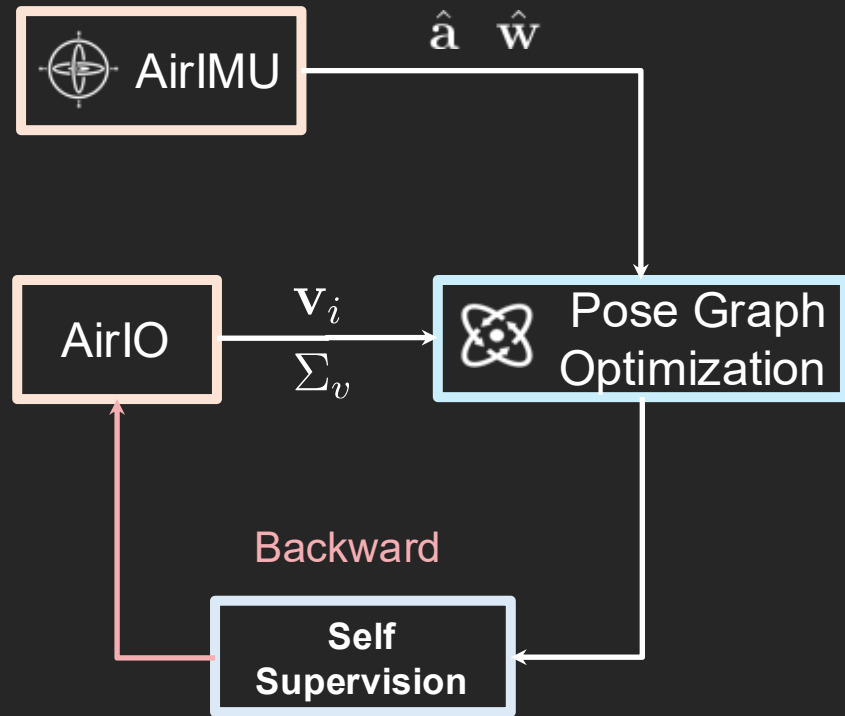


Pretraining
(w/o GT label)

Trajectory Evaluation: MACIO-pretrained to EuRoC MH04

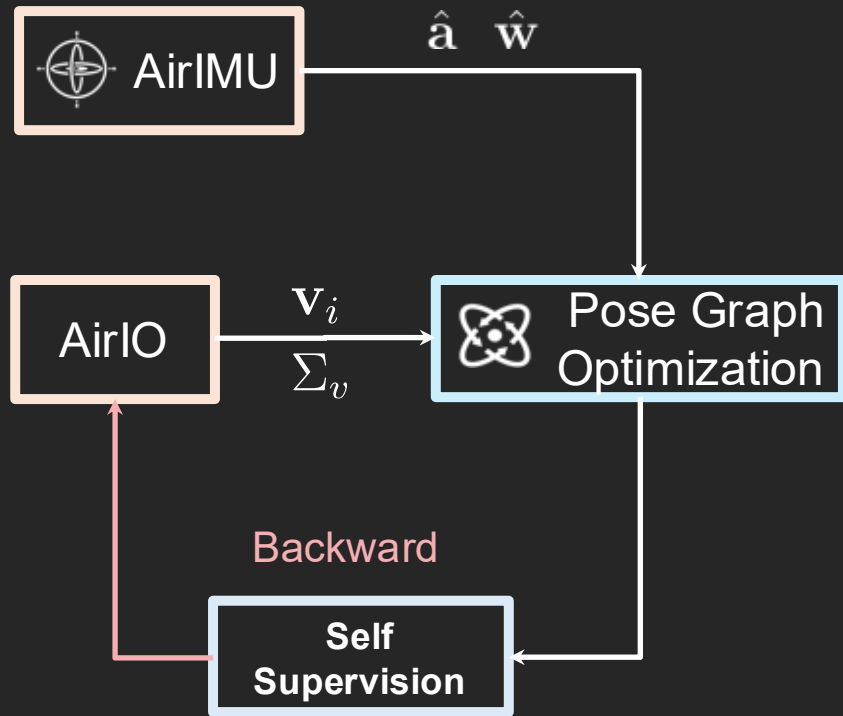


Method: Stage 2 – Adaptation on Velocity



Imperative Learning, Bi-level Optimization

Method: Stage 2 – Adaptation on Covariance



 Pose Graph Optimization

Fisher Information matrix to calculate covariance

$$\Sigma^* = \Lambda^{-1} = (J^T W J)^{-1}$$

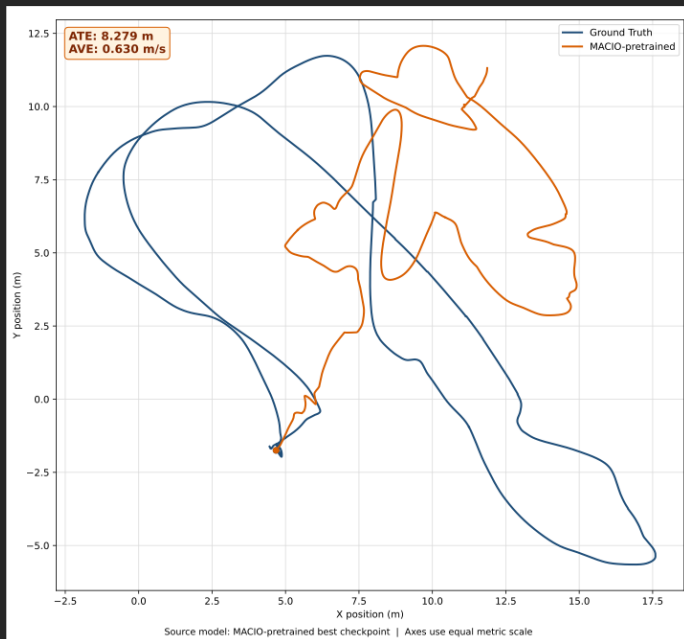
KL Divergence of PGO and the network

$$D_{\text{KL}} \left(\mathcal{N}(v^*, \Sigma^*) \parallel \mathcal{N}(\hat{v}, \hat{\Sigma}) \right)$$

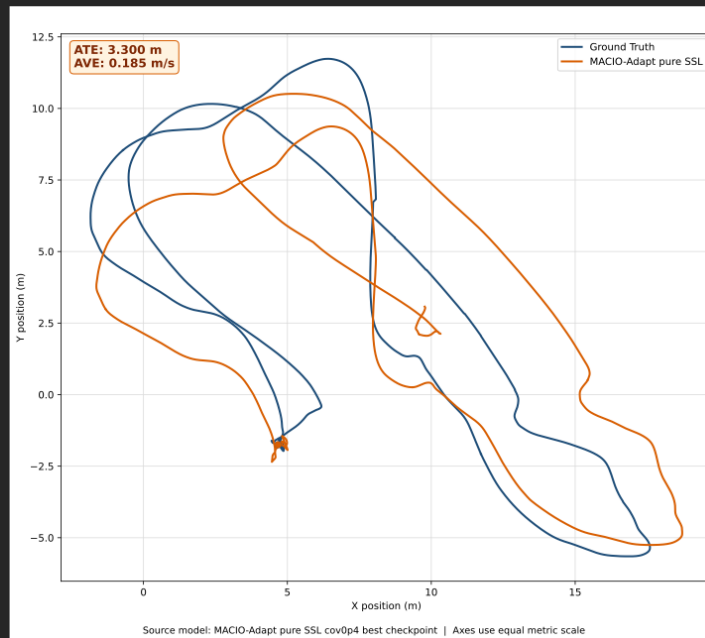
$$\mathcal{L}_{\text{KL}} = \frac{1}{2} \sum_d \left[\frac{(\hat{v}_d - v_d^*)^2}{\hat{\Sigma}_d} + \log \hat{\Sigma}_d - \log \Sigma_d^* - 1 + \frac{\Sigma_d^*}{\hat{\Sigma}_d} \right]$$

Results on Velocity estimation

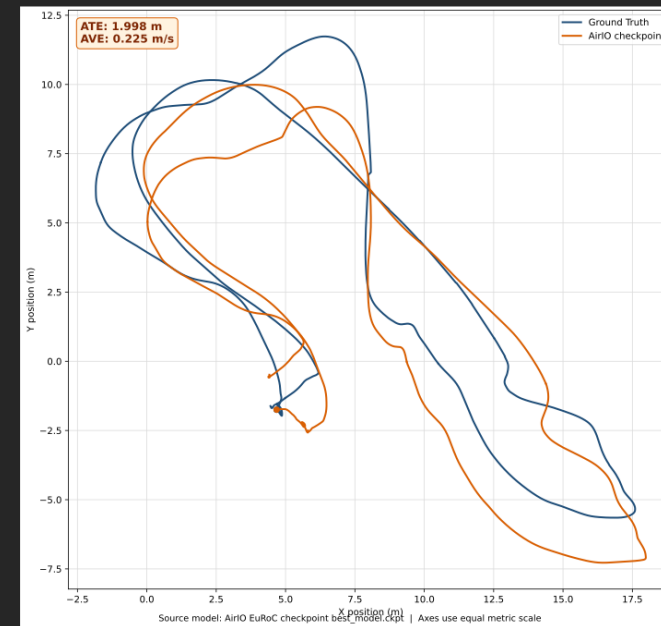
Pretraining on EuRoC



Adaptation on EuRoC

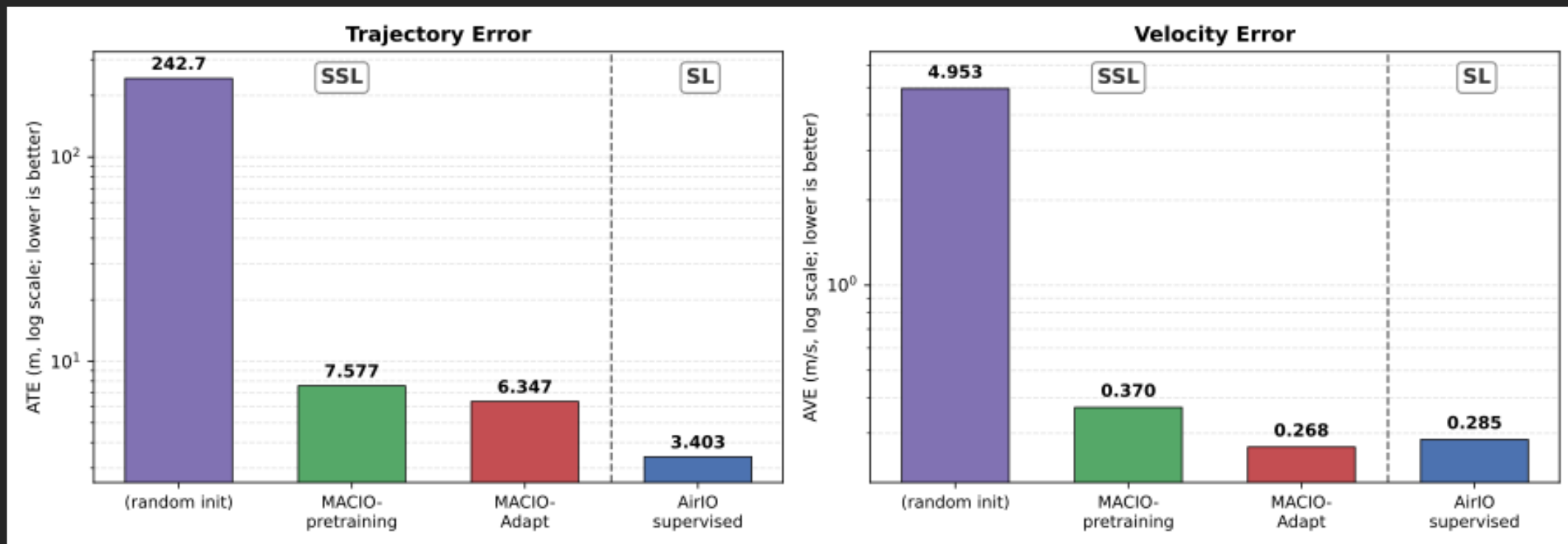


Supervised Learning



Self-supervision approximates the performance of supervised learning

Results on Velocity estimation



Self-supervision approximates the performance of supervised learning

MAC-IO: Self-supervision from unlabeled data on larger datasets



Calibration Set

Training Set

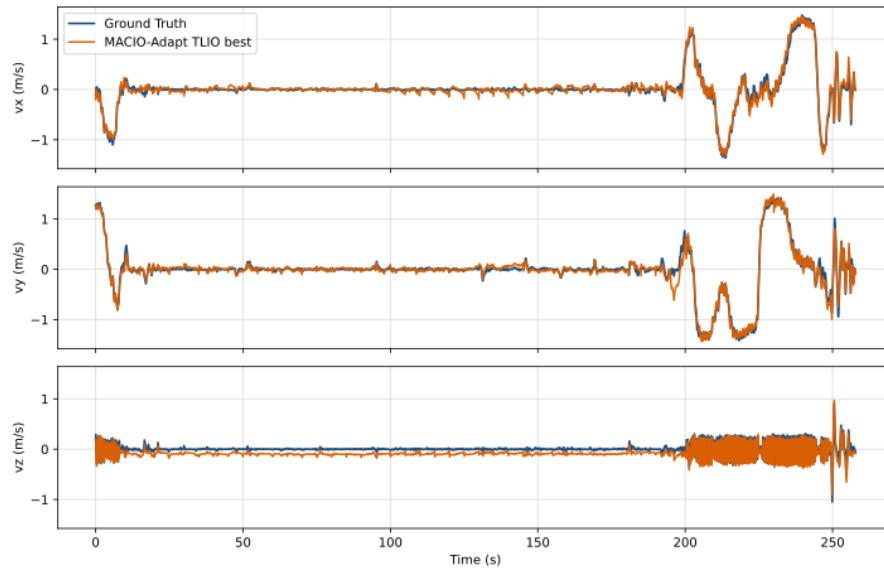
Testing Set

- Few sequences for calibration or training AirIMU
- Apply self-supervision without ground truth label
- Totally not seen before for testing

MAC-IO: Self-supervision from unlabeled data on larger datasets



Trajectory Evaluation: MACIO-Adapt TLIO Best on 643495372649567: Velocity Comparison

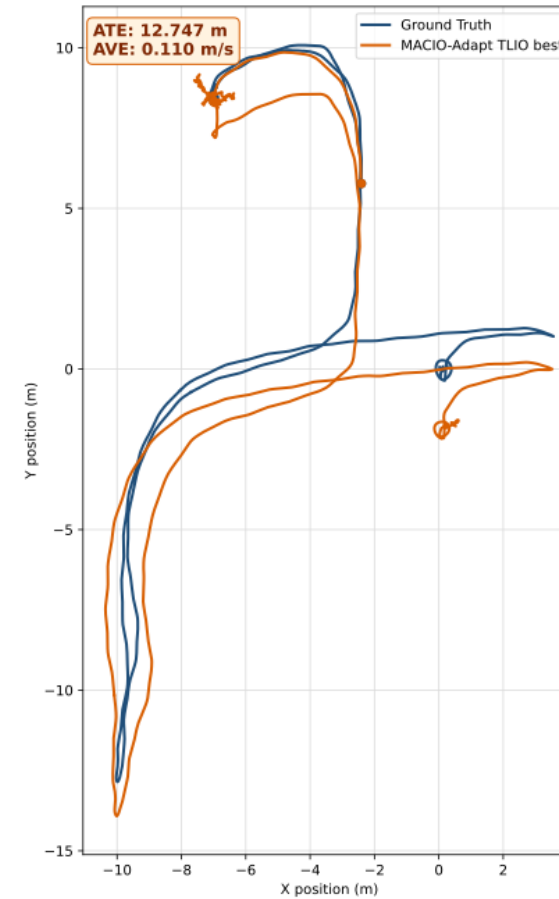


All velocity components share the same y-axis limits

Self-supervision AVE: 0.161

TLIO (network) AVE: 0.119

Trajectory Evaluation: MACIO-Adapt TLIO Best on 643495372649567



Source model: TLIO MACIO-Adapt best_model.ckpt epoch 175 | Axes use equal metric scale

MAC-IO: Adaptation on Unlabeled data



Labeled Data

Unlabeled Data

- We have limited training data on platform
- Each Robot has Distinct Structure and Hardware Setup

Racing Drone



VR Glasses



Legged Robot



MAC-IO Adaptation: Covariance anchoring



Labeled Data

Unlabeled Data

Except for self-supervision loss, using the labeled data to add “anchoring”

$$L_{cov} = (v - \hat{v})^T \widehat{\Sigma}_f^{-1} (v - \hat{v}) + \log(\det \widehat{\Sigma}_v)$$

$$L_v = ||v - \hat{v}||_2$$

$$L = L_v + \alpha L_{cov} + \beta L_{SSL}$$

Racing Drone



VR Glasses

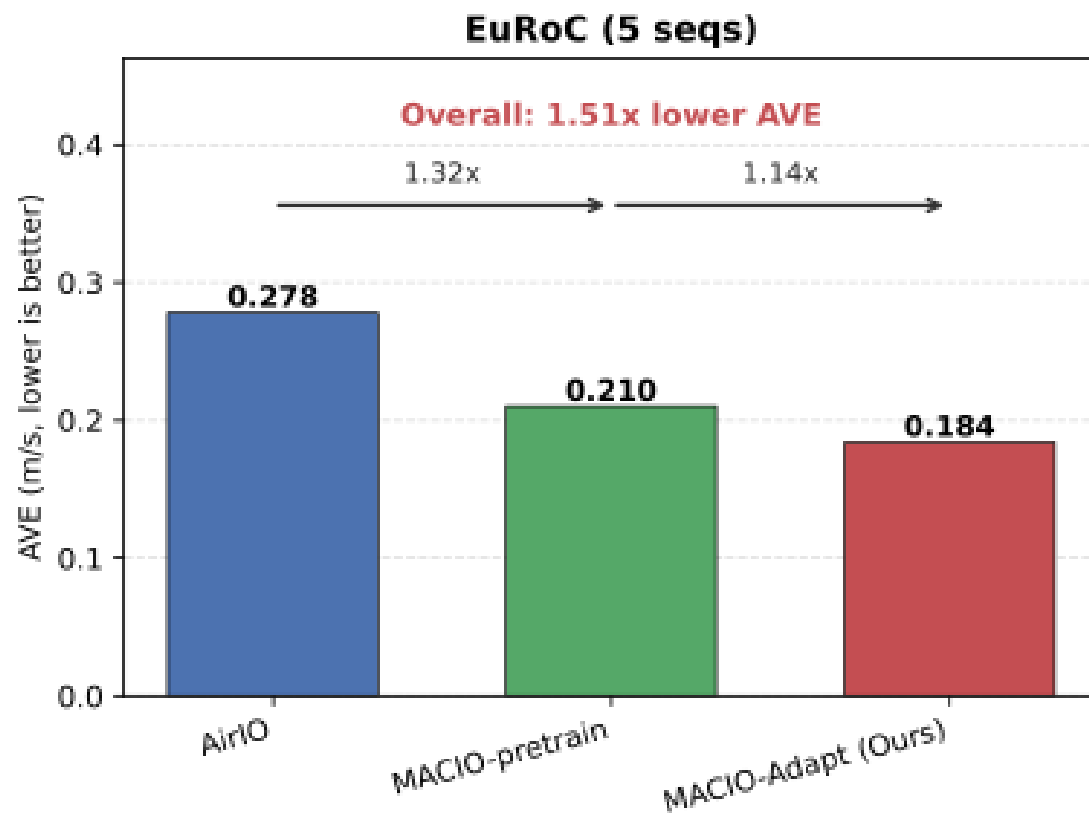
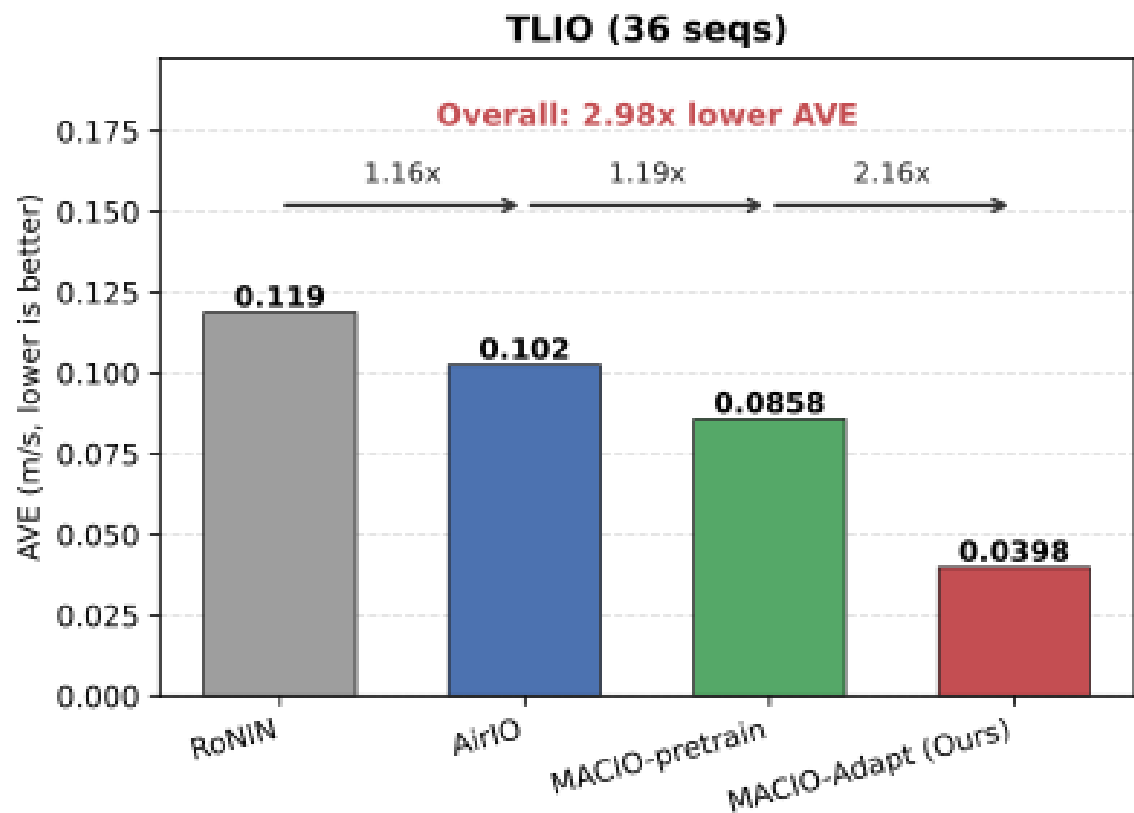


Legged Robot



Quantitative Result : TLIO and EuRoC

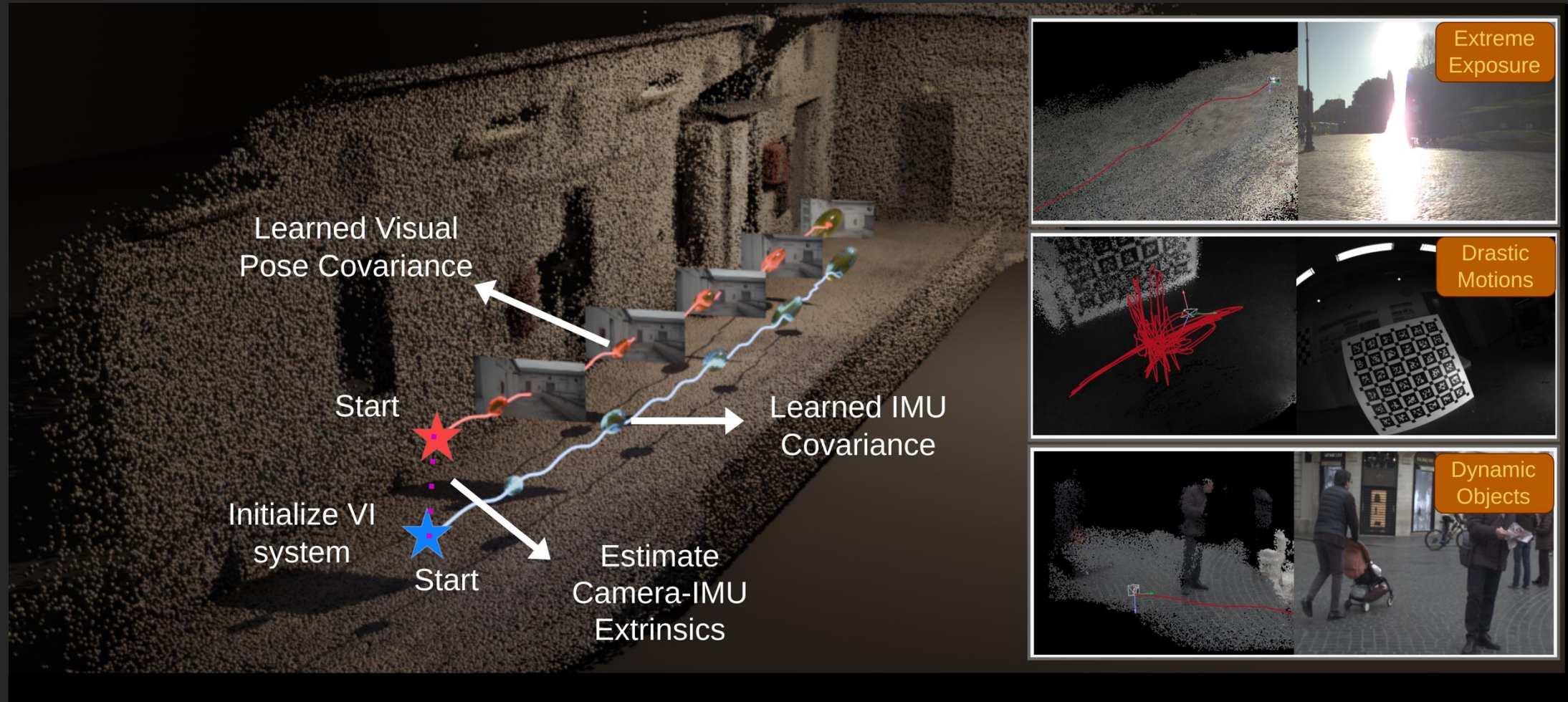
Step-by-step AVE improvement: RoNIN --> AirIO --> MACIO-pretrain --> MACIO-Adapt



AVE: ~ 3 cm/s

Introducing **MAC-I²**

Learned **M**etrics-**A**ware **C**ovariance for Robust
Visual-**I**nertial **I**nitialization and Calibration



Visual-Inertial Initialization and Calibration

We define the state as:

$$\mathcal{X} = [x_0, x_1, \dots, x_n, g^{c_0}, b^g, b^a, T_c^b],$$
$$x_k = \begin{bmatrix} T_{b_k}^{c_0} \\ v_{b_k}^{b_k} \end{bmatrix},$$

Gravity Acc Bias

Gyro Bias

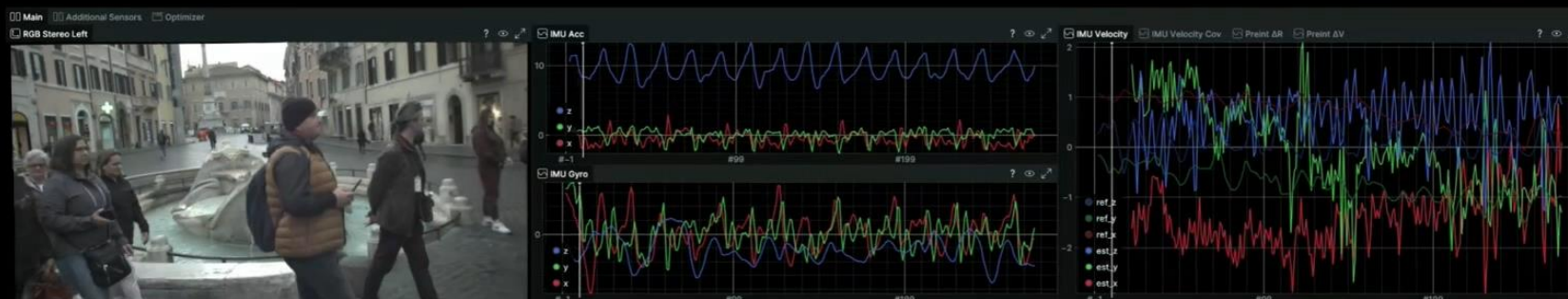
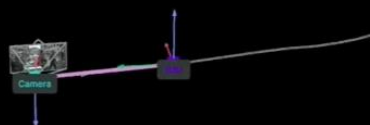
Body Pose Velocity

Cam-IMU
Extrinsic

Background: Initialization for VIO in challenging environment

MAC-I² Initialization: Dynamic Objects

Initializing



Geometry-based methods are prone to **losing tracking**

Human-tuned uncertainty are **suboptimal** and **not generalizable**

Dynamic Objects



 VINS-Fusion

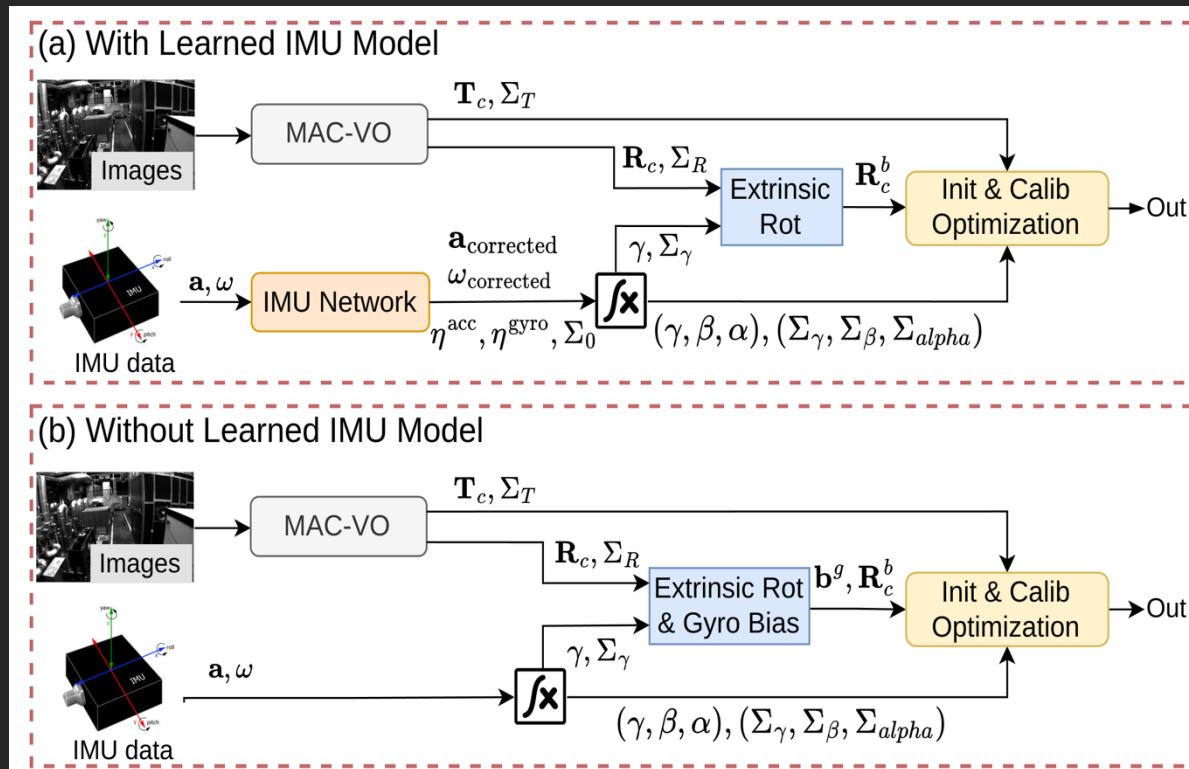
Extreme Exposure



 VINS-Fusion

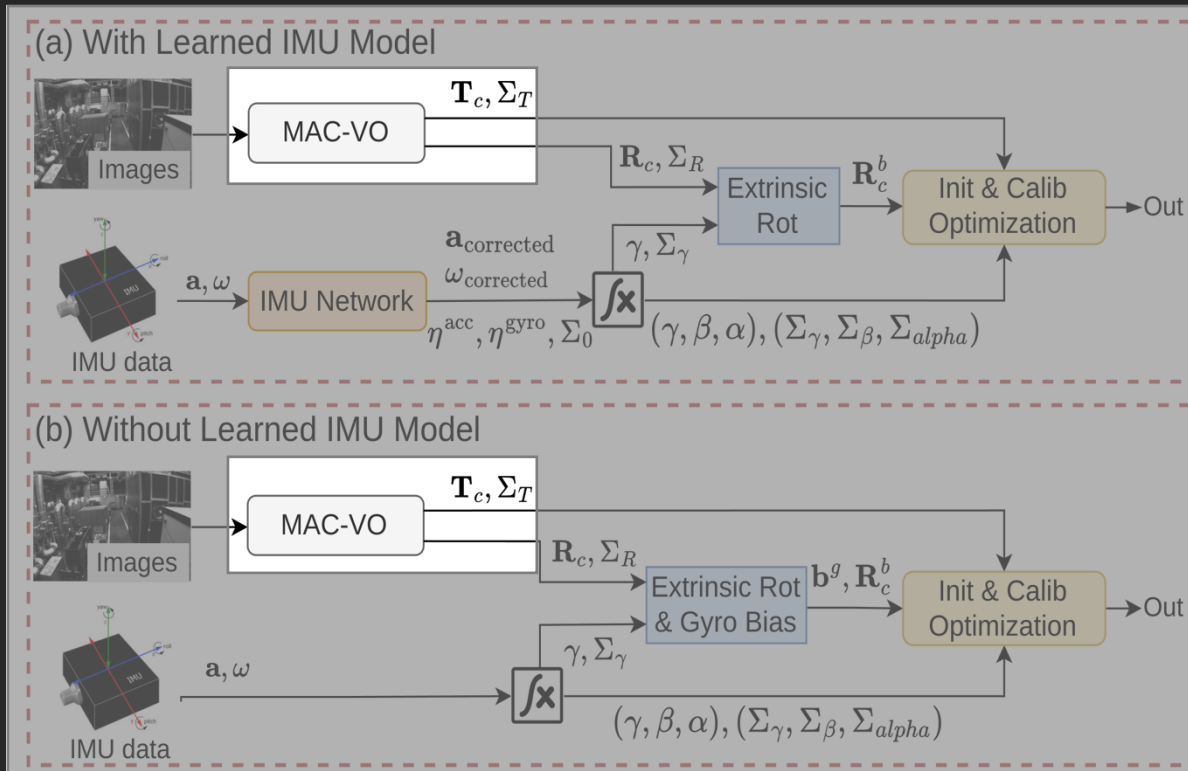
*ORB-SLAM3 failed to initialize in both cases

Contributions of MAC-I²



- A Different framework
- Metrics-aware visual pose covariance
- Metrics-aware IMU

Camera Motion Estimation and Pose Covariance



Recap pose graph optimization as MAC-VO:

$$T_{c_k}^{c_{k-1}*} = \underset{T_{c_k}^{c_{k-1}}}{\operatorname{argmin}} \sum_i |p_{k-1,i}^c - T_{c_k}^{c_{k-1}} p_{k,i}^c|_{\Sigma_i}^2,$$

$$\Sigma_i = \Sigma_{i,k-1}^p + R_{c_k}^{c_{k-1}} \Sigma_{i,k}^p (R_{c_k}^{c_{k-1}})^T,$$

Left perturbation model is used to derive the pose covariance:

$$r_{i,k}(\delta\xi) = p_{k-1,i}^c - \operatorname{Exp}(\delta\xi) T_{c_k}^{c_{k-1}} p_{k,i}^c$$

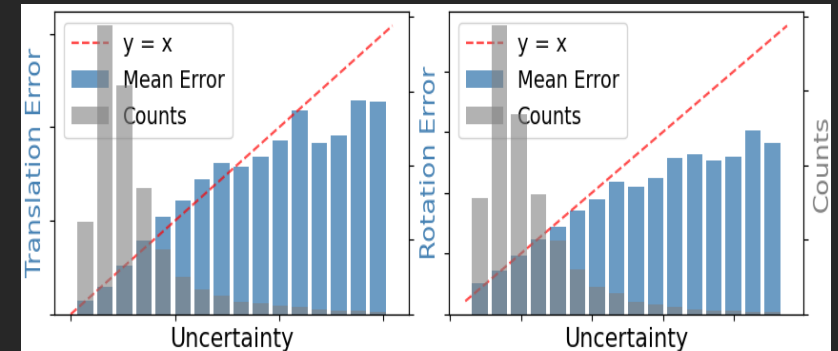
$$J_{i,k} = \frac{\partial r_{i,k}(\delta\xi)}{\partial \delta\xi} = \begin{bmatrix} -I_{3 \times 3} & (T_{c_k}^{c_{k-1}} p_{k,i}^c)^\wedge \end{bmatrix},$$

$$H_k \approx \sum_i (J_{i,k})^T \Sigma_i^{-1} J_{i,k}, \quad \Sigma_{T_{c_k}^{c_{k-1}}} \approx H_k^{-1}$$

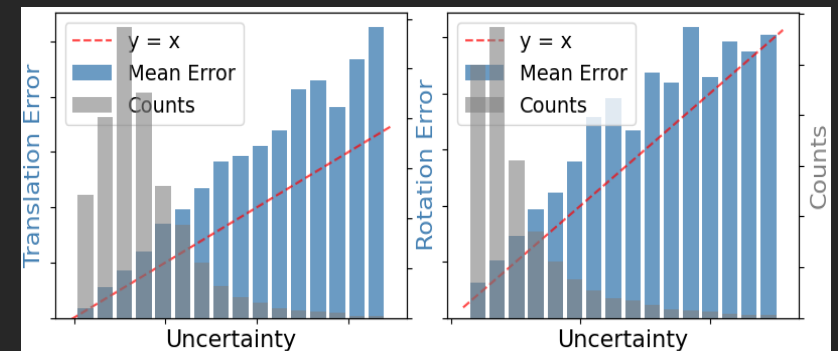
Visual Pose Covariance Evaluation

The model is entirely trained on a synthetic dataset TartanAir

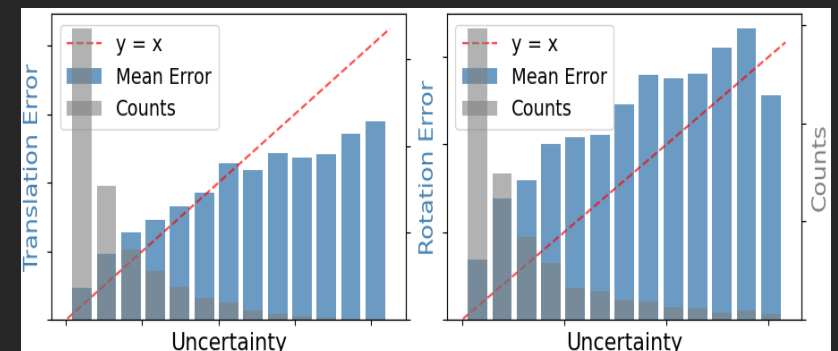
The model generalize across sensor types, motion regimes, and diverse scenes



TartanAir v2



EuRoC

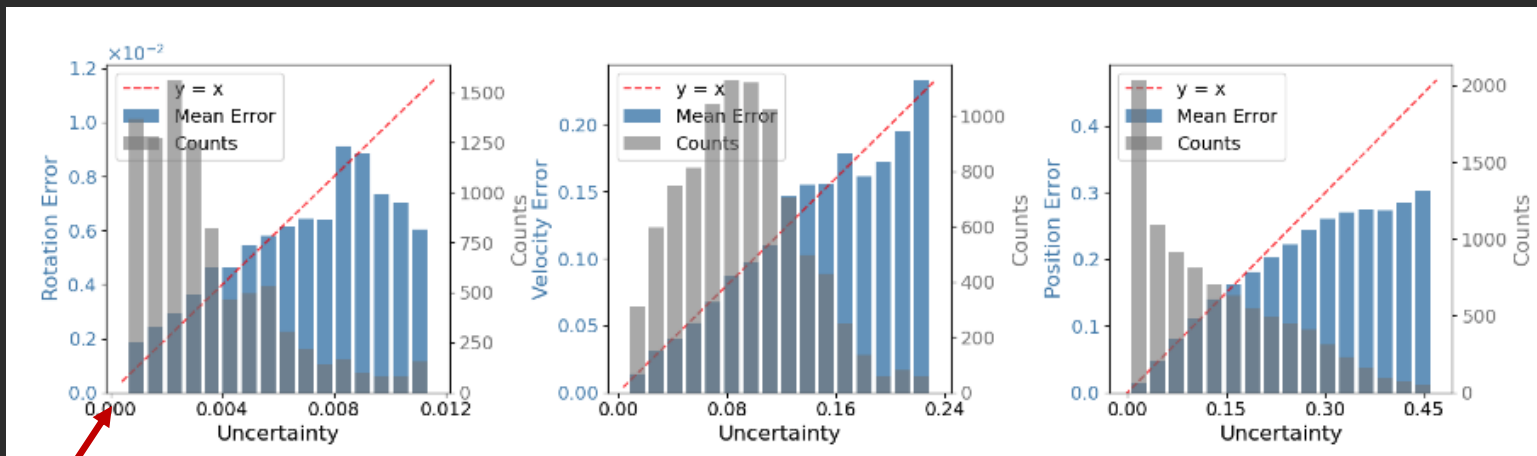
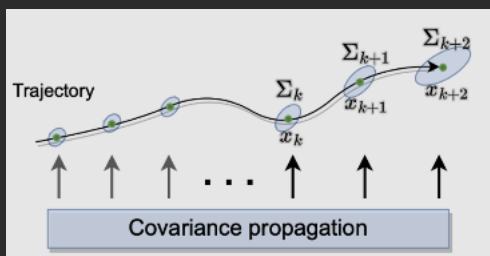


TUM-VI

Metrics-aware Covariance from AirIMU

Metrics-aware IMU covariance estimation on unseen EuRoC sequences

AirIMU



In unseen data, the estimated uncertainty for velocity rotation and position are following the error in the same scale.

Experiments on Initialization

Our method outperform SOTA method in accuracy and also achieving 100% success rate

TABLE I: Detailed initialization results for the 10KFs setting in EuRoC dataset. Only even-ordered trajectory is shown due to page limit. G.Dir Error $> 5^\circ$ or Vel RMSE > 0.5 m/s is treated as failure.

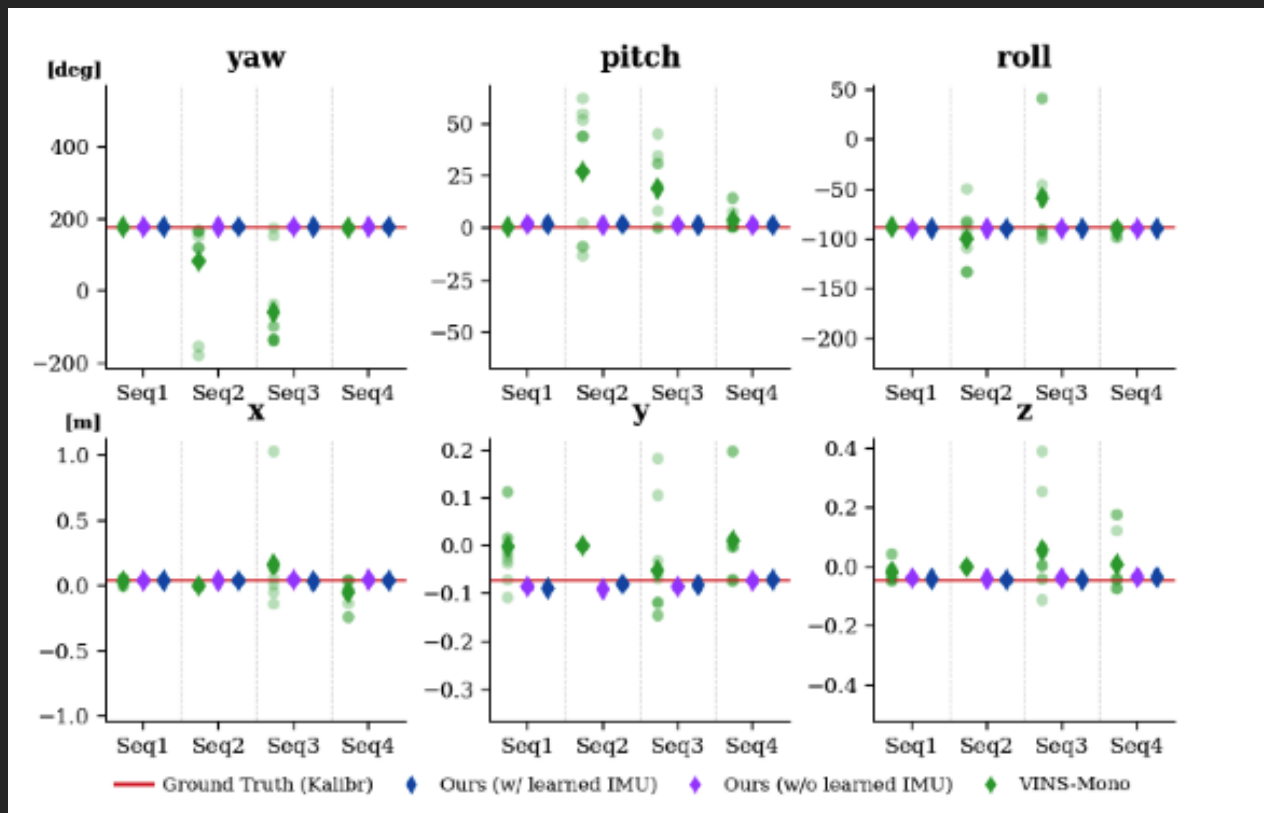
Trajectory	MH02		MH04		V101		V103		V202		Avg.		SR
	Vel(m/s)	G.Dir($^\circ$)	Vel(m/s)	G.Dir($^\circ$)	Vel(m/s)	G.Dir($^\circ$)	Vel(m/s)	G.Dir($^\circ$)	Vel(m/s)	G.Dir($^\circ$)	Vel(m/s)	G.Dir($^\circ$)	
 VINS-Mono	0.136	2.651	0.266	3.225	0.052	2.824	-	-	0.251	3.803	0.176	3.126	0.36
 DRT-1	0.049	0.927	0.106	0.944	0.047	1.011	0.095	1.304	0.054	0.896	0.070	1.016	0.74
 VINS-Fusion	0.120	2.734	0.217	2.960	0.017	2.874	0.183	2.867	0.231	3.844	0.154	3.056	0.48
 ORB-SLAM 3	0.146	2.188	0.229	2.406	0.087	2.111	0.290	2.554	0.325	3.000	0.215	2.452	0.61
 Stereo-NEC	0.148	2.203	0.226	2.439	0.090	2.158	0.273	2.462	0.330	2.839	0.213	2.420	0.60
 MAC-I ² w/o Learned IMU	<u>0.011</u>	<u>0.820</u>	0.022	<u>0.800</u>	0.010	<u>0.872</u>	0.025	<u>1.149</u>	0.020	0.804	0.018	<u>0.889</u>	1.00
 MAC-I ² w/ Learned IMU	0.010	0.166	<u>0.023</u>	0.208	0.010	0.332	0.025	0.581	<u>0.024</u>	0.804	0.018	0.418	1.00

 Monocular method.  Stereo method. - The initialization fails on all segments, or succeeds on too few segments to be statistically meaningful.

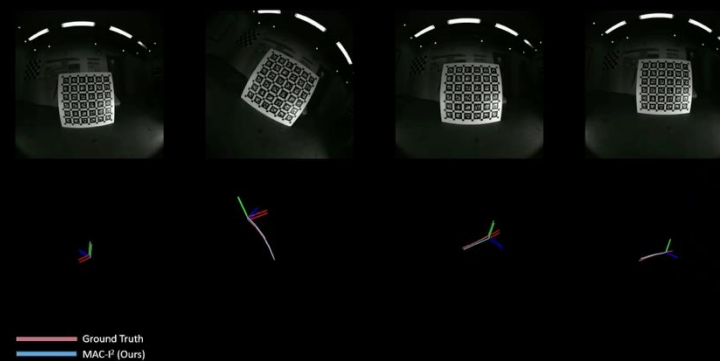


Success in very dark environment on our payload

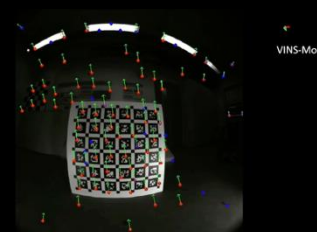
Experiments on Extrinsic Calibration



MAC-I² Calibration: Drastic Motions



Geometry-based methods are prone to **losing tracking**
 Human-tuned uncertainty are **suboptimal** and **not generalizable**

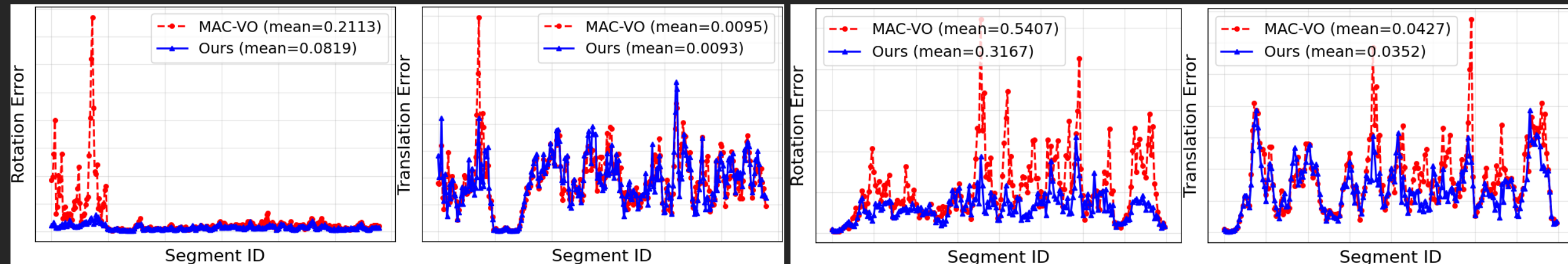


Drastic Motions During Online Calibration

MAC-VIO vs MAC-VO

		MH02	MH04	V101	V103	V202	Avg.
ATE (m)	MAC-VO	.2751	.5629	.1876	.1641	.2359	.2851
	MAC-VIO (w/o Learned IMU)	.2807	.4795	.1562	.2239	.1985	.2678
	MAC-VIO (w/ Learned IMU)	.2281	.2986	.0682	.1897	.1654	.1900

MAC-VIO achieves **33% improvement** on Average Trajectory Error (ATE) over the vision-only MAC-VO baseline



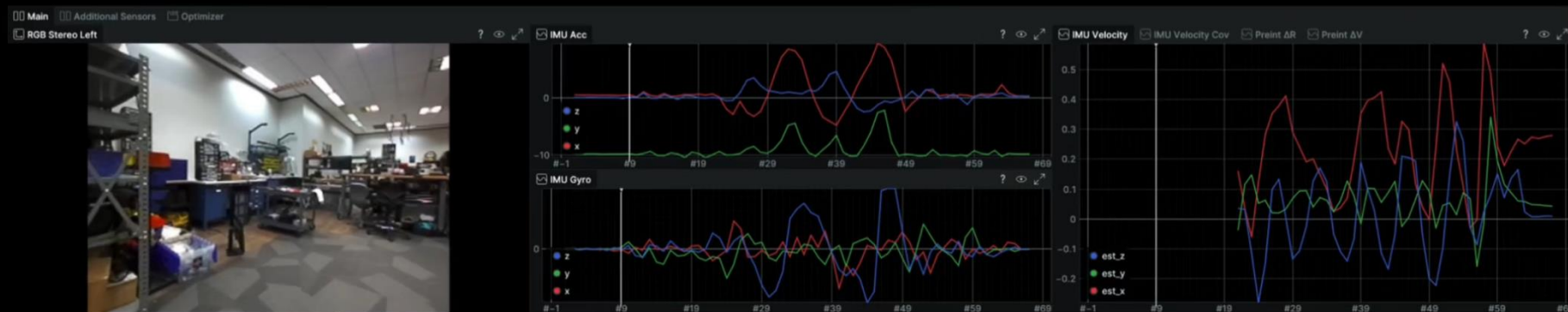
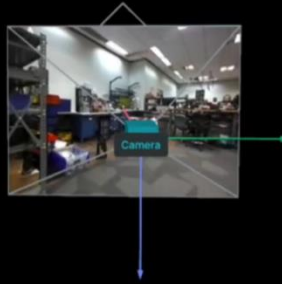
Pose error on MH02 (w/ Learned IMU)

Pose error on V103 (w/ Learned IMU)

MAC-I² + MAC-VIO: On NVIDIA Jetson Thor with ZED 2i Camera



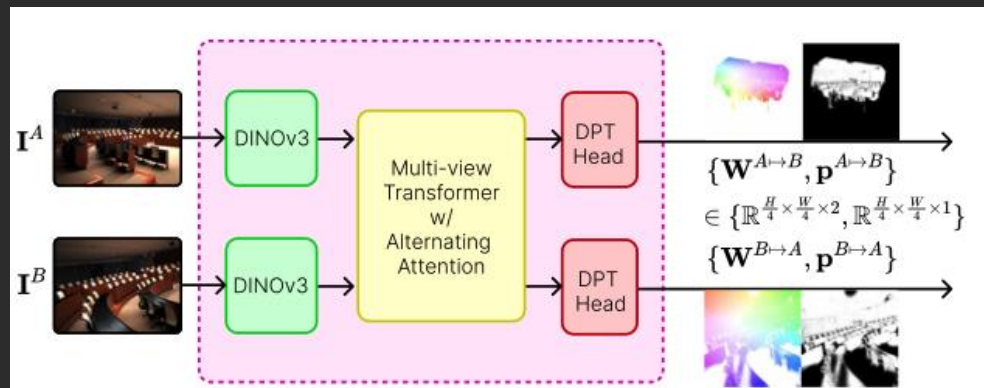
Initializing



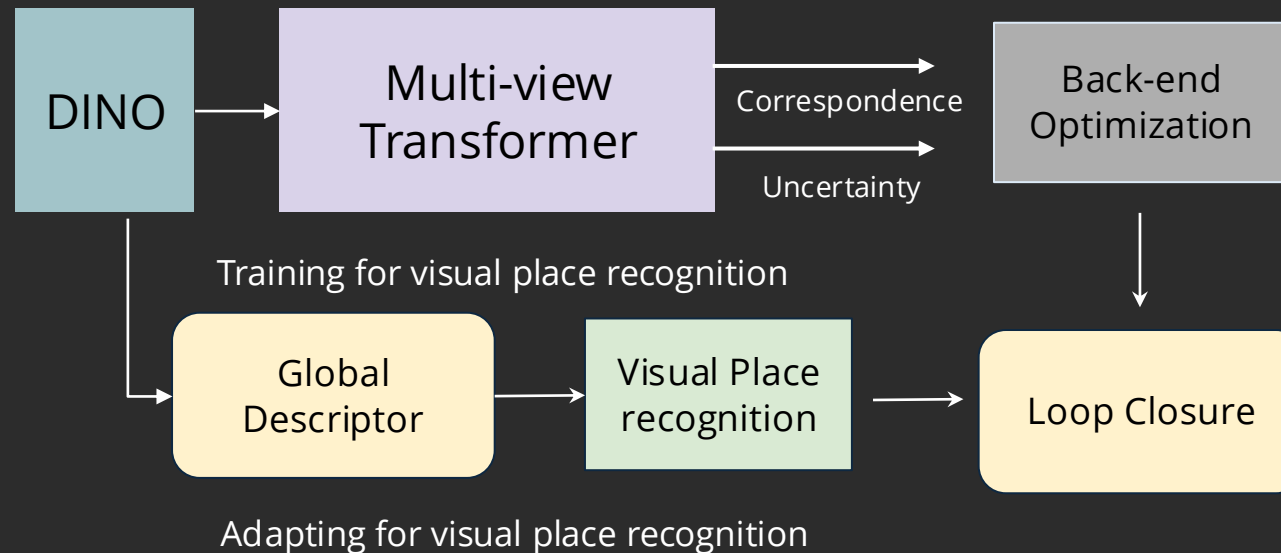
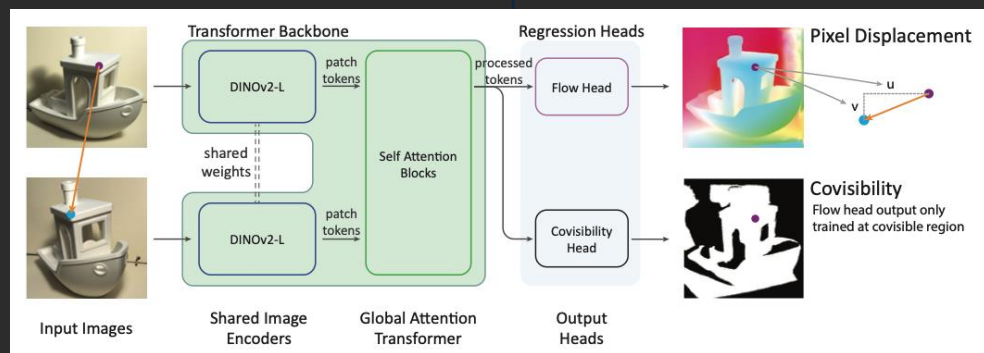
MAC-SLAM: Uncertainty-aware Spatial Perception with semantics and language correspondence



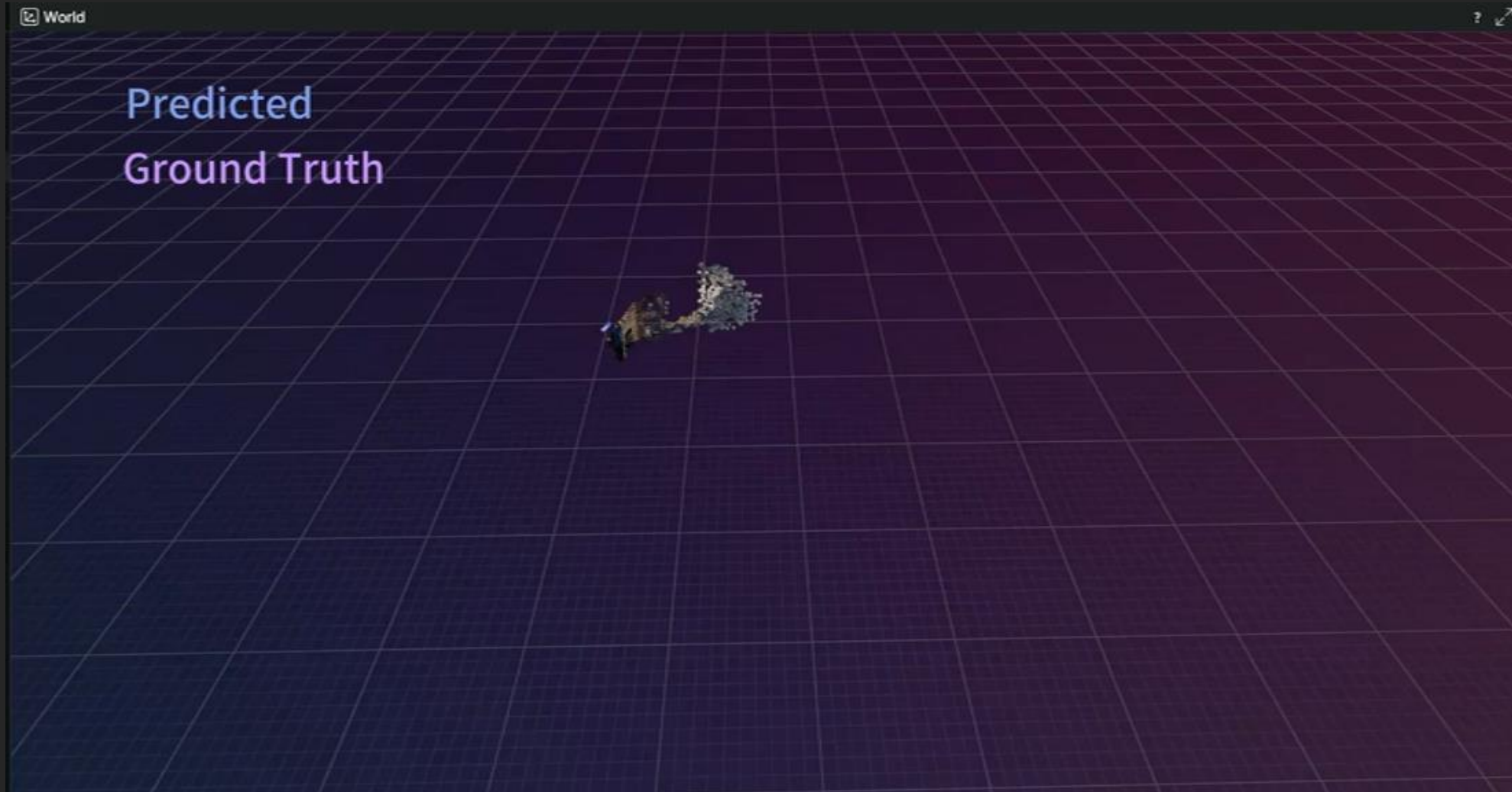
RoMa v2: Harder Better Faster Denser Feature Matching



UFM: A Simple Path towards Unified Dense Correspondence with Flow



MAC-SLAM: Uncertainty-aware Spatial Perception with semantics and language correspondence



Visual Perception

[Part. 1]



[1] AirDOS



[6] TartanAir

[2] MACVO: Learning Stereo Visual Odometry



[3] QuantMACVO: Quantization for Efficient Learning VO



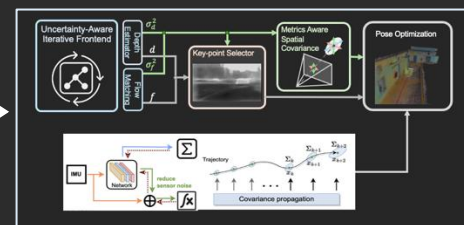
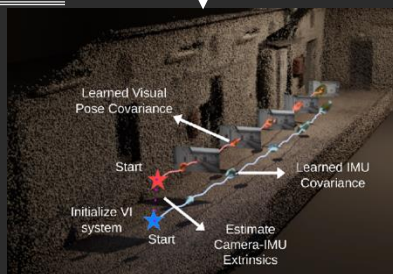
MAC-Thermal

Uncertainty-aware Spatial Perception

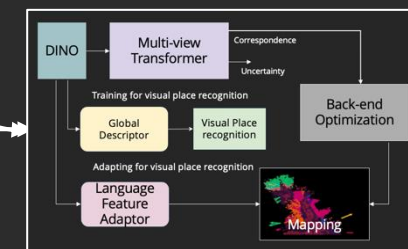
Sensor Fusion & Adaptation

[Part. 3]

[5] MAC-I²: Metrics-aware covariance for IMU initialization and Calibration



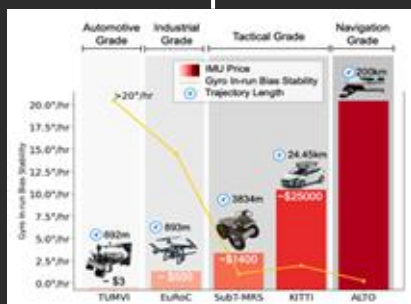
MACVO



MAC-SLAM

Inertial Perception

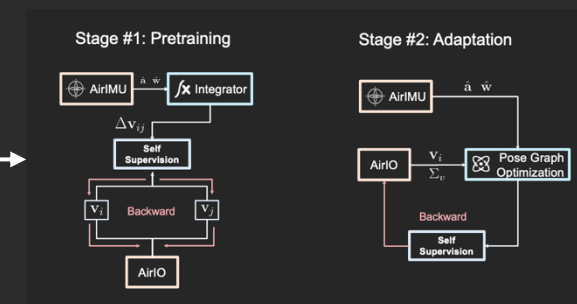
[Part. 2]



[4] AirIMU: Learning IMU sensor model

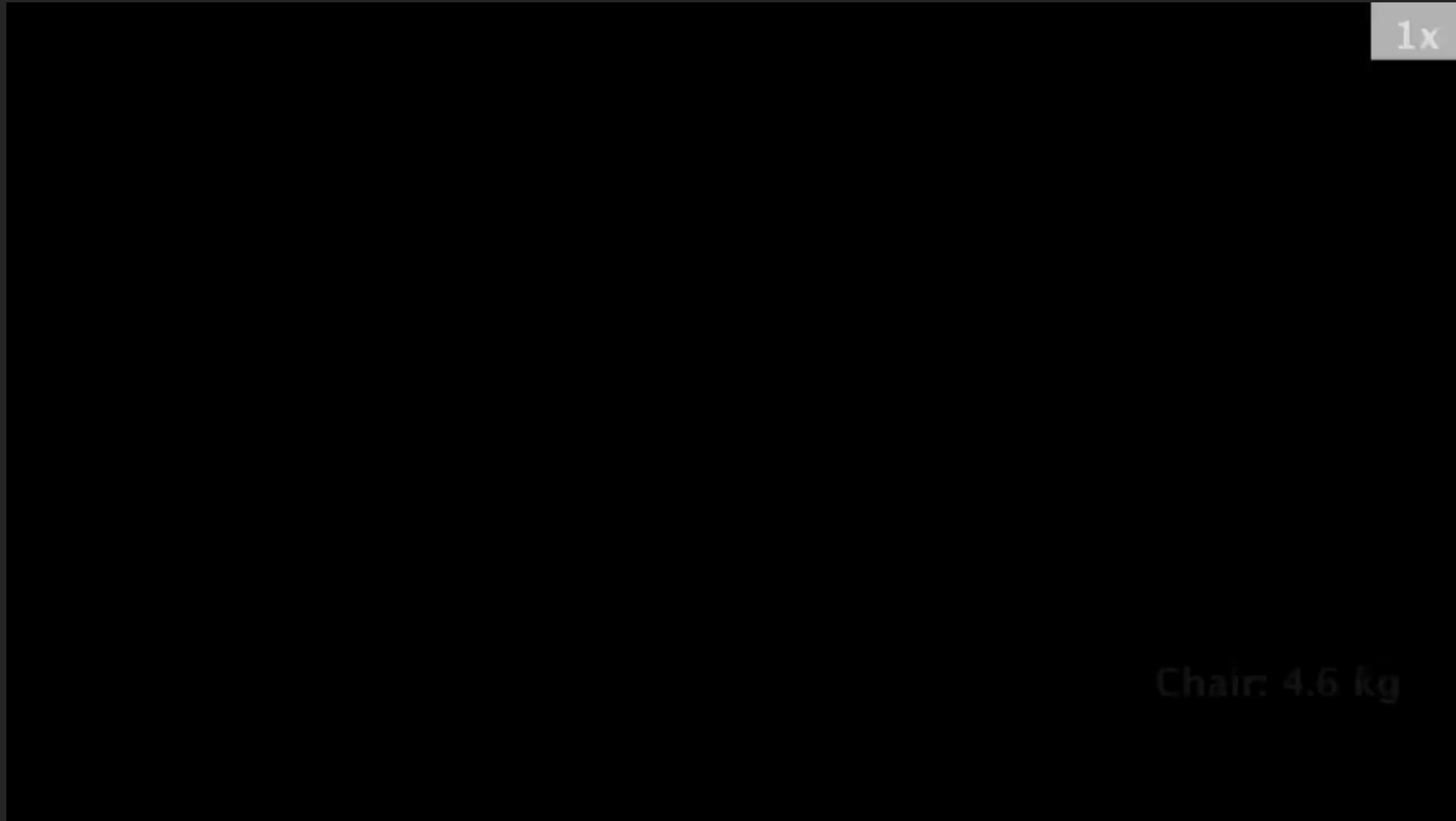


[4] AirIO: Learning IO with enhanced observability



MAC-IO

Future Direction: Reformulate the Objective of State Estimation



OmniRetarget: Interaction-Preserving Data Generation for Humanoid
Whole-Body Loco-Manipulation and Scene Interaction

Moving Toward Spatial Memory

256 Frames Input



...



Co-Me Accelerated VGGT

00:00.00

Original VGGT

00:00.00

[14] Co-Me: Confidence-Guided Token Merging for Visual Geometric Transformers

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- [2] Qiu, Yuheng, et al. "MACVO: MAC-VO: Metrics-Aware Covariance for Learning-based Stereo Visual Odometry." *ICRA* (2025).
- [3] Qiu, Yuheng, et al. "AirIMU: Learning Uncertainty Propagation for Inertial Odometry." *arXiv preprint* (2023).
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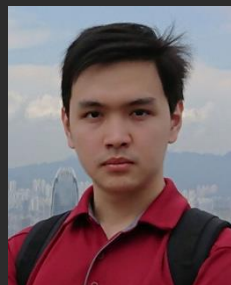
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Bowen Li



Seungchan Kim



Carl Cort



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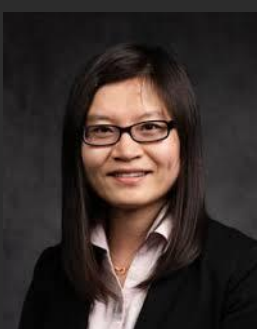


Omar

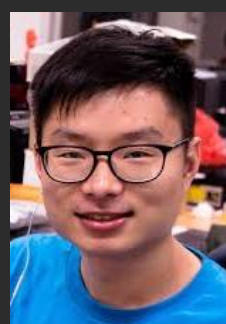
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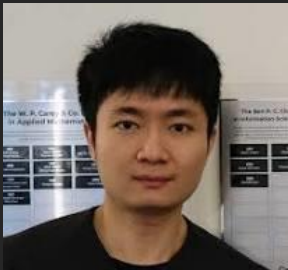
Teams from Amazon FAR



Pieter Abbeel



Rocky Duan



Guanya Shi



Karen Liu



Carmelo Sferrazza



Koushil Sreenath

Limitation of MAC-IO

Training is slow



Integration bias may be problematic

Covariance adaptation assumes Independent input

Outlier Rejection



Trajectory	MAC-VO (Baseline)	w/ Outlier Rejection
EuRoC_MH01	0.001	0.001
EuRoC_MH02	0.0009	0.0009
EuRoC_MH03	0.0034	0.0034
EuRoC_MH04	0.0038	0.0038
EuRoC_MH05	0.0025	0.0025
EuRoC_V101	0.0024	0.0024
EuRoC_V102	0.0026	0.0026
EuRoC_V103	0.0029	0.0029
EuRoC_V201	0.0009	0.0009
EuRoC_V202	0.0019	0.0019
EuRoC_V203	0.0056	0.0061
Average	0.0025	0.0026

Outlier Rejection



Trajectory	MAC-VO (Baseline)	w/ Outlier Rejection
H000	0.0118	0.0113
H001	0.0022	0.0021
H002	0.0179	0.0146
H003	0.0078	0.0077
H004	0.0015	0.0015
H005	0.0127	0.0128
H006	0.0069	0.0062
H007	0.0232	0.0224
Average	0.0105	0.0098

VIO Fusion



VINS-Mono (w/o Loop)	VINS-Fusion (w/o Loop)	ORB-SLAM3 V-I (w/o Loop)	MAC-VO2 (apr27-2026.yaml)	(A b
0.0032	0.0031	0.0036	<u>0.001</u>	
0.0026	0.0029	0.0034	<u>0.0009</u>	
0.0071	0.0073	0.0049	<u>0.0034</u>	
0.007	0.008	0.0054	<u>0.0038</u>	
0.0066	0.0068	0.0054	<u>0.0025</u>	
<i>0.0021</i>	0.0029	<i>0.0021</i>	0.0024	
0.0059	0.0061	0.0069	<u>0.0026</u>	
0.0075	0.0072	0.0046	<u>0.0029</u>	
0.0032	0.0028	0.0027	<u>0.0009</u>	
0.0047	0.0052	0.003	<u>0.0019</u>	
0.0109	0.0095	0.0047	0.0061	
0.005527273	0.005618182	0.004245455	0.002581818	